

Single-Variable Calculus

Complete Worked Solutions · The Curious Family Library, Module Two

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Single-Variable Calculus — Complete Worked Solutions. The Curious Family Library · Mathematics
· Module Two.

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These are the worked solutions to the problems in *Single-Variable Calculus* (Module Two); the main text is a separate volume.

Typeset in Libertinus with pandoc and LaTeX. First edition, 2026.

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A complete worked solution to every problem in *Single-Variable Calculus* (Module Two). Solutions are grouped by chapter and by the book's six problem tiers — **Fluency, Practice, Concept & counterexample, ♦ Diamond, Teach it, and Looking ahead**. In keeping with the book, each solution is solved with only the tools available by that point: the early chapters work *from the definition*, not from rules proved later.

Read a solution only after a real attempt. The goal is never the answer; it is to see the *method*, so you can rebuild it yourself next time.

Chapter 1

Patterns That Go Somewhere

Fluency

1. Plug in $n = 1, 2, 3, 4$. (a) $a_n = 3n - 1$: 2, 5, 8, 11. (b) $a_n = \left(\frac{1}{2}\right)^n$: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$. (c)

$a_n = \frac{n}{n+1}$: $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$. (d) $a_n = \frac{(-1)^n}{n}$: $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}$.

2. Check whether successive terms differ by a constant (arithmetic) or share a constant ratio (geometric). (a) 7, 10, 13, 16: differences all +3. **Arithmetic**, $d = 3$.

(b) 3, -6, 12, -24: ratios all $\times(-2)$. **Geometric**, $r = -2$. (c) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$: differences $-\frac{1}{2}, -\frac{1}{6}, -\frac{1}{12}$ (not constant); ratios $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ (not constant). **Neither**. (d) 2, 5, 10, 17: differences 3, 5, 7 (not constant) — these are $n^2 + 1$; ratios not constant. **Neither**.

3. (a) Arithmetic, $a_n = a_1 + (n-1)d$: $a_{15} = 4 + (15-1)(5) = 4 + 70 = \boxed{74}$. (b) Geometric, $a_n = a_1 r^{n-1}$: $a_6 = 2 \cdot 3^5 = 2 \cdot 243 = \boxed{486}$.

4. (a) $\frac{1}{n} \rightarrow 0$. (b) $\frac{2n-1}{n} = 2 - \frac{1}{n} \rightarrow 2$. (c) $(-1)^n$ bounces $-1, 1, -1, \dots$: **no limit** (oscillates). (d) $(0.8)^n$: $|r| < 1$ so $\rightarrow 0$. (e) $n^2 \rightarrow \infty$ (diverges, no finite limit). (f) $\frac{(-1)^n}{n}$: magnitude $\frac{1}{n} \rightarrow 0$, so $\rightarrow 0$.

5. Divide top and bottom by the highest power of n present, then send each $\frac{1}{n^k} \rightarrow 0$.

(a) $\frac{3n+1}{n+2} = \frac{3 + \frac{1}{n}}{1 + \frac{2}{n}} \rightarrow \frac{3+0}{1+0} = \boxed{3}$. (b) $\frac{n^2-1}{2n^2+n} = \frac{1 - \frac{1}{n^2}}{2 + \frac{1}{n}} \rightarrow \frac{1-0}{2+0} = \boxed{\frac{1}{2}}$. (c)

$\frac{5n}{n^2+1} = \frac{\frac{5}{n}}{1 + \frac{1}{n^2}} \rightarrow \frac{0}{1} = \boxed{0}$. (d) $\frac{2n^3}{n^2+1} = \frac{2n}{1 + \frac{1}{n^2}}$: numerator $2n \rightarrow \infty$, denominator $\rightarrow 1$, so $\boxed{\text{diverges to } \infty}$.

6. Geometric rule: $|r| < 1 \Rightarrow \rightarrow 0$; $r = 1 \Rightarrow \rightarrow 1$; $|r| > 1 \Rightarrow$ blow up; $r = -1 \Rightarrow$ oscillate. $r = \frac{1}{3}$: $|r| < 1 \Rightarrow \rightarrow 0$. $r = -\frac{1}{2}$: $|r| = \frac{1}{2} < 1 \Rightarrow \rightarrow 0$. $r = 1$: $\rightarrow 1$. $r = -1$: oscillates (1, -1, 1, ...), no limit. $r = 2$: $|r| > 1 \Rightarrow$ blows up.

Practice

7. Highest power is n^2 ; divide top and bottom by n^2 :

$$\frac{n^2 - 3n}{2n^2 + 1} = \frac{1 - \frac{3}{n}}{2 + \frac{1}{n^2}} \rightarrow \frac{1 - 0}{2 + 0} = \boxed{\frac{1}{2}}.$$

8. Rationalize by the conjugate $\sqrt{n^2 + n} + n$:

$$\sqrt{n^2 + n} - n = \frac{(n^2 + n) - n^2}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n}.$$

Divide top and bottom by n (and $\sqrt{n^2 + n} = n\sqrt{1 + \frac{1}{n}}$):

$$= \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{\sqrt{1} + 1} = \boxed{\frac{1}{2}}.$$

9. Rationalize, mirroring Fight one:

$$n \left(\sqrt{1 + \frac{1}{n}} - 1 \right) = n \cdot \frac{\left(1 + \frac{1}{n}\right) - 1}{\sqrt{1 + \frac{1}{n}} + 1} = n \cdot \frac{1/n}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}.$$

$$\boxed{\frac{1}{2}}$$

10. Use $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$. (a) $x = 3$: $\left(1 + \frac{3}{n}\right)^n \rightarrow \boxed{e^3}$. (b) $x = -2$: $\left(1 - \frac{2}{n}\right)^n \rightarrow \boxed{e^{-2}}$.

(c) Write $\left(1 + \frac{1}{n}\right)^{3n} = \left[\left(1 + \frac{1}{n}\right)^n\right]^3 \rightarrow e^3$, so $\boxed{e^3}$.

11. Gauge: compute $k = \lim(\text{exponent}) \times (\text{base} - 1)$; the answer is e^k . - $\left(1 + \frac{1}{n}\right)^{n^2}$: $k = n^2 \cdot \frac{1}{n} = n \rightarrow \infty$, so limit = $e^\infty = \boxed{\infty}$ (diverges). - $\left(1 + \frac{1}{n^2}\right)^n$: $k = n \cdot \frac{1}{n^2} = \frac{1}{n} \rightarrow 0$, so limit = $e^0 = \boxed{1}$. - $\left(1 + \frac{5}{n}\right)^n$: $k = n \cdot \frac{5}{n} = 5$, so limit = $\boxed{e^5}$.

12. $a_1 = 1$, $a_{n+1} = \sqrt{12 + a_n}$. $a_2 = \sqrt{12 + 1} = \sqrt{13} \approx 3.606$. $a_3 = \sqrt{12 + \sqrt{13}} = \sqrt{15.606} \approx 3.950$. Let L satisfy the rule: $L = \sqrt{12 + L} \Rightarrow L^2 = 12 + L \Rightarrow L^2 - L - 12 = 0 \Rightarrow (L - 4)(L + 3) = 0$. Terms are positive, so reject -3 .

$$\boxed{L = 4}$$

13. With $a_1 = 1$, $r = 0.9$ the n -th term is $a_n = a_1 r^{n-1} = 0.9^{n-1}$. Want $0.9^{n-1} < 0.001$. Take logs (any base); since $\ln(0.9) < 0$, divide and flip:

$$n - 1 > \frac{\ln(0.001)}{\ln(0.9)} = \frac{-6.9078}{-0.10536} \approx 65.56, \quad \text{so } n - 1 \geq 66, \text{ i.e. } n \geq 67.$$

The first term below 0.001 is the 67th ($a_{66} = 0.9^{65} \approx 0.00106$, still above; $a_{67} = 0.9^{66} \approx 0.000956$, below).

14. Split the fraction:

$$\frac{n + (-1)^n}{n} = \frac{n}{n} + \frac{(-1)^n}{n} = 1 + \frac{(-1)^n}{n}.$$

The first piece is 1; the second has magnitude $\frac{1}{n} \rightarrow 0$ (problem 4f). Sum of limits:

$$\rightarrow 1 + 0 = \boxed{1}.$$

Concept & counterexample

15. (a) **False.** $a_n \rightarrow L$ guarantees the terms are within 0.01 of L only *eventually* — for all $n > N$. Early terms may be far away (e.g. $a_n = \frac{1}{n} \rightarrow 0$ has $a_1 = 1$, not within 0.01 of 0). (b) **False.** Bounded does not imply convergent: $a_n = (-1)^n$ is bounded by 1 but oscillates and has no limit. (c) **False.** $a_n \rightarrow 0$ says nothing about monotonicity: $a_n = \frac{(-1)^n}{n}$ converges to 0 while alternating sign, never eventually decreasing. (d) **False.** Limits are unique — a sequence cannot converge to two different values (proved in problem 28).

16. $a_n = (-1)^n$ gives $-1, 1, -1, 1, \dots$: bounded between -1 and 1 , but it never gets and stays near any single number, so it does not converge. (Any oscillating bounded sequence works, e.g. $\sin n$.)

17. $a_n = n$ ($1, 2, 3, \dots$) is increasing but unbounded, so it diverges to ∞ . The missing hypothesis is **bounded above** — Monotone Convergence requires both “increasing” and “bounded above,” and this example has only the first.

18. $a_n = 5 + \frac{(-1)^n}{n} = 5 - 1, 5 + \frac{1}{2}, 5 - \frac{1}{3}, \dots$. The correction term $\frac{(-1)^n}{n} \rightarrow 0$, so $a_n \rightarrow 5$, while the alternating sign puts terms below 5 (odd n) and above 5 (even n) forever.

19. **The error.** “Gets arbitrarily close infinitely often” is **not** the definition of a

limit. The definition demands that *past some cutoff N , every term stays within ε* – not merely that infinitely many do. The sequence $(-1)^n$ is near 1 only on even seats and near -1 on odd seats, so for a small ε (say $\varepsilon = \frac{1}{2}$) no single N makes *all* later terms within ε of 1 (the odd terms equal -1). What is true: 1 and -1 are **subsequential limits** (limits of sub-lists), but the sequence itself has **no limit**. A convergent sequence has exactly one limit.

20. Yes, it is possible. Take $a_n = \frac{1}{n}$. Then $a_n \rightarrow 0$, yet the running totals are the harmonic sums $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$, which grow without bound. (Sketch: grouping $\frac{1}{3} + \frac{1}{4} > \frac{1}{2}$, $\frac{1}{5} + \cdots + \frac{1}{8} > \frac{1}{2}$, each block of doubling length adds more than $\frac{1}{2}$, so the totals exceed any ceiling.) This previews infinite series in Chapter 12: terms shrinking to 0 does **not** force the sum to be finite.

21. - Drop “bounded above”: $a_n = n$ is increasing but unbounded – it diverges to ∞ , no limit. So “bounded above” is load-bearing. - **Drop “increasing”:** $a_n = (-1)^n$ is bounded above (by 1) but not increasing – it oscillates, no limit. So “increasing” is load-bearing. Each hypothesis alone is insufficient; the principle needs both.

◆ Diamond

22. Let $\varepsilon > 0$ be given. We need N so that $n > N \Rightarrow \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon$. Now $\frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}$. By the Archimedean property there is an integer $N \geq \frac{1}{\varepsilon}$. Then for every $n > N \geq \frac{1}{\varepsilon}$ we have $n > \frac{1}{\varepsilon}$, i.e. $\frac{1}{n} < \varepsilon$. Since ε was arbitrary, $\frac{1}{n} \rightarrow 0$. ■

23. Let $\varepsilon > 0$. We need $\left| \frac{1}{n^2} - 0 \right| = \frac{1}{n^2} < \varepsilon$, i.e. $n^2 > \frac{1}{\varepsilon}$, i.e. $n > \frac{1}{\sqrt{\varepsilon}}$. Choose an integer $N \geq \frac{1}{\sqrt{\varepsilon}}$. Then for $n > N$, $n^2 > \frac{1}{\varepsilon}$, so $\frac{1}{n^2} < \varepsilon$. Hence $\frac{1}{n^2} \rightarrow 0$. ■ (Alternatively: for $n \geq 1$, $\frac{1}{n^2} \leq \frac{1}{n}$, and $\frac{1}{n} \rightarrow 0$, so squeeze between 0 and $\frac{1}{n}$.)

24. First simplify the distance:

$$\left| \frac{n}{n+1} - 1 \right| = \left| \frac{n - (n+1)}{n+1} \right| = \left| \frac{-1}{n+1} \right| = \frac{1}{n+1}.$$

Let $\varepsilon > 0$. We need $\frac{1}{n+1} < \varepsilon \iff n+1 > \frac{1}{\varepsilon} \iff n > \frac{1}{\varepsilon} - 1$. Choose an integer $N \geq \frac{1}{\varepsilon}$ (so certainly $N > \frac{1}{\varepsilon} - 1$). Then for $n > N$, $\frac{1}{n+1} < \frac{1}{n} \leq \frac{1}{N} \leq \varepsilon$... more directly $n+1 > n > \frac{1}{\varepsilon}$ gives $\frac{1}{n+1} < \varepsilon$. Hence $\frac{n}{n+1} \rightarrow 1$. ■

25. Distance: $\left| \frac{2n+1}{n} - 2 \right| = \left| \frac{2n+1-2n}{n} \right| = \frac{1}{n}$. Let $\varepsilon > 0$; we need $\frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}$. Choose integer $N \geq \frac{1}{\varepsilon}$. Then $n > N \Rightarrow \frac{1}{n} < \varepsilon$. Hence $\frac{2n+1}{n} \rightarrow 2$. ■

26. Given $a_n \rightarrow 5$. - $\frac{1}{n} \rightarrow 0$ (workhorse fact / problem 22). - By the **constant-multiple law**, $3a_n \rightarrow 3 \cdot 5 = 15$. - By the **constant-multiple law** again, $-\frac{1}{n} \rightarrow -1 \cdot 0 = 0$ (or: $\frac{1}{n} \rightarrow 0$). - By the **sum law**, $3a_n - \frac{1}{n} \rightarrow 15 + 0 = 15$.

$$\lim \left(3a_n - \frac{1}{n} \right) = 15$$

27. The two walls are $a_n = -\frac{1}{n}$ and $c_n = \frac{1}{n}$. Since $-1 \leq \sin n \leq 1$ for all n , dividing by $n > 0$ gives

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}.$$

Both walls $\rightarrow 0$ (problem 22), so by the **Squeeze Theorem** $\frac{\sin n}{n} \rightarrow 0$. $\sin n$ never needed a limit of its own because the squeeze only uses that it is *trapped* in $[-1, 1]$ — the walls, not the wild middle, carry the term to 0. ■

28. **Uniqueness of limits.** Suppose, for contradiction, $a_n \rightarrow L$ and $a_n \rightarrow M$ with $L \neq M$. Set $\varepsilon = \frac{|L-M|}{3} > 0$. Because $a_n \rightarrow L$, there is N_1 with $|a_n - L| < \varepsilon$ for $n > N_1$. Because $a_n \rightarrow M$, there is N_2 with $|a_n - M| < \varepsilon$ for $n > N_2$. For any $n > \max(N_1, N_2)$ both hold, so by the triangle inequality

$$|L - M| \leq |L - a_n| + |a_n - M| < \varepsilon + \varepsilon = \frac{2|L - M|}{3}.$$

This says $|L-M| < \frac{2}{3}|L-M|$, i.e. $1 < \frac{2}{3}$, impossible (since $|L-M| > 0$). Contradiction. Hence $L = M$: a sequence has at most one limit. This is exactly why **15(d)** is false. ■

29. Sequence $a_1 = 1$, $a_{n+1} = \sqrt{12 + a_n}$ (problem 12). **Bounded above by 4** (induction): $a_1 = 1 < 4$. If $a_n < 4$, then $a_{n+1} = \sqrt{12 + a_n} < \sqrt{12 + 4} = \sqrt{16} = 4$. So $a_n < 4$ for all n . **Increasing:** we show $a_{n+1} > a_n$, i.e. $\sqrt{12 + a_n} > a_n$. Both sides nonnegative, so square: $12 + a_n > a_n^2$, i.e. $a_n^2 - a_n - 12 < 0$, i.e. $(a_n - 4)(a_n + 3) < 0$. This holds whenever $-3 < a_n < 4$, which is guaranteed since $0 < a_n < 4$. So each term exceeds the last. **Converges:** increasing and bounded above $\Rightarrow$ by Monotone Convergence the limit L exists. Taking limits of both sides of the rule (as in problem 12) gives $L = 4$. ■

30. **Bernoulli's inequality** $(1+h)^n \geq 1+nh$ for $h > -1$, $n \geq 1$, by induction on n . Base $n = 1$: $(1+h)^1 = 1+h \geq 1+1 \cdot h$, equality. ✓ Step: assume $(1+h)^n \geq 1+nh$.

Since $h > -1$, $1 + h > 0$, so multiplying the inequality by $1 + h$ preserves it:

$$(1 + h)^{n+1} \geq (1 + nh)(1 + h) = 1 + nh + h + nh^2 = 1 + (n + 1)h + nh^2.$$

Because $nh^2 \geq 0$, $1 + (n + 1)h + nh^2 \geq 1 + (n + 1)h$. Therefore $(1 + h)^{n+1} \geq 1 + (n + 1)h$.

By induction the inequality holds for all $n \geq 1$. ■

31. Show $\left(1 + \frac{1}{n}\right)^n < 3$ for all n . Expand by the binomial theorem:

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^n \frac{1}{k!} \cdot \frac{n(n-1) \cdots (n-k+1)}{n^k}.$$

Bound each term by $\frac{1}{k!}$: the fraction $\frac{n(n-1) \cdots (n-k+1)}{n^k} = \prod_{j=0}^{k-1} \frac{n-j}{n} \leq 1$ (each factor ≤ 1), so the k -th term $\leq \frac{1}{k!}$. Hence

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

Bound $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$ for $k \geq 1$: since $k! = 1 \cdot 2 \cdot 3 \cdots k \geq 2^{k-1}$ (each of the $k - 1$ factors $2, \dots, k$ is ≥ 2). Thus for $k \geq 1$, $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$, giving

$$\sum_{k=0}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \frac{1}{2^{k-1}} = 1 + \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}}\right).$$

Sum the geometric bound: $1 + \frac{1}{2} + \cdots + \frac{1}{2^{n-1}} = \frac{1 - (1/2)^n}{1 - 1/2} = 2\left(1 - \frac{1}{2^n}\right) < 2$. Therefore

$$\left(1 + \frac{1}{n}\right)^n < 1 + 2 = 3$$

for every n . ■

32. Show $\sqrt[n]{2^n + 3^n} \rightarrow 3$. Factor out the dominant 3^n :

$$\sqrt[n]{2^n + 3^n} = \sqrt[n]{3^n \left(1 + \left(\frac{2}{3}\right)^n\right)} = 3 \left(1 + \left(\frac{2}{3}\right)^n\right)^{1/n}.$$

Bound the bracketed factor. Since $0 \leq \left(\frac{2}{3}\right)^n \leq 1$, we have $1 \leq 1 + \left(\frac{2}{3}\right)^n \leq 2$, so

$$1 = 1^{1/n} \leq \left(1 + \left(\frac{2}{3}\right)^n\right)^{1/n} \leq 2^{1/n}.$$

The right wall $2^{1/n} \rightarrow 1$ (it is $\sqrt[n]{2}$; e.g. by the $n^{1/n} \rightarrow 1$ idea, or since $2^{1/n} = e^{(\ln 2)/n}$ with exponent $\rightarrow 0$), and the left wall is 1. By the Squeeze Theorem the bracket $\rightarrow 1$, hence

$$\sqrt[n]{2^n + 3^n} = 3 \cdot \left(1 + \left(\frac{2}{3}\right)^n\right)^{1/n} \rightarrow 3 \cdot 1 = 3. \quad \blacksquare$$

Teach it

33. To a friend: “Saying a limit is ‘what the sequence gets closer and closer to’ sounds right, but it’s unsafe because a sequence can edge closer for a while and then walk away — ‘closer for a while’ isn’t a promise. Take 1, 0.5, 0.9, 0.99, 1.5, 0.5, 1.5, 0.5, ...: at first it crowds toward 1, but then it settles into bouncing between 0.5 and 1.5 and never homes in on anything. The honest definition fixes this: past some point in the list, *every* term must stay within whatever tiny tolerance you name of the target. That word *every* is what ‘gets closer and closer’ quietly leaves out.”

34. Worked solution to $\lim_{n \rightarrow \infty} \frac{3n^2 + 2n}{n^2 - 5}$. We have a ratio of polynomials, and plugging in “ $n = \infty$ ” gives the indeterminate $\frac{\infty}{\infty}$ — so we can’t just substitute. The standard move is to *divide top and bottom by the highest power of n that appears*, here n^2 . This is legal because for $n \geq 1$ we are dividing by a nonzero quantity, and it leaves the value of the fraction unchanged:

$$\frac{3n^2 + 2n}{n^2 - 5} = \frac{\frac{3n^2 + 2n}{n^2}}{\frac{n^2 - 5}{n^2}} = \frac{3 + \frac{2}{n}}{1 - \frac{5}{n^2}}.$$

Now the point of the trick shows: every term carrying a $\frac{1}{n}$ heads to 0 as n grows. Concretely $\frac{2}{n} \rightarrow 0$ and $\frac{5}{n^2} \rightarrow 0$ (the workhorse fact $\frac{1}{n^k} \rightarrow 0$). By the limit laws — the quotient of the limits, since the bottom limit isn’t 0 — the expression approaches

$$\frac{3 + 0}{1 - 0} = 3.$$

So the limit is $\boxed{3}$. In words: the lower-degree pieces ($2n$ and -5) become negligible next to the n^2 terms, and what survives is the ratio of leading coefficients, $\frac{3}{1}$.

35. Debug. The flawed step is treating “ $\left(1 + \frac{1}{n}\right)^n$ ” as “(lim of the base)^{fixed power}” — i.e. letting the base go to 1 *while freezing the exponent*. But the exponent is **not**

fixed; it grows to ∞ at the same time the base shrinks to 1, and you may not take the limit in one slot while holding the other. This is the indeterminate form 1^∞ , whose value depends on the *race* between base and exponent. **Correction (one sentence):** Because the exponent grows with n , the form is 1^∞ (indeterminate), and the true limit is e , not 1.

36. Design a problem. *Problem:* Let $a_1 = 1$ and $a_{n+1} = \sqrt{c + a_n}$. Choose c so the limit is exactly 5, and prove convergence. *Finding c :* a limit $L = 5$ must satisfy $L = \sqrt{c + L}$, so $25 = c + 5$, giving $\boxed{c = 20}$. So use $a_{n+1} = \sqrt{20 + a_n}$. *Solution key* (four steps): 1. **Bounded above by 5** (induction): $a_1 = 1 < 5$; if $a_n < 5$ then $a_{n+1} = \sqrt{20 + a_n} < \sqrt{20 + 5} = \sqrt{25} = 5$. 2. **Increasing:** $a_{n+1} > a_n \iff \sqrt{20 + a_n} > a_n \iff 20 + a_n > a_n^2 \iff a_n^2 - a_n - 20 < 0 \iff (a_n - 5)(a_n + 4) < 0$, true for $0 < a_n < 5$ (guaranteed by Step 1). 3. **Converges:** increasing and bounded above $\Rightarrow$ a limit L exists (Monotone Convergence). 4. **Solve for L :** taking limits of both sides, $L = \sqrt{20 + L} \Rightarrow L^2 - L - 20 = 0 \Rightarrow (L - 5)(L + 4) = 0 \Rightarrow L = 5$ (reject -4 , terms positive). ■

37. Bounded vs. convergent. *Bounded* means the terms stay inside some fixed window — there's a ceiling and a floor they never cross. *Convergent* means more: the terms actually home in on one specific number and stay arbitrarily close to it. Convergence forces boundedness, but boundedness does **not** force convergence — a bounded sequence still has room to wander forever inside its window. Example: $a_n = (-1)^n$ stays trapped in $[-1, 1]$ (bounded) yet keeps jumping between -1 and 1 , never settling, so it does not converge.

Looking ahead

38. Factor the numerator and cancel the common factor $x - 2$ (valid for $x \neq 2$, which is all that matters as $x \rightarrow 2$):

$$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2 \quad (x \neq 2).$$

As $x \rightarrow 2$ this approaches $2 + 2 = \boxed{4}$.

39. With $h = 0.001$:

$$\frac{2^{0.001} - 1}{0.001} \approx \frac{1.000693 - 1}{0.001} \approx 0.693 (< 1), \quad \frac{3^{0.001} - 1}{0.001} \approx \frac{1.0010992 - 1}{0.001} \approx 1.099 (> 1).$$

The “slope = 1” base gives ratio exactly 1; since base 2 undershoots ($0.693 < 1$) and base 3 overshoots ($1.099 > 1$), that special base lies **between 2 and 3**. (It is $e \approx 2.718$.)

40. Newton’s iteration is $x \leftarrow x - \frac{f(x)}{f'(x)}$. If it settles to a value L , then L is a fixed point of the rule, so applying the rule leaves it unchanged:

$$L = L - \frac{f(L)}{f'(L)} \implies \frac{f(L)}{f'(L)} = 0 \implies \boxed{f(L) = 0}$$

(assuming $f'(L) \neq 0$). So the limit must be a **root** of f — exactly what Newton’s method hunts for.

Chapter 2

Getting Arbitrarily Close

Fluency

1. Each is continuous at the target point, so substitution gives the limit.

(a) $\lim_{x \rightarrow 1} (x^3 - 2x + 1) = 1^3 - 2(1) + 1 = 1 - 2 + 1.$

0

(b) $\lim_{x \rightarrow -2} (3x + 1) = 3(-2) + 1 = -6 + 1.$

-5

(c) $\lim_{x \rightarrow 0} \cos x = \cos 0.$

1

(d) $\lim_{x \rightarrow 9} \sqrt{x} = \sqrt{9}.$

3

2. Substitution gives $\frac{0}{0}$ in each; factor and cancel the common factor (legal since the canceled factor is nonzero on the approach).

(a) $\frac{x^2 - 5x + 6}{x - 2} = \frac{(x - 2)(x - 3)}{x - 2} = x - 3 \rightarrow 2 - 3.$

-1

(b) $\frac{x^3 - 1}{x - 1} = \frac{(x - 1)(x^2 + x + 1)}{x - 1} = x^2 + x + 1 \rightarrow 1 + 1 + 1.$

3

$$(c) \frac{x^2 - x - 6}{x^2 - 9} = \frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \rightarrow \frac{5}{6}.$$

$$\boxed{\frac{5}{6}}$$

3. Substitution gives $\frac{0}{0}$; multiply by the conjugate to clear the root.

$$(a) \frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} = \frac{x - 4}{(x - 4)(\sqrt{x} + 2)} = \frac{1}{\sqrt{x} + 2} \rightarrow \frac{1}{2 + 2}.$$

$$\boxed{\frac{1}{4}}$$

$$(b) \frac{\frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}}{\frac{1}{1+1}} = \frac{(x+1)-1}{x(\sqrt{x+1}+1)} = \frac{x}{x(\sqrt{x+1}+1)} = \frac{1}{\sqrt{x+1}+1} \rightarrow \frac{1}{1+1}.$$

$$\boxed{\frac{1}{2}}$$

4. Divide top and bottom by the highest power of x present.

$$(a) \frac{3x^3 - x}{2x^3 + 5} = \frac{3 - 1/x^2}{2 + 5/x^3} \rightarrow \frac{3 - 0}{2 + 0}.$$

$$\boxed{\frac{3}{2}}$$

$$(b) \frac{5x + 1}{x^2 + 2} = \frac{5/x + 1/x^2}{1 + 2/x^2} \rightarrow \frac{0 + 0}{1 + 0}. \text{ (Denominator grows faster.)}$$

$$\boxed{0}$$

$$(c) \frac{4x^2 - 3}{2x^2 + x} = \frac{4 - 3/x^2}{2 + 1/x} \rightarrow \frac{4 - 0}{2 + 0}.$$

$$\boxed{2}$$

$$(d) \frac{2x^2 + 1}{x + 1} = \frac{2x + 1/x}{1 + 1/x}. \text{ The numerator } 2x \rightarrow \infty \text{ while the denominator } \rightarrow 1. \\ \text{(Top degree exceeds bottom degree.)}$$

$$\boxed{+\infty \text{ (the limit does not exist as a finite number; it diverges to } +\infty)}$$

5. $f(x) = x$ for $x < 1$, $f(x) = x + 2$ for $x \geq 1$.

- From the left, use $f(x) = x$: $\lim_{x \rightarrow 1^-} f = 1$.
- From the right, use $f(x) = x + 2$: $\lim_{x \rightarrow 1^+} f = 1 + 2 = 3$.

The one-sided limits exist but disagree ($1 \neq 3$), so the two-sided limit does not exist.

$$\lim_{x \rightarrow 1^-} f = 1, \quad \lim_{x \rightarrow 1^+} f = 3, \quad \lim_{x \rightarrow 1} f \text{ does not exist (a jump).}$$

6. One line each.

- $\frac{1}{x}$ at $x = 2$: defined ($\frac{1}{2}$) and no break nearby, so $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2} = f(2)$ — **continuous**.
- $\frac{1}{x}$ at $x = 0$: $f(0)$ is undefined and the function blows up (an infinite discontinuity) — **not continuous**.
- $\sqrt{x}$ at $x = -1$: $\sqrt{-1}$ is not a real number, so $f(-1)$ doesn't exist (the point isn't in the domain) — **not continuous there**.

$$\text{continuous at 2; not at 0; } \sqrt{x} \text{ not continuous at } -1.$$

7. Each function is continuous (polynomials; $\cos x - x$ is a sum of continuous functions). Check the sign at the endpoints; a sign change forces a root by the IVT (the value 0 lies between $f(a)$ and $f(b)$).

- (a) $f(x) = x^3 + x - 1$: $f(0) = -1 < 0$, $f(1) = 1 + 1 - 1 = 1 > 0$. Sign change $\Rightarrow$ a root in $(0, 1)$. root exists in $[0, 1]$.
- (b) $f(x) = \cos x - x$: $f(0) = 1 - 0 = 1 > 0$, $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Sign change $\Rightarrow$ a root in $(0, 1)$. root exists in $[0, 1]$.
- (c) $f(x) = x^5 - 3x + 1$: $f(0) = 1 > 0$, $f(1) = 1 - 3 + 1 = -1 < 0$. Sign change $\Rightarrow$ a root in $(0, 1)$. root exists in $[0, 1]$.

8. The EVT guarantees a max and a min only when f is **continuous** on a **closed, bounded** interval $[a, b]$. Check both.

- (a) x^2 on $[-1, 3]$: continuous, interval closed. Yes. (It attains min 0 at $x = 0$ and max 9 at $x = 3$.)
- (b) $\frac{1}{x}$ on $(0, 1]$: the interval is **not closed** (open at 0), and $\frac{1}{x} \rightarrow +\infty$ as $x \rightarrow 0^+$, so there is no maximum. No.

- (c) x on $[0, 5)$: the interval is **not closed** (open at 5); values climb toward 5 but never reach it, so no maximum. No.
- (d) Any continuous function on $[2, 7]$: continuous on a closed bounded interval – the EVT's hypotheses are exactly met. Yes.

Practice

9. Substitution gives $\frac{0}{0}$. Factor a difference of cubes on top and a difference of squares on the bottom; cancel the common $(x - 3)$:

$$\frac{x^3 - 27}{x^2 - 9} = \frac{(x - 3)(x^2 + 3x + 9)}{(x - 3)(x + 3)} = \frac{x^2 + 3x + 9}{x + 3} \rightarrow \frac{9 + 9 + 9}{6} = \frac{27}{6}.$$

$$\boxed{\frac{9}{2}}$$

10. Combine the small fractions first: $\frac{1}{x} - \frac{1}{3} = \frac{3-x}{3x}$. Then

$$\frac{\frac{1}{x} - \frac{1}{3}}{x - 3} = \frac{3 - x}{3x(x - 3)} = \frac{-(x - 3)}{3x(x - 3)} = \frac{-1}{3x} \rightarrow \frac{-1}{3 \cdot 3}.$$

$$\boxed{-\frac{1}{9}}$$

11. This is $\infty - \infty$. Rationalize using the conjugate $\sqrt{x^2 + 3x} + x$:

$$\sqrt{x^2 + 3x} - x = \frac{(x^2 + 3x) - x^2}{\sqrt{x^2 + 3x} + x} = \frac{3x}{\sqrt{x^2 + 3x} + x}.$$

Divide top and bottom by x (for $x > 0$, $\sqrt{x^2 + 3x}/x = \sqrt{1 + 3/x}$):

$$= \frac{3}{\sqrt{1 + 3/x} + 1} \rightarrow \frac{3}{\sqrt{1 + 1} + 1} = \frac{3}{2}.$$

$$\boxed{\frac{3}{2}}$$

12. Same move with conjugate $\sqrt{4x^2 + x} + 2x$:

$$\sqrt{4x^2 + x} - 2x = \frac{(4x^2 + x) - 4x^2}{\sqrt{4x^2 + x} + 2x} = \frac{x}{\sqrt{4x^2 + x} + 2x}.$$

Divide by x (for $x > 0$, $\sqrt{4x^2 + x}/x = \sqrt{4 + 1/x}$):

$$= \frac{1}{\sqrt{4 + 1/x} + 2} \rightarrow \frac{1}{\sqrt{4} + 2} = \frac{1}{4}.$$

$$\boxed{\frac{1}{4}}$$

13. Each piece is continuous on its own; only the junctions $x = 1$ and $x = 3$ can break. For continuity the pieces must agree there.

At $x = 1$: left/value piece x^2 gives 1; middle piece $ax + b$ gives $a + b$. Match:

$$a + b = 1.$$

At $x = 3$: middle piece $ax + b$ gives $3a + b$; right piece gives 4. Match:

$$3a + b = 4.$$

Subtract: $2a = 3 \Rightarrow a = \frac{3}{2}$, then $b = 1 - \frac{3}{2} = -\frac{1}{2}$.

$$\boxed{a = \frac{3}{2}, \quad b = -\frac{1}{2}.}$$

(Check: at $x = 1$, $\frac{3}{2} + (-\frac{1}{2}) = 1 = 1^2$ ✓; at $x = 3$, $\frac{9}{2} - \frac{1}{2} = 4$ ✓.)

14. Three-part test at $x = 1$.

- **Value:** $f(1) = 2(1) = 2$ (from the $x \geq 1$ piece). Exists.
- **Left limit:** for $x < 1$, $f(x) = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1 \rightarrow 1 + 1 = 2$.
- **Right limit:** for $x > 1$, $f(x) = 2x \rightarrow 2$.

Both one-sided limits equal 2, so $\lim_{x \rightarrow 1} f = 2$, and this equals $f(1) = 2$. All three line up.

Yes, f is continuous at $x = 1$.

15. $f(x) = x^3 + x - 1$, root known to lie in $[0, 1]$.

- **Step 1 — midpoint 0.5:** $f(0.5) = 0.125 + 0.5 - 1 = -0.375 < 0$. Since $f(0) = -1 < 0$ and $f(1) = 1 > 0$, the sign change is between 0.5 and 1. Keep $[0.5, 1]$.
- **Step 2 — midpoint 0.75:** $f(0.75) = 0.421875 + 0.75 - 1 = 0.171875 > 0$. Now $f(0.5) < 0$ and $f(0.75) > 0$: the sign change is in $[0.5, 0.75]$.

The root is trapped in $[0.5, 0.75]$ (width 0.25).

16. Model temperature as a function $T(t)$ that is continuous in time t , with $T(6 \text{ a.m.}) = 40$ and $T(2 \text{ p.m.}) = 75$. The value 60 lies between 40 and 75. By the IVT, a continuous function on $[6 \text{ a.m.}, 2 \text{ p.m.}]$ takes every value between its endpoint values, so there is some moment c with $T(c) = 60$.

Why it can't say *when*: the IVT is an **existence** theorem — it guarantees a crossing point exists but uses no information about *where*. Temperature could pass 60° once, or wiggle up and down across it many times; many different time-graphs share the same two endpoint readings, so the endpoints alone cannot pin down the time(s).

A moment with $T = 60^\circ$ must exist (IVT); the theorem gives existence, not location.

Concept & counterexample

17.

- (a) **False.** The limit ignores the value at a . Counterexample: $f(x) = \frac{x^2 - 1}{x - 1}$ has $\lim_{x \rightarrow 1} f = 2$ but $f(1)$ doesn't even exist. (Or define $f(1) = 10$: limit still $2 \neq 10$.) False.
- (b) **False.** Knowing the value says nothing about the approach. Counterexample: $f(x) = 1$ for $x \neq a$ and $f(a) = 5$. Then $f(a) = 5$ but $\lim_{x \rightarrow a} f = 1$. False.
- (c) **False.** Both one-sided limits can exist yet disagree, in which case the two-sided limit fails. Counterexample: $f(x) = \frac{|x|}{x}$ at $a = 0$ has left limit -1 , right limit $+1$, both existing, but no two-sided limit. False.
- (d) **False.** Two non-limits can cancel. Counterexample: at $a = 0$ let $f(x) = \frac{|x|}{x}$ (no limit) and $g(x) = -\frac{|x|}{x}$ (no limit). Then $f + g = 0$ everywhere except 0, so $\lim_{x \rightarrow 0} (f + g) = 0$ exists. False.

18. Build a removable discontinuity: take a function whose approach value is 4 but whose actual value at 2 is 10:

$$f(x) = \begin{cases} x + 2, & x \neq 2, \\ 10, & x = 2. \end{cases}$$

Then $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$, while $f(2) = 10$. The limit exists but disagrees with the value, so this is a **removable** discontinuity.

$f(x) = x + 2$ for $x \neq 2$, $f(2) = 10$; removable discontinuity.
--

19. Need a jump at 0 and a removable hole at 1. One clean piecewise build:

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & 0 \leq x < 1, \\ 5, & x = 1, \\ 1, & x > 1. \end{cases}$$

- At $x = 0$: left limit 0, right limit 1 — both exist, disagree: a **jump**. - At $x = 1$: $\lim_{x \rightarrow 1} f = 1$ (both sides) but $f(1) = 5$: a **removable hole**.

Sketch: a flat segment at height 0 for $x < 0$; a flat segment at height 1 for $x \geq 0$ (other than $x = 1$), with a step up at 0; and an isolated dot at $(1, 5)$ above an open circle at $(1, 1)$.

Function above: jump at 0, removable hole at 1.

20. The error: “ $\frac{0}{0}$ undefined $\Rightarrow$ limit does not exist” confuses the *value* of the expression at $x = 3$ with the *limit*. $\frac{0}{0}$ is an **indeterminate form**, not a verdict — it signals a hidden common factor to cancel, not that the limit fails. Correctly:

$$\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3 \rightarrow 3 + 3.$$

The limit is 6; $\frac{0}{0}$ means “rewrite,” not “no limit.”
--

21. Reuse the cancellation idea multiplicatively. At $a = 0$ let

$$f(x) = \frac{|x|}{x}, \quad g(x) = \frac{|x|}{x}.$$

Each has no limit at 0 (left -1 , right $+1$). But their product is $\left(\frac{|x|}{x}\right)^2 = 1$ for all $x \neq 0$, so $\lim_{x \rightarrow 0} f(x)g(x) = 1$ exists.

$$f = g = \frac{|x|}{x}; \quad fg \equiv 1 \text{ for } x \neq 0, \text{ so } \lim_{x \rightarrow 0} fg = 1.$$

(Equally, $g(x) = -\frac{|x|}{x}$ gives product $\equiv -1$.)

22. A hole can be repaired **only when it is a removable discontinuity** — i.e. only when $\lim_{x \rightarrow 2} f(x)$ exists (both one-sided limits agree on a finite value L). Then defining $f(2) = L$ — *and nothing else* — makes f continuous at 2.

They are **wrong** that “any number you like” works: continuity requires $f(2)$ to equal that limit L exactly; any other choice leaves a removable discontinuity. And if the two-sided limit doesn’t exist (a jump or a blow-up at 2), **no** choice of $f(2)$ can make it continuous.

$$\text{Right only if } \lim_{x \rightarrow 2} f \text{ exists, and only with } f(2) = \lim_{x \rightarrow 2} f.$$

23.

- (a) **False.** Same-sign endpoints don’t forbid a root in between. Counterexample: $f(x) = x^2 - \frac{1}{4}$ on $[-1, 1]$ has $f(-1) = f(1) = \frac{3}{4} > 0$ (same sign) yet roots at $x = \pm \frac{1}{2}$ inside. (The IVT gives a *sufficient* condition for a root, not a necessary one.) False.
- (b) **False.** On an open interval the peak can sit at a missing endpoint. Counterexample: $f(x) = x$ on $(0, 1)$ is continuous but attains no maximum (values approach 1, never reach it). False.
- (c) **True.** If f is continuous on $[a, b]$ and takes a positive value and a negative value there, then 0 lies between two attained values, so by the IVT f equals 0 somewhere in the interval. True.
- (d) **False.** The IVT only guarantees *at least one* crossing; it counts nothing. Example: $f(x) = \sin x$ on $[0, 10\pi]$ has many roots, while the IVT applied to

a single sign change promises just one. False.

24. Break continuity (the EVT also needs a closed bounded interval, but here $[0, 1]$ is already closed and bounded — so the only hypothesis left to violate is continuity). Example:

$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 0, & x = 1. \end{cases}$$

Values climb toward 1 as $x \rightarrow 1^-$ but the supremum 1 is never attained (the function drops to 0 at the endpoint), so **no maximum** exists. The broken hypothesis is **continuity** (the jump at $x = 1$).

Why any such example must be discontinuous: $[0, 1]$ is closed and bounded, so if f were continuous the EVT would *force* a maximum to exist. Failing to attain a max on a closed bounded interval is therefore only possible when continuity fails.

$f(x) = x \text{ on } [0, 1), f(1) = 0; \text{ breaks continuity (a jump).}$

◆ Diamond

25. Claim: $\lim_{x \rightarrow 2} (3x - 1) = 5$.

Let $\varepsilon > 0$ be given. We need $\delta > 0$ so that $0 < |x - 2| < \delta \Rightarrow |(3x - 1) - 5| < \varepsilon$.
Work backward:

$$|(3x - 1) - 5| = |3x - 6| = 3|x - 2|.$$

This is $< \varepsilon$ exactly when $|x - 2| < \frac{\varepsilon}{3}$. **Choose** $\delta = \frac{\varepsilon}{3}$. Then $0 < |x - 2| < \delta$ gives

$$|(3x - 1) - 5| = 3|x - 2| < 3\delta = 3 \cdot \frac{\varepsilon}{3} = \varepsilon.$$

This works for every $\varepsilon > 0$, so the limit is 5. ■

$\delta = \varepsilon/3.$

26. Claim: $\lim_{x \rightarrow 3} x^2 = 9$.

Let $\varepsilon > 0$. Factor the target: $|x^2 - 9| = |x - 3||x + 3|$. The factor $|x + 3|$ must be bounded; restrict $\delta \leq 1$ first. Then $|x - 3| < 1$ means $2 < x < 4$, so $5 < x + 3 < 7$,

giving $|x + 3| < 7$. Hence

$$|x^2 - 9| = |x - 3| |x + 3| < 7|x - 3|.$$

To force this below ε , also require $|x - 3| < \frac{\varepsilon}{7}$. **Choose** $\delta = \min(1, \frac{\varepsilon}{7})$. Then $0 < |x - 3| < \delta$ gives both $|x + 3| < 7$ and $|x - 3| < \frac{\varepsilon}{7}$, so

$$|x^2 - 9| < 7|x - 3| < 7 \cdot \frac{\varepsilon}{7} = \varepsilon. \quad \blacksquare$$

$\delta = \min(1, \varepsilon/7).$

27. Claim: $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$.

Let $\varepsilon > 0$. Compute the target:

$$\left| \frac{1}{x} - \frac{1}{2} \right| = \left| \frac{2 - x}{2x} \right| = \frac{|x - 2|}{2|x|}.$$

The denominator $|x|$ must be kept away from 0. Restrict $\delta \leq 1$: then $|x - 2| < 1$ means $1 < x < 3$, so $|x| > 1$, i.e. $\frac{1}{|x|} < 1$. Hence

$$\left| \frac{1}{x} - \frac{1}{2} \right| = \frac{|x - 2|}{2|x|} < \frac{|x - 2|}{2}.$$

To force this below ε , require $|x - 2| < 2\varepsilon$. **Choose** $\delta = \min(1, 2\varepsilon)$. Then $0 < |x - 2| < \delta$ gives $|x| > 1$ and $|x - 2| < 2\varepsilon$, so

$$\left| \frac{1}{x} - \frac{1}{2} \right| < \frac{|x - 2|}{2} < \frac{2\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

$\delta = \min(1, 2\varepsilon).$

28. Sum law. Suppose $\lim_{x \rightarrow a} f = L$ and $\lim_{x \rightarrow a} g = M$. Let $\varepsilon > 0$.

Apply each hypothesis with tolerance $\frac{\varepsilon}{2}$: - there is $\delta_1 > 0$ with $0 < |x - a| < \delta_1 \Rightarrow |f(x) - L| < \frac{\varepsilon}{2}$; - there is $\delta_2 > 0$ with $0 < |x - a| < \delta_2 \Rightarrow |g(x) - M| < \frac{\varepsilon}{2}$.

Choose $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - a| < \delta$ both conditions hold, and by the triangle inequality

$$|(f + g)(x) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\lim_{x \rightarrow a}(f + g) = L + M$. ■

29. Squeeze-by-zero lemma. Suppose $|f(x)| \leq M$ for all x near a (with $M \geq 0$) and $g(x) \rightarrow 0$. Show $f(x)g(x) \rightarrow 0$.

If $M = 0$ then $f \equiv 0$ near a and $fg \equiv 0$, done. Take $M > 0$. Let $\varepsilon > 0$. Since $g \rightarrow 0$, there is $\delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |g(x)| < \frac{\varepsilon}{M}$ (and x is close enough that $|f(x)| \leq M$ holds). Then

$$|f(x)g(x) - 0| = |f(x)| |g(x)| \leq M \cdot |g(x)| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

So $f(x)g(x) \rightarrow 0$. ■

One-line application. With $f(x) = \sin \frac{1}{x}$ (so $|f(x)| \leq 1 = M$) and $g(x) = x^2 \rightarrow 0$, the lemma gives $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$.

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0.$$

30. Squeeze Theorem. Suppose $g(x) \leq f(x) \leq h(x)$ for all x near a , and $\lim_{x \rightarrow a} g = \lim_{x \rightarrow a} h = L$. Show $\lim_{x \rightarrow a} f = L$.

Let $\varepsilon > 0$. Choose: $-\delta_1$ with $0 < |x - a| < \delta_1 \Rightarrow |g(x) - L| < \varepsilon$, i.e. $L - \varepsilon < g(x)$; $-\delta_2$ with $0 < |x - a| < \delta_2 \Rightarrow |h(x) - L| < \varepsilon$, i.e. $h(x) < L + \varepsilon$; $-\delta_3$ such that the sandwich $g(x) \leq f(x) \leq h(x)$ holds for $0 < |x - a| < \delta_3$.

Choose $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then for $0 < |x - a| < \delta$,

$$L - \varepsilon < g(x) \leq f(x) \leq h(x) < L + \varepsilon,$$

so $L - \varepsilon < f(x) < L + \varepsilon$, i.e. $|f(x) - L| < \varepsilon$. Hence $\lim_{x \rightarrow a} f = L$. ■

31. Claim: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$.

Multiply by the conjugate $\frac{1 + \cos x}{1 + \cos x}$ and use $1 - \cos^2 x = \sin^2 x$:

$$\frac{1 - \cos x}{x} = \frac{(1 - \cos x)(1 + \cos x)}{x(1 + \cos x)} = \frac{1 - \cos^2 x}{x(1 + \cos x)} = \frac{\sin^2 x}{x(1 + \cos x)}.$$

Split into known limits:

$$= \frac{\sin x}{x} \cdot \frac{\sin x}{1 + \cos x}.$$

As $x \rightarrow 0$: $\frac{\sin x}{x} \rightarrow 1$, and $\frac{\sin x}{1 + \cos x} \rightarrow \frac{0}{1 + 1} = 0$. The product tends to $1 \cdot 0 = 0$.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

32. Manufacture the $\frac{\sin(\text{stuff})}{\text{stuff}}$ shape in each.

(a) $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x}$. As $x \rightarrow 0$, $5x \rightarrow 0$, so $\frac{\sin 5x}{5x} \rightarrow 1$. Limit = $5 \cdot 1$.

$$\boxed{5}$$

(b) $\frac{\sin 2x}{\sin 3x} = \frac{\sin 2x}{2x} \cdot \frac{3x}{\sin 3x} \cdot \frac{2x}{3x} = \frac{\sin 2x}{2x} \cdot \frac{1}{\frac{\sin 3x}{3x}} \cdot \frac{2}{3}$. As $x \rightarrow 0$ both sine ratios $\rightarrow 1$, leaving $1 \cdot 1 \cdot \frac{2}{3}$.

$$\boxed{\frac{2}{3}}$$

(c) $\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x}$. As $x \rightarrow 0$, $\frac{\sin x}{x} \rightarrow 1$ and $\frac{1}{\cos x} \rightarrow \frac{1}{1} = 1$. Limit = $1 \cdot 1$.

$$\boxed{1}$$

33. **Fixed point.** Let $f : [0, 1] \rightarrow [0, 1]$ be continuous. Define $g(x) = f(x) - x$, which is continuous on $[0, 1]$ (difference of continuous functions). Examine the endpoints, using that f 's outputs lie in $[0, 1]$:

$$g(0) = f(0) - 0 = f(0) \geq 0, \quad g(1) = f(1) - 1 \leq 1 - 1 = 0.$$

If $g(0) = 0$ then $f(0) = 0$, a fixed point; if $g(1) = 0$ then $f(1) = 1$, a fixed point. Otherwise $g(0) > 0$ and $g(1) < 0$, a sign change, so by the IVT there is $c \in (0, 1)$ with $g(c) = 0$, i.e. $f(c) = c$. In every case a fixed point exists. ■

$$\boxed{\text{Some } c \in [0, 1] \text{ has } f(c) = c.}$$

Teach it

34. (Model answer, to the classmate.)

$\frac{0}{0}$ doesn't mean the limit fails — it's a flag that the expression has a *common factor* you haven't canceled yet, not a final answer. A limit asks what the function approaches *near* the point, not its value *at* it, so a hole there is fine. For example, $\frac{x^2 - 9}{x - 3}$ gives $\frac{0}{0}$ at $x = 3$, but factoring turns it into $\frac{(x - 3)(x + 3)}{x - 3} = x + 3$, which sails to 6 as $x \rightarrow 3$. The rule is: hit $\frac{0}{0}$, then *rewrite* (factor and cancel, or rationalize) and try again — only after that do you decide whether the limit exists.”

35. (Model worked solution.)

Problem. Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$.

Solution. Substituting $x = 0$ gives $\frac{\sqrt{4} - 2}{0} = \frac{0}{0}$ — indeterminate, so there is genuine work to do. The obstruction is the difference of a square root and a number on top; we remove it by multiplying top and bottom by the **conjugate** $\sqrt{x+4} + 2$. Why this helps: $(\sqrt{x+4} - 2)(\sqrt{x+4} + 2)$ is a difference of squares, $(x+4) - 4 = x$, which clears the root and exposes a factor of x that cancels the troublesome x in the denominator.

$$\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}.$$

Now the form is no longer $\frac{0}{0}$, so substitution is safe:

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{\sqrt{4} + 2} = \frac{1}{4}.$$

Takeaway for a new problem: when a $\frac{0}{0}$ limit contains $\sqrt{} \pm (\text{number})$, multiply by its conjugate; the difference of squares clears the root and usually cancels the factor causing the $\frac{0}{0}$.

$\frac{1}{4}$

36. **The error:** a two-sided limit exists *only* if the left- and right-hand limits **agree**; you may not “average” disagreeing one-sided limits. Here the left limit is -1 and the right limit is $+1$; because $-1 \neq +1$ the two-sided limit **does not exist**

at all — averaging them to 0 invents a value the function never approaches (note $\frac{|x|}{x}$ never equals 0).

What is actually true: $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$, $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = +1$, and therefore $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist (a jump).

No averaging: the limit DNE because the one-sided limits $-1, +1$ disagree.

37. (Designed problem + full key.)

Problem. Let

$$f(x) = \begin{cases} \frac{x^2 - 1}{x + 1}, & x \neq -1 \text{ and } x < 2, \\ 3, & x = -1, \\ x + 5, & x \geq 2. \end{cases}$$

Classify the discontinuity of f at $x = -1$ and at $x = 2$, and give the one-sided limits at $x = 2$.

Solution key. - **At $x = -1$ (removable hole).** For $x \neq -1$, $\frac{x^2 - 1}{x + 1} = \frac{(x - 1)(x + 1)}{x + 1} = x - 1$, so $\lim_{x \rightarrow -1} f = -1 - 1 = -2$. But $f(-1) = 3 \neq -2$. The two-sided limit exists yet disagrees with the value: a **removable discontinuity** (repairable by setting $f(-1) = -2$). - **At $x = 2$ (jump).** Left side uses $x - 1$: $\lim_{x \rightarrow 2^-} f = 2 - 1 = 1$. Right side uses $x + 5$: $\lim_{x \rightarrow 2^+} f = 2 + 5 = 7$. Both exist but $1 \neq 7$, so the two-sided limit fails: a **jump discontinuity** (of size $7 - 1 = 6$).

$x = -1$: removable hole; $x = 2$: jump, left limit 1, right limit 7.

38. (Model answer, to a newcomer.)

“A limit, $\lim_{x \rightarrow a} f(x)$, asks where the function is *headed* as the input creeps toward a — it watches the journey, not the destination. The value $f(a)$ is a totally separate question: what the function actually is exactly at a . They usually match, but not always: for $f(x) = \frac{x^2 - 1}{x - 1}$, the value $f(1)$ doesn’t exist (you’d divide by zero), yet everywhere nearby the function equals $x + 1$, so it is clearly *heading toward* 2 —

the limit is 2 even though there's no value at the point. 'Where it's going' and 'where it is' are different questions, and calculus is built on the gap between them."

39. (*Model answer, to a friend.*)

"Picture a hiker climbing from an altitude of 1,000 ft at dawn to 5,000 ft at noon. Because altitude changes *continuously* — she can't teleport from one height to another — she must pass through *every* altitude in between, so at some instant she stood at exactly 3,000 ft. That's the Intermediate Value Theorem: a continuously changing quantity going from one value to another cannot skip any value along the way. The one hypothesis that makes it work is **continuity** — if the hiker could blink instantly upward (a jump), she could leap over 3,000 ft and the guarantee would vanish."

Hypothesis that makes the IVT work: continuity.

Looking ahead

40. $f(x) = x^2$, at $x = 2$. Expand the difference quotient:

$$\frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 4}{h} = \frac{4 + 4h + h^2 - 4}{h} = \frac{4h + h^2}{h} = 4 + h \quad (h \neq 0).$$

Then $\lim_{h \rightarrow 0} (4 + h) = 4$.

4 (the slope of x^2 at $x = 2$).

41. General point a :

$$\frac{(a+h)^2 - a^2}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h \quad (h \neq 0).$$

Then $\lim_{h \rightarrow 0} (2a + h) = 2a$.

$2a$ (the slope of x^2 at every point — this is $\frac{d}{dx}x^2$).

42. $f(x) = \frac{1}{x}$. Combine the small fractions first (problem 10's move):

$$\frac{\frac{1}{a+h} - \frac{1}{a}}{h} = \frac{\frac{a-(a+h)}{a(a+h)}}{h} = \frac{-h}{h a(a+h)} = \frac{-1}{a(a+h)} \quad (h \neq 0).$$

Then $\lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = \frac{-1}{a \cdot a} = -\frac{1}{a^2}$.

$$\boxed{-\frac{1}{a^2}.$$

43. $f(x) = \sqrt{x}$. Rationalize with the conjugate $\sqrt{a+h} + \sqrt{a}$:

$$\frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{(a+h) - a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{h}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}} \quad (h \neq 0).$$

Then $\lim_{h \rightarrow 0} \frac{1}{\sqrt{a+h} + \sqrt{a}} = \frac{1}{2\sqrt{a}}$ (for $a > 0$).

$$\boxed{\frac{1}{2\sqrt{a}}.$$

44. Newton's method blindly follows the tangent line, so it can fail wherever the tangent points it astray — whereas bisection, which only ever keeps a bracket around a sign change, cannot.

A concrete failure to predict: **at (or near) a point where the slope is 0, the tangent line is (nearly) horizontal, so it crosses the axis very far away — flinging the next guess wildly off, possibly outside the region of interest or away from the root entirely** (it can overshoot, diverge to infinity, or land near another root). Bisection never suffers this: it only needs a sign change and a midpoint, both always available, so it cannot fly off — it just halves the interval every step, guaranteed.

(Other acceptable failures: the iterates can cycle/oscillate between two values forever; or a guess can land where the slope is 0 and the tangent never meets the axis, so the next step is undefined.)

A flat (near-zero) slope sends the tangent's next guess far away — Newton can overshoot/diverge

Chapter 3

The Slope of a Single Point

Every derivative below is computed **from the definition** — write the difference quotient $\frac{f(a+h) - f(a)}{h}$, simplify until the h in the denominator cancels, then let $h \rightarrow 0$. No differentiation rules are used; those arrive in Chapter 4.

Fluency

1. (a) $f(x) = x^2$.

$$\frac{(a+h)^2 - a^2}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h \xrightarrow{h \rightarrow 0} 2a.$$

$$\boxed{f'(x) = 2x}$$

(b) $f(x) = 3x^2$.

$$\frac{3(a+h)^2 - 3a^2}{h} = \frac{3(2ah + h^2)}{h} = \frac{3h(2a + h)}{h} = 3(2a + h) \xrightarrow{h \rightarrow 0} 6a.$$

$$\boxed{f'(x) = 6x}$$

(c) $f(x) = x^2 + 5$. The constant +5 cancels in the subtraction:

$$\frac{[(a+h)^2 + 5] - [a^2 + 5]}{h} = \frac{2ah + h^2}{h} = 2a + h \xrightarrow{h \rightarrow 0} 2a.$$

$$\boxed{f'(x) = 2x}$$

(Adding a constant shifts the graph up but never tilts it, so the slope is unchanged.)

2. (a) $f(x) = 5$.

$$\frac{5-5}{h} = \frac{0}{h} = 0 \xrightarrow{h \rightarrow 0} 0.$$

$$\boxed{f'(x) = 0}$$

A flat line has slope 0.

(b) $f(x) = 7x - 2$.

$$\frac{[7(a+h)-2] - [7a-2]}{h} = \frac{7h}{h} = 7 \xrightarrow{h \rightarrow 0} 7.$$

$$\boxed{f'(x) = 7}$$

A line is its own slope.

(c) $f(x) = -x$.

$$\frac{-(a+h) - (-a)}{h} = \frac{-h}{h} = -1 \xrightarrow{h \rightarrow 0} -1.$$

$$\boxed{f'(x) = -1}$$

3. (a) $f(x) = x^3$. Expand $(a+h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$:

$$\frac{(a+h)^3 - a^3}{h} = \frac{3a^2h + 3ah^2 + h^3}{h} = 3a^2 + 3ah + h^2 \xrightarrow{h \rightarrow 0} 3a^2.$$

$$\boxed{f'(x) = 3x^2}$$

(b) $f(x) = x^4$. Expand $(a+h)^4 = a^4 + 4a^3h + 6a^2h^2 + 4ah^3 + h^4$:

$$\frac{(a+h)^4 - a^4}{h} = \frac{4a^3h + 6a^2h^2 + 4ah^3 + h^4}{h} = 4a^3 + 6a^2h + 4ah^2 + h^3 \xrightarrow{h \rightarrow 0} 4a^3.$$

$$\boxed{f'(x) = 4x^3}$$

4. $f(x) = 3x^2 - 4x + 1$. Build the quotient at a :

$$\frac{[3(a+h)^2 - 4(a+h) + 1] - [3a^2 - 4a + 1]}{h}.$$

The numerator: $3(2ah + h^2) - 4h = 6ah + 3h^2 - 4h$. So

$$\frac{6ah + 3h^2 - 4h}{h} = 6a + 3h - 4 \xrightarrow{h \rightarrow 0} 6a - 4.$$

$$\boxed{f'(x) = 6x - 4}$$

At $x = 2$: $f'(2) = 6(2) - 4 = \boxed{8}$.

5. (a) $f(x) = \frac{1}{x}$. Combine the top over the common denominator $a(a+h)$:

$$\frac{\frac{1}{a+h} - \frac{1}{a}}{h} = \frac{\frac{a-(a+h)}{a(a+h)}}{h} = \frac{-h}{h a(a+h)} = \frac{-1}{a(a+h)} \xrightarrow{h \rightarrow 0} \frac{-1}{a^2}.$$

$$\boxed{f'(x) = -\frac{1}{x^2}}$$

(b) $f(x) = \frac{1}{x^2}$. Common denominator $a^2(a+h)^2$:

$$\frac{\frac{1}{(a+h)^2} - \frac{1}{a^2}}{h} = \frac{\frac{a^2-(a+h)^2}{a^2(a+h)^2}}{h} = \frac{-2ah - h^2}{h a^2(a+h)^2} = \frac{-2a - h}{a^2(a+h)^2} \xrightarrow{h \rightarrow 0} \frac{-2a}{a^4} = \frac{-2}{a^3}.$$

$$\boxed{f'(x) = -\frac{2}{x^3}}$$

6. $f(x) = \sqrt{x}$. Rationalize by multiplying top and bottom by the conjugate $\sqrt{a+h} + \sqrt{a}$:

$$\frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{(a+h) - a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}} \xrightarrow{h \rightarrow 0} \frac{1}{2\sqrt{a}}.$$

$$\boxed{f'(x) = \frac{1}{2\sqrt{x}}}$$

At $x = 9$: $f'(9) = \frac{1}{2\sqrt{9}} = \frac{1}{6} = \boxed{\frac{1}{6}}$.

7. From Problem 1(a), $f'(x) = 2x$ for $f(x) = x^2$. - At $x = 0$: slope = $2(0) = \boxed{0}$. - At $x = 1$: slope = $2(1) = \boxed{2}$. - At $x = -2$: slope = $2(-2) = \boxed{-4}$.

The slope is zero where $2x = 0$, i.e. at $\boxed{x = 0}$ — the bottom of the parabola, where the curve is momentarily flat.

8. Tangent line formula: $y = f(a) + f'(a)(x - a)$ with $f(x) = x^2$, $f'(x) = 2x$.

(a) At $x = 1$: $f(1) = 1$, $f'(1) = 2$.

$$y = 1 + 2(x - 1) = \boxed{2x - 1}.$$

Check: passes through $(1, 1)$ since $2(1) - 1 = 1$. ✓

(b) At $x = 3$: $f(3) = 9$, $f'(3) = 6$.

$$y = 9 + 6(x - 3) = \boxed{6x - 9}.$$

Check: passes through $(3, 9)$ since $6(3) - 9 = 9$. ✓

9. $s(t) = -5t^2 + 30t$. From the definition:

$$\frac{s(t+h) - s(t)}{h} = \frac{[-5(t+h)^2 + 30(t+h)] - [-5t^2 + 30t]}{h}.$$

Numerator: $-5(2th + h^2) + 30h = -10th - 5h^2 + 30h$. So

$$\frac{-10th - 5h^2 + 30h}{h} = -10t - 5h + 30 \xrightarrow{h \rightarrow 0} -10t + 30.$$

$$\boxed{v(t) = -10t + 30}$$

Then $v(0) = \boxed{30 \text{ m/s}}$ (launch speed). And $v = 0$ when $-10t + 30 = 0$, i.e. $\boxed{t = 3 \text{ s}}$ — the top of the flight.

Practice

10. $f(x) = \frac{1}{x+1}$. Common denominator $(a+h+1)(a+1)$:

$$\frac{\frac{1}{a+h+1} - \frac{1}{a+1}}{h} = \frac{\frac{(a+1)-(a+h+1)}{(a+h+1)(a+1)}}{h} = \frac{-h}{h(a+h+1)(a+1)} = \frac{-1}{(a+h+1)(a+1)} \xrightarrow{h \rightarrow 0} \frac{-1}{(a+1)^2}.$$

$$\boxed{f'(x) = -\frac{1}{(x+1)^2}}$$

11. $f(x) = \sqrt{x+3}$. Rationalize with conjugate $\sqrt{a+h+3} + \sqrt{a+3}$:

$$\frac{\sqrt{a+h+3} - \sqrt{a+3}}{h} = \frac{(a+h+3) - (a+3)}{h(\sqrt{a+h+3} + \sqrt{a+3})} = \frac{1}{\sqrt{a+h+3} + \sqrt{a+3}} \xrightarrow{h \rightarrow 0} \frac{1}{2\sqrt{a+3}}.$$

$$f'(x) = \frac{1}{2\sqrt{x+3}}$$

12. $f(x) = x^2 + \frac{1}{x}$. Keep both pieces in one quotient:

$$\frac{\left[(a+h)^2 + \frac{1}{a+h}\right] - \left[a^2 + \frac{1}{a}\right]}{h} = \underbrace{\frac{(a+h)^2 - a^2}{h}}_{=2a+h} + \underbrace{\frac{\frac{1}{a+h} - \frac{1}{a}}{h}}_{=\frac{-1}{a(a+h)}}.$$

As $h \rightarrow 0$: the first piece $\rightarrow 2a$, the second $\rightarrow -\frac{1}{a^2}$.

$$f'(x) = 2x - \frac{1}{x^2}$$

13. *Tangent to $y = \frac{1}{x}$ at $x = 2$.* Here $f(2) = \frac{1}{2}$ and (Problem 5a) $f'(x) = -\frac{1}{x^2}$, so $f'(2) = -\frac{1}{4}$.

$$y = \frac{1}{2} - \frac{1}{4}(x-2) = -\frac{1}{4}x + 1 \quad \Longrightarrow \quad \boxed{y = -\frac{1}{4}x + 1}.$$

Check: at $x = 2$, $-\frac{1}{4}(2) + 1 = \frac{1}{2}$. ✓

Tangent to $y = \sqrt{x}$ at $x = 4$. Here $f(4) = 2$ and (Problem 6) $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$.

$$y = 2 + \frac{1}{4}(x-4) = \frac{1}{4}x + 1 \quad \Longrightarrow \quad \boxed{y = \frac{1}{4}x + 1}.$$

Check: at $x = 4$, $\frac{1}{4}(4) + 1 = 2$. ✓

14. $f(x) = x^3 - 3x$. In one quotient:

$$\frac{[(a+h)^3 - 3(a+h)] - [a^3 - 3a]}{h} = \frac{(3a^2h + 3ah^2 + h^3) - 3h}{h} = 3a^2 + 3ah + h^2 - 3 \xrightarrow{h \rightarrow 0} 3a^2 - 3.$$

$$f'(x) = 3x^2 - 3$$

Horizontal tangent where $f'(x) = 0$: $3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow \boxed{x = \pm 1}$. The points are $(1, -2)$ and $(-1, 2)$ – the local valley and peak you'll formalize in Chapter 8.

15. $s(t) = t^2 - 6t + 5$. From the definition:

$$\frac{[(t+h)^2 - 6(t+h) + 5] - [t^2 - 6t + 5]}{h} = \frac{2th + h^2 - 6h}{h} = 2t + h - 6 \xrightarrow{h \rightarrow 0} 2t - 6.$$

$$\boxed{v(t) = 2t - 6}$$

At rest when $v(t) = 0$: $2t - 6 = 0 \Rightarrow \boxed{t = 3}$. At $t = 1$: $v(1) = 2(1) - 6 = -4 < 0$, so the particle is moving in the **negative direction (left)**.

16. For $y = x^2$, $f'(x) = 2x$. Set the slope equal to 10:

$$2x = 10 \implies x = 5, \quad y = 5^2 = 25.$$

$$\boxed{(5, 25)}$$

Check: slope at $x = 5$ is $2(5) = 10$. ✓

17. Use the tangent line to $f(x) = \sqrt{x}$ at $a = 100$ as a stand-in for the curve near there.

$$f(100) = 10, \quad f'(100) = \frac{1}{2\sqrt{100}} = \frac{1}{20} = 0.05.$$

Tangent: $y = 10 + 0.05(x - 100)$. Plug in $x = 101$:

$$\sqrt{101} \approx 10 + 0.05(101 - 100) = 10 + 0.05 = \boxed{10.05}.$$

(The true value is 10.0499..., so the linear approximation is right to four decimals — a preview of Chapter 7.)

Concept & counterexample

18. (a) **True.** At a *fixed* point a , $f'(a)$ is a single number (the slope there). But since the recipe works at every a , collecting all those numbers gives a new **function** f' . Both statements hold; they describe the same object at two levels (one input vs. all inputs).

(b) **False.** Continuity is necessary but not sufficient for differentiability. $f(x) = |x|$ is continuous everywhere yet has no derivative at 0 (a corner). Differentiability is strictly stronger than continuity.

(c) **True.** $f'(a) = 0$ means the slope of the tangent line at a is 0, i.e. the tangent

is horizontal. (The converse direction — that every peak/valley of a differentiable function has $f' = 0$ — is Chapter 8.)

(d) **False.** The chapter's theorem says differentiable $\Rightarrow$ continuous. So at any point where f is *not* continuous it cannot be differentiable. (Contrapositive of the theorem.)

19. The error: they set $h = 0$ *too early*, while h was still in the denominator. The derivative is a **limit** as $h \rightarrow 0$, not the value at $h = 0$. After cancelling the h ,

$$\frac{2ah + h^2}{h} = \frac{h(2a + h)}{h} = 2a + h \quad (\text{valid for } h \neq 0),$$

and only *now* do you let $h \rightarrow 0$, giving $\lim_{h \rightarrow 0}(2a + h) = 2a$. The $\frac{0}{0}$ form is an *invitation to cancel*, not a verdict that the limit fails. The derivative of x^2 is $2x$.

20. Construction: take $f(x) = |x|$ (continuous everywhere — you can draw it without lifting your pen). At $a = 0$ the difference quotient is

$$\frac{f(0 + h) - f(0)}{h} = \frac{|h| - 0}{h} = \frac{|h|}{h}.$$

- **Right** ($h \rightarrow 0^+$, so $|h| = h$): $\frac{|h|}{h} = +1$. - **Left** ($h \rightarrow 0^-$, so $|h| = -h$): $\frac{|h|}{h} = -1$.

The one-sided limits are $+1$ and -1 — they disagree, so the two-sided limit (the derivative) does not exist at 0 . Geometrically, the graph has a sharp **corner** there, with no single tangent slope. (Any “V-shaped” function works, e.g. $f(x) = |x - 0|$ shifted or scaled.)

21. No — it is not possible. The chapter proves the implication *differentiable* $\Rightarrow$ *continuous*. Its contrapositive is *not continuous* $\Rightarrow$ *not differentiable*. So if f were not continuous at 0 , it could not be differentiable at 0 . The arrow only runs one way: differentiability forces continuity, never the reverse. (That is why $|x|$ — continuous but not differentiable — is allowed, while “differentiable but not continuous” is forbidden.)

22. Yes, it is possible, and the function must be constant. If the slope is 0 everywhere, the graph never rises or falls — a perfectly horizontal line $f(x) = c$. Intuitively, a curve with zero rate of change at every instant cannot move, so its output stays fixed. (The rigorous proof — that $f' \equiv 0$ on an interval forces f constant — comes from the Mean Value Theorem in Chapter 6. The guess: $f(x) =$

c.)

23. As f rises, levels at a peak, then falls, the slope f' goes positive $\rightarrow 0 \rightarrow$ negative: - **Increasing region** (left of peak): $f' > 0$ (positive). - **At the peak**: $f' = 0$ (horizontal tangent). - **Decreasing region** (right of peak): $f' < 0$ (negative).

So the graph of f' starts above the axis, **crosses zero** at the x -value of the peak (descending through the axis), and continues below it. (A clean instance: $f(x) = -x^2$ has $f'(x) = -2x$, a straight line crossing zero at the peak $x = 0$.)

◆ Diamond

24. ◆ Differentiable $\Rightarrow$ continuous. Suppose $f'(a)$ exists. To show f is continuous at a we must show $f(a+h) \rightarrow f(a)$, equivalently $f(a+h) - f(a) \rightarrow 0$, as $h \rightarrow 0$. Write the change in f as the difference quotient times h (legal for $h \neq 0$):

$$f(a+h) - f(a) = \underbrace{\frac{f(a+h) - f(a)}{h}}_{\rightarrow f'(a)} \cdot h.$$

By the product law for limits, as $h \rightarrow 0$ the first factor tends to the finite number $f'(a)$ and the second factor tends to 0, so the product tends to $f'(a) \cdot 0 = 0$. Hence $f(a+h) - f(a) \rightarrow 0$, i.e. $\lim_{h \rightarrow 0} f(a+h) = f(a)$, which is exactly continuity at a . ■

(The converse is false: $|x|$ is continuous but not differentiable at 0, so differentiability sits strictly above continuity.)

25. ◆ $|x|$ is not differentiable at 0. Let $f(x) = |x|$, $a = 0$, $f(0) = 0$. The difference quotient is

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

Take the two one-sided limits: - **Right** $h \rightarrow 0^+$: here $h > 0$ so $|h| = h$, giving $\frac{h}{h} = 1$.

Thus $\lim_{h \rightarrow 0^+} \frac{|h|}{h} = +1$. - **Left** $h \rightarrow 0^-$: here $h < 0$ so $|h| = -h$, giving $\frac{-h}{h} = -1$. Thus $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$.

Since $+1 \neq -1$, the left- and right-hand limits disagree, so $\lim_{h \rightarrow 0} \frac{|h|}{h}$ does not exist.

Therefore $f'(0)$ does not exist. ■ (This is exactly the sign-function jump of $\frac{|x|}{x}$)

from Chapter 2, reappearing as the slope of $|x|$.)

26. ♦♦ The power rule from the binomial theorem. Let $f(x) = x^n$ for a positive integer n . The binomial theorem gives

$$(a + h)^n = a^n + n a^{n-1} h + \binom{n}{2} a^{n-2} h^2 + \cdots + h^n.$$

Subtract a^n and divide by h . The leading a^n cancels, and **every remaining term contains at least one factor of h** , so a single h divides out of each:

$$\frac{(a + h)^n - a^n}{h} = \frac{n a^{n-1} h + \binom{n}{2} a^{n-2} h^2 + \cdots + h^n}{h} = n a^{n-1} + \binom{n}{2} a^{n-2} h + \cdots + h^{n-1}.$$

Now let $h \rightarrow 0$. Every term *except the first* still carries at least one factor of h and therefore vanishes; only $n a^{n-1}$ survives:

$$f'(a) = \lim_{h \rightarrow 0} \left[n a^{n-1} + \binom{n}{2} a^{n-2} h + \cdots + h^{n-1} \right] = n a^{n-1}.$$

$$\frac{d}{dx} x^n = n x^{n-1}$$

This matches every case computed by hand: $n = 2 \Rightarrow 2x$, $n = 3 \Rightarrow 3x^2$, $n = 4 \Rightarrow 4x^3$. The “everything but one term survives” mechanism is the same one behind Chapter 1’s $n^{1/n}$ and Chapter 7’s Taylor coefficients. ■

27. ♦♦ A vertical tangent: $f(x) = x^{1/3}$. With $f(0) = 0$, the difference quotient at 0 is

$$\frac{f(0 + h) - f(0)}{h} = \frac{h^{1/3} - 0}{h} = \frac{h^{1/3}}{h} = h^{1/3-1} = h^{-2/3} = \frac{1}{h^{2/3}}.$$

As $h \rightarrow 0$ (from either side), $h^{2/3} = (h^{1/3})^2 \geq 0$ shrinks to 0 through positive values, so $\frac{1}{h^{2/3}} \rightarrow +\infty$. The difference quotient blows up rather than settling on a number, so the limit is not a (finite) real number:

$f'(0)$ does not exist.

Geometric difference from $|x|$. Both fail to be differentiable at 0, but for opposite reasons: $-|x|$ — **a corner**. The two one-sided slopes exist and are finite (+1 and

-1) but *disagree*; the graph kinks abruptly. $-x^{1/3}$ — a **vertical tangent**. The one-sided slopes *agree* (both $+\infty$) but are *infinite*; the curve passes smoothly through the origin while its tangent line stands straight up (vertical), and a vertical line has no finite slope.

So one failure is a disagreement of finite slopes, the other a single but infinite slope.

■

28. ♦♦ One-sided derivatives. Define

$$f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}, \quad f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}.$$

Claim: $f'(a)$ exists $\iff$ both $f'_+(a)$ and $f'_-(a)$ exist and are equal; and then $f'(a)$ equals their common value.

Proof. Let $g(h) = \frac{f(a+h) - f(a)}{h}$. By definition $f'(a) = \lim_{h \rightarrow 0} g(h)$, and $f'_+(a)$, $f'_-(a)$ are the right- and left-hand limits of the same g . A basic fact about limits (Chapter 2) is that a two-sided limit exists and equals L **iff** both one-sided limits exist and equal L :

$$\lim_{h \rightarrow 0} g(h) = L \iff \lim_{h \rightarrow 0^+} g(h) = L \text{ and } \lim_{h \rightarrow 0^-} g(h) = L.$$

Applying this to g gives precisely the claim: $f'(a)$ exists and equals L exactly when $f'_+(a) = f'_-(a) = L$. ■

Re-explaining the $|x|$ corner. For $f(x) = |x|$ at $a = 0$ we found $f'_+(0) = +1$ and $f'_-(0) = -1$. Both one-sided derivatives exist, but $+1 \neq -1$, so by the criterion the two-sided derivative $f'(0)$ fails to exist — the corner is exactly a *mismatch of one-sided derivatives*. (These one-sided difference quotients are the tool behind Chapter 6's Fermat's lemma.)

Teach it

29. (Model answer.) A derivative is the slope of a curve at one single point. To get it, pick your point and a nearby second point, and draw the straight **secant** line through the two — its slope, $\frac{f(a+h)-f(a)}{h}$, is the *average* rate of change over that little gap. Now slide the second point toward the first by letting $h \rightarrow 0$: the secant pivots and settles onto the **tangent** line that just grazes the curve at your point, and its slope is the *instantaneous* rate of change — the derivative. So shrinking h

is what turns an average over an interval into a rate at a single instant.

30. Debug. What they wrote, $\frac{(x+h)^2 - x^2}{h}$, is only the **difference quotient** — step 1 of three. They left out (i) simplifying until the h cancels and (ii) taking the limit $h \rightarrow 0$; as written it is not even a number (it still contains h). The correct final two steps:

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \quad (\text{cancel } h),$$

$$\frac{d}{dx} x^2 = \lim_{h \rightarrow 0} (2x + h) = 2x \quad (\text{let } h \rightarrow 0).$$

So the derivative is $2x$, not the unsimplified quotient.

31. (Model worked solution.)

Find $\frac{d}{dx} \frac{1}{x}$ from the definition. Let $f(x) = \frac{1}{x}$.

Step 1 — the difference quotient.

$$\frac{f(a+h) - f(a)}{h} = \frac{\frac{1}{a+h} - \frac{1}{a}}{h}.$$

Step 2 — simplify until h cancels. The numerator is a difference of two fractions; **I combine them over a common denominator first**, because until the top is a single fraction there is no factor of h visible to cancel against the h in the bottom — and that cancellation is the whole point (it is what dissolves the $\frac{0}{0}$). With common denominator $a(a+h)$:

$$\frac{1}{a+h} - \frac{1}{a} = \frac{a - (a+h)}{a(a+h)} = \frac{-h}{a(a+h)}.$$

Now divide by h (i.e. multiply by $\frac{1}{h}$); the $-h$ on top meets the h on the bottom and cancels:

$$\frac{1}{h} \cdot \frac{-h}{a(a+h)} = \frac{-1}{a(a+h)}.$$

Step 3 — let $h \rightarrow 0$.

$$f'(a) = \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = \frac{-1}{a \cdot a} = -\frac{1}{a^2}.$$

$$\boxed{\frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}}$$

To redo it for $\frac{1}{x+2}$: same three steps, common denominator $(a+2)(a+h+2)$, giving

$$\frac{1}{a+h+2} - \frac{1}{a+2} = \frac{-h}{(a+h+2)(a+2)} \Rightarrow \frac{-1}{(a+h+2)(a+2)} \xrightarrow{h \rightarrow 0} -\frac{1}{(x+2)^2}.$$

32. (Model “design a problem” with full key.)

Problem. From the definition, find the derivative of $f(x) = 2x^2 - 3x$, and evaluate it at $x = 1$.

Solution key. Step 1 – difference quotient.

$$\frac{f(a+h) - f(a)}{h} = \frac{[2(a+h)^2 - 3(a+h)] - [2a^2 - 3a]}{h}.$$

Step 2 – simplify until h cancels. Expand the numerator: $2(a^2 + 2ah + h^2) - 3a - 3h - 2a^2 + 3a = 4ah + 2h^2 - 3h$. Then

$$\frac{4ah + 2h^2 - 3h}{h} = \frac{h(4a + 2h - 3)}{h} = 4a + 2h - 3.$$

Step 3 – let $h \rightarrow 0$.

$$f'(a) = \lim_{h \rightarrow 0} (4a + 2h - 3) = 4a - 3.$$

$$\boxed{f'(x) = 4x - 3}$$

At $x = 1$: $f'(1) = 4(1) - 3 = 1$. (Clean integer answer, as designed.)

33. (Model answer.) A function is **continuous** at a point if its graph has no break there – you can draw through the point without lifting your pen. It is **differentiable** at a point if, on top of that, the graph has a single well-defined tangent slope there – if you zoom in, it looks like one straight line. Differentiability is the stronger demand: every differentiable function is continuous, but not conversely. The separating example is $|x|$ at 0: you can draw it unbroken (continuous), but at the origin it makes a sharp **corner**, and the slope coming from the left (-1) disagrees with the slope from the right ($+1$), so there is no single tangent – not differentiable. Smoothness asks for more than mere connectedness.

Looking ahead

34. The pattern from $\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x^3 = 3x^2$, $\frac{d}{dx}x^4 = 4x^3$ is: **bring the exponent down as a multiplier and lower the exponent by one**. Predictions:

$$\boxed{\frac{d}{dx}x^5 = 5x^4}, \quad \boxed{\frac{d}{dx}x^{10} = 10x^9}.$$

Doing x^{10} from the definition would mean expanding $(a+h)^{10}$ — eleven terms — just to watch ten of them die in the limit. That tedium is exactly why Chapter 4 proves the **power rule** $\frac{d}{dx}x^n = nx^{n-1}$ once and for all (the derivation is Problem 26).

35. From $\frac{d}{dx}(3x^2 - 4x + 1) = 6x - 4$ — where the $3x^2$ contributed $6x$, the $-4x$ contributed -4 , and the $+1$ contributed 0 — the derivative was taken **term by term**. Prediction (the **sum rule**):

$$\boxed{\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)}.$$

Reasoning: the difference quotient of a sum splits into the sum of the two difference quotients,

$$\frac{[f(a+h) + g(a+h)] - [f(a) + g(a)]}{h} = \frac{f(a+h) - f(a)}{h} + \frac{g(a+h) - g(a)}{h},$$

and the limit of a sum is the sum of the limits, giving $f'(a) + g'(a)$. (Likewise a constant multiple pulls out: $\frac{d}{dx}[cf] = cf'$.)

36. For $f(x) = e^x$, the difference quotient at x uses $e^{x+h} = e^x e^h$:

$$\frac{e^{x+h} - e^x}{h} = \frac{e^x e^h - e^x}{h} = e^x \cdot \frac{e^h - 1}{h}.$$

So the derivative is the limit

$$\frac{d}{dx}e^x = \lim_{h \rightarrow 0} e^x \cdot \frac{e^h - 1}{h} = e^x \cdot \lim_{h \rightarrow 0} \frac{e^h - 1}{h},$$

since e^x is constant in h and factors out of the limit. The whole answer hinges on

the single piece

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h}.$$

Chapter 5 shows this limit equals 1, which is exactly what makes $\frac{d}{dx}e^x = e^x \cdot 1 = e^x$ — the function that is its own derivative.

Chapter 4

A Toolkit for Slopes

Fluency

1. Power rule, $\frac{d}{dx}x^n = nx^{n-1}$ (with sum/constant-multiple where needed).

(a) $\frac{d}{dx}x^9 = 9x^8$.

(b) $\frac{d}{dx}4x^6 = 4 \cdot 6x^5 = \boxed{24x^5}$.

(c) $\frac{d}{dx}x^{100} = \boxed{100x^{99}}$.

(d) 7 is a constant, so $\frac{d}{dx}7 = \boxed{0}$.

(e) $\frac{d}{dx}x = 1 \cdot x^0 = \boxed{1}$.

2. Power rule with n negative or fractional. Rewrite as a single power, bring the exponent down, subtract one.

(a) $\frac{d}{dx}x^{-3} = -3x^{-4} = \boxed{-\frac{3}{x^4}}$.

(b) $\frac{1}{x^5} = x^{-5}$, so $\frac{d}{dx}x^{-5} = -5x^{-6} = \boxed{-\frac{5}{x^6}}$.

(c) $\frac{d}{dx}x^{1/3} = \frac{1}{3}x^{-2/3} = \boxed{\frac{1}{3x^{2/3}}} = \frac{1}{3\sqrt[3]{x^2}}$.

(d) $\sqrt[4]{x} = x^{1/4}$, so $\frac{d}{dx}x^{1/4} = \frac{1}{4}x^{-3/4} = \boxed{\frac{1}{4x^{3/4}}} = \frac{1}{4\sqrt[4]{x^3}}$.

(e) $\frac{d}{dx}x^{-1/2} = -\frac{1}{2}x^{-3/2} = \boxed{-\frac{1}{2x^{3/2}}} = -\frac{1}{2\sqrt{x^3}}$.

(f) $\frac{1}{\sqrt{x}} = x^{-1/2}$ — same as (e): $\boxed{-\frac{1}{2x^{3/2}}}$.

3. Rewrite as powers first, then power rule + sum rule.

(a) $\frac{1}{x^2} + \sqrt{x} = x^{-2} + x^{1/2}$. Then

$$\frac{d}{dx} (x^{-2} + x^{1/2}) = -2x^{-3} + \frac{1}{2}x^{-1/2} = \boxed{-\frac{2}{x^3} + \frac{1}{2\sqrt{x}}}.$$

(b) Split the fraction: $\frac{x^3-1}{x} = x^2 - x^{-1}$. Then

$$\frac{d}{dx} (x^2 - x^{-1}) = 2x + x^{-2} = \boxed{2x + \frac{1}{x^2}}.$$

(c) Combine into one power: $x^2\sqrt{x} = x^2 \cdot x^{1/2} = x^{5/2}$. Then

$$\frac{d}{dx} x^{5/2} = \frac{5}{2}x^{3/2} = \boxed{\frac{5}{2}x^{3/2}} = \frac{5}{2}x\sqrt{x}.$$

4. Sum / constant-multiple + power rule, term by term.

$$(a) \frac{d}{dx}(5x^3 - 2x^2 + 7x - 4) = \boxed{15x^2 - 4x + 7}.$$

$$(b) \frac{d}{dx} \left(\frac{1}{3}x^6 + 4x \right) = \frac{1}{3} \cdot 6x^5 + 4 = \boxed{2x^5 + 4}.$$

$$(c) \frac{d}{dx}(9x^5 - x^4 + 2x - 11) = \boxed{45x^4 - 4x^3 + 2}.$$

(d) Expand first: $(2x-1)(x+3) = 2x^2 + 5x - 3$. Then $\frac{d}{dx}(2x^2 + 5x - 3) = \boxed{4x + 5}$.
(Check by product rule: $(2)(x+3) + (2x-1)(1) = 2x + 6 + 2x - 1 = 4x + 5$.
✓)

5. Product rule $(fg)' = f'g + fg'$.

$$(a) f = x^2, g = e^x: \frac{d}{dx}(x^2 e^x) = 2x e^x + x^2 e^x = \boxed{e^x(x^2 + 2x)} = xe^x(x + 2).$$

$$(b) f = x, g = \sin x: \frac{d}{dx}(x \sin x) = (1) \sin x + x \cos x = \boxed{\sin x + x \cos x}.$$

$$(c) f = x^2 + 1, g = x^3 - 2: \frac{d}{dx}[(x^2 + 1)(x^3 - 2)] = (2x)(x^3 - 2) + (x^2 + 1)(3x^2) \\ = 2x^4 - 4x + 3x^4 + 3x^2 = \boxed{5x^4 + 3x^2 - 4x}. \text{ (Check: expand } (x^2 + 1)(x^3 - 2) = \\ x^5 + x^3 - 2x^2 - 2, \text{ derivative } 5x^4 + 3x^2 - 4x. \checkmark)}$$

$$(d) f = x^3, g = \cos x: \frac{d}{dx}(x^3 \cos x) = 3x^2 \cos x + x^3(-\sin x) = \\ \boxed{3x^2 \cos x - x^3 \sin x}.$$

$$(e) f = e^x, g = \cos x: \frac{d}{dx}(e^x \cos x) = e^x \cos x + e^x(-\sin x) = \boxed{e^x(\cos x - \sin x)}.$$

6. Product rule, then evaluate at $x = 0$.

$$(a) \ f = x + 1, \ g = e^x: \frac{d}{dx}[(x + 1)e^x] = (1)e^x + (x + 1)e^x = (x + 2)e^x. \text{ At } x = 0: (0 + 2)e^0 = \boxed{2}.$$

$$(b) \ f = x^2, \ g = \sin x: \frac{d}{dx}(x^2 \sin x) = 2x \sin x + x^2 \cos x. \text{ At } x = 0: 2(0) \sin 0 + 0^2 \cos 0 = \boxed{0}.$$

7. Quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$

$$(a) \ f = x, \ g = x^2 + 1: \frac{(1)(x^2 + 1) - x(2x)}{(x^2 + 1)^2} = \boxed{\frac{1 - x^2}{(x^2 + 1)^2}}.$$

$$(b) \ f = x^2, \ g = x - 1: \frac{(2x)(x - 1) - x^2(1)}{(x - 1)^2} = \frac{2x^2 - 2x - x^2}{(x - 1)^2} = \boxed{\frac{x^2 - 2x}{(x - 1)^2}}.$$

$$(c) \ f = \sin x, \ g = x: \frac{(\cos x)x - \sin x(1)}{x^2} = \boxed{\frac{x \cos x - \sin x}{x^2}}.$$

$$(d) \ f = e^x, \ g = x: \frac{e^x \cdot x - e^x \cdot 1}{x^2} = \boxed{\frac{e^x(x - 1)}{x^2}}.$$

$$(e) \ f = x - 1, \ g = x + 1: \frac{(1)(x + 1) - (x - 1)(1)}{(x + 1)^2} = \frac{x + 1 - x + 1}{(x + 1)^2} = \boxed{\frac{2}{(x + 1)^2}}.$$

8. Quotient rule with trig.

$$(a) \ \tan x = \frac{\sin x}{\cos x}: \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \boxed{\sec^2 x}.$$

$$(b) \ f = \cos x, \ g = x: \frac{(-\sin x)x - \cos x(1)}{x^2} = \boxed{\frac{-x \sin x - \cos x}{x^2}}.$$

$$(c) \ \frac{1}{\sin x}: f = 1, \ g = \sin x: \frac{(0) \sin x - 1 \cdot \cos x}{\sin^2 x} = \boxed{-\frac{\cos x}{\sin^2 x}} = -\csc x \cot x.$$

9. Chain rule on powers: $\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x).$

$$(a) \ \frac{d}{dx}(x^2 + 1)^5 = 5(x^2 + 1)^4 \cdot 2x = \boxed{10x(x^2 + 1)^4}.$$

$$(b) \ \frac{d}{dx}(1 - x^2)^{10} = 10(1 - x^2)^9 \cdot (-2x) = \boxed{-20x(1 - x^2)^9}.$$

$$(c) \ \frac{d}{dx}(3x + 2)^7 = 7(3x + 2)^6 \cdot 3 = \boxed{21(3x + 2)^6}.$$

$$(d) \ \frac{d}{dx}(x^3 - x)^4 = 4(x^3 - x)^3 \cdot (3x^2 - 1) = \boxed{4(3x^2 - 1)(x^3 - x)^3}.$$

10. Chain rule on roots: $\sqrt{u} = u^{1/2}$, so $\frac{d}{dx}\sqrt{u} = \frac{u'}{2\sqrt{u}}$.

$$(a) \quad \frac{d}{dx}\sqrt{3x+2} = \frac{3}{2\sqrt{3x+2}} = \boxed{\frac{3}{2\sqrt{3x+2}}}.$$

$$(b) \quad \frac{d}{dx}\sqrt{x^2+1} = \frac{2x}{2\sqrt{x^2+1}} = \boxed{\frac{x}{\sqrt{x^2+1}}}.$$

$$(c) \quad \frac{d}{dx}\sqrt{1-x^2} = \frac{-2x}{2\sqrt{1-x^2}} = \boxed{-\frac{x}{\sqrt{1-x^2}}}.$$

$$(d) \quad \frac{1}{\sqrt{x^2+4}} = (x^2+4)^{-1/2}; \quad \frac{d}{dx}(x^2+4)^{-1/2} = -\frac{1}{2}(x^2+4)^{-3/2} \cdot 2x = \boxed{-\frac{x}{(x^2+4)^{3/2}}}.$$

11. Chain rule on exponentials: $\frac{d}{dx}e^u = e^u u'$.

$$(a) \quad \frac{d}{dx}e^{4x} = e^{4x} \cdot 4 = \boxed{4e^{4x}}.$$

$$(b) \quad \frac{d}{dx}e^{-x} = e^{-x} \cdot (-1) = \boxed{-e^{-x}}.$$

$$(c) \quad \frac{d}{dx}e^{x^2} = e^{x^2} \cdot 2x = \boxed{2xe^{x^2}}.$$

$$(d) \quad \frac{d}{dx}e^{-x^2/2} = e^{-x^2/2} \cdot (-x) = \boxed{-xe^{-x^2/2}}.$$

$$(e) \quad \frac{d}{dx}e^{3x+1} = e^{3x+1} \cdot 3 = \boxed{3e^{3x+1}}.$$

12. Chain rule on trig.

$$(a) \quad \frac{d}{dx}\sin(3x) = \cos(3x) \cdot 3 = \boxed{3\cos(3x)}.$$

$$(b) \quad \frac{d}{dx}\cos(x^2) = -\sin(x^2) \cdot 2x = \boxed{-2x\sin(x^2)}.$$

$$(c) \quad \frac{d}{dx}\sin(x^2+1) = \cos(x^2+1) \cdot 2x = \boxed{2x\cos(x^2+1)}.$$

$$(d) \quad \frac{d}{dx}\cos(5x) = -\sin(5x) \cdot 5 = \boxed{-5\sin(5x)}.$$

$$(e) \quad \frac{d}{dx}\tan(2x) = \sec^2(2x) \cdot 2 = \boxed{2\sec^2(2x)}.$$

13. Trig basics (with sum/constant-multiple).

$$(a) \quad \frac{d}{dx}\cos x = \boxed{-\sin x}.$$

$$(b) \quad \frac{d}{dx}\tan x = \boxed{\sec^2 x}.$$

$$(c) \quad \frac{d}{dx}(3\sin x - 2\cos x) = 3\cos x - 2(-\sin x) = \boxed{3\cos x + 2\sin x}.$$

$$(d) \frac{d}{dx}(x + \cos x) = \boxed{1 - \sin x}.$$

$$(e) \frac{d}{dx}(x^2 + \sin x) = \boxed{2x + \cos x}.$$

14. Powers of trig (chain rule: write $\sin^2 x = (\sin x)^2$, etc.).

$$(a) \frac{d}{dx} \sin^2 x = 2 \sin x \cdot \cos x = \boxed{2 \sin x \cos x} (= \sin 2x).$$

$$(b) \frac{d}{dx} \cos^3 x = 3 \cos^2 x \cdot (-\sin x) = \boxed{-3 \cos^2 x \sin x}.$$

$$(c) \frac{d}{dx} \tan^2 x = 2 \tan x \cdot \sec^2 x = \boxed{2 \tan x \sec^2 x}.$$

15. Mixed – identify the rule first.

$$(a) \textbf{Product: } \frac{d}{dx}(x^2 e^x) = 2x e^x + x^2 e^x = \boxed{e^x(x^2 + 2x)}.$$

$$(b) \textbf{Quotient: } \frac{d}{dx} \frac{x}{\cos x} = \frac{(1) \cos x - x(-\sin x)}{\cos^2 x} = \boxed{\frac{\cos x + x \sin x}{\cos^2 x}}.$$

$$(c) \textbf{Chain: } \frac{d}{dx} \sin(x^3) = \cos(x^3) \cdot 3x^2 = \boxed{3x^2 \cos(x^3)}.$$

$$(d) \textbf{Chain: } \frac{d}{dx}(x^2 + x)^6 = 6(x^2 + x)^5(2x + 1) = \boxed{6(2x + 1)(x^2 + x)^5}.$$

$$(e) \textbf{Sum + chain: } \frac{d}{dx}(e^x + e^{-x}) = e^x - e^{-x} = \boxed{e^x - e^{-x}}.$$

16. Evaluate the derivative at the given point.

$$(a) f(x) = x^3 - x \Rightarrow f'(x) = 3x^2 - 1. \text{ At } x = 2: 3(4) - 1 = \boxed{11}.$$

$$(b) f(x) = e^x \sin x \Rightarrow f'(x) = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x) \text{ (product rule). At } x = 0: e^0(\sin 0 + \cos 0) = 1 \cdot (0 + 1) = \boxed{1}.$$

$$(c) f(x) = \frac{1}{x} = x^{-1} \Rightarrow f'(x) = -x^{-2} = -\frac{1}{x^2}. \text{ At } x = \frac{1}{2}: -\frac{1}{(1/2)^2} = -\frac{1}{1/4} = \boxed{-4}.$$

$$(d) f(x) = \sqrt{x} \Rightarrow f'(x) = \frac{1}{2\sqrt{x}}. \text{ At } x = 4: \frac{1}{2\sqrt{4}} = \frac{1}{4} = \boxed{\frac{1}{4}}.$$

17. Tangent line: $y - f(a) = f'(a)(x - a)$.

$$(a) y = (x^2 - 1)^3 \text{ at } x = 1. f(1) = (1 - 1)^3 = 0. f'(x) = 3(x^2 - 1)^2 \cdot 2x = 6x(x^2 - 1)^2, \text{ so } f'(1) = 6(1)(0)^2 = 0. \text{ Tangent: } \boxed{y = 0} \text{ (horizontal).}$$

$$(b) y = e^x \text{ at } x = 0. f(0) = 1, f'(x) = e^x, f'(0) = 1. \text{ Tangent: } y - 1 = 1(x - 0) \Rightarrow \boxed{y = x + 1}.$$

$$(c) y = \sin x \text{ at } x = 0. f(0) = 0, f'(x) = \cos x, f'(0) = 1. \text{ Tangent: } \boxed{y = x}.$$

$$(d) y = \frac{1}{x} \text{ at } x = 1. f(1) = 1, f'(x) = -\frac{1}{x^2}, f'(1) = -1. \text{ Tangent: } y - 1 =$$

$$-1(x-1) \Rightarrow \boxed{y = -x + 2}.$$

18. Differentiate, then simplify.

$$(a) \frac{e^x}{x} \text{ (quotient): } \frac{e^x x - e^x}{x^2} = \boxed{\frac{e^x(x-1)}{x^2}}.$$

$$(b) x^2 \cos x \text{ (product): } 2x \cos x + x^2(-\sin x) = \boxed{2x \cos x - x^2 \sin x} = x(2 \cos x - x \sin x).$$

$$(c) \sqrt{x} e^x = x^{1/2} e^x \text{ (product): } \frac{1}{2} x^{-1/2} e^x + x^{1/2} e^x = e^x \left(\frac{1}{2\sqrt{x}} + \sqrt{x} \right) = \boxed{\frac{e^x(1+2x)}{2\sqrt{x}}}.$$

$$(d) \frac{\sin x}{1 + \cos x} \text{ (quotient):}$$

$$\frac{(\cos x)(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2} = \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \boxed{\frac{1}{1 + \cos x}}.$$

(Used $\sin^2 x + \cos^2 x = 1$, then cancelled $1 + \cos x$.)

19. Second derivatives: differentiate twice.

$$(a) f = x^4 \Rightarrow f' = 4x^3 \Rightarrow f'' = \boxed{12x^2}.$$

$$(b) f = \sin x \Rightarrow f' = \cos x \Rightarrow f'' = \boxed{-\sin x}.$$

$$(c) f = e^{2x} \Rightarrow f' = 2e^{2x} \Rightarrow f'' = \boxed{4e^{2x}}.$$

$$(d) f = x^3 - 2x \Rightarrow f' = 3x^2 - 2 \Rightarrow f'' = \boxed{6x}.$$

20. Higher-order derivatives.

$$(a) \text{ Third derivative of } x^5: \frac{d}{dx} x^5 = 5x^4 \rightarrow 20x^3 \rightarrow \boxed{60x^2}.$$

$$(b) \text{ Fourth derivative of } \sin x: \sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \boxed{\sin x}.$$

(Back to start after four – a length-4 cycle.)

$$(c) \text{ Second derivative of } \frac{1}{x} = x^{-1}: \frac{d}{dx} x^{-1} = -x^{-2} \rightarrow 2x^{-3} = \boxed{\frac{2}{x^3}}.$$

Practice

21. $f(x) = x^2(x^3 + 1)^4$ – a chain inside a product. Product rule with $g = x^2$, $h = (x^3 + 1)^4$; the second factor needs the chain rule, $h' = 4(x^3 + 1)^3 \cdot 3x^2 =$

$$12x^2(x^3 + 1)^3.$$

$$f'(x) = 2x(x^3 + 1)^4 + x^2 \cdot 12x^2(x^3 + 1)^3 = 2x(x^3 + 1)^4 + 12x^4(x^3 + 1)^3.$$

Factor out $2x(x^3 + 1)^3$:

$$f'(x) = 2x(x^3 + 1)^3[(x^3 + 1) + 6x^3] = \boxed{2x(x^3 + 1)^3(7x^3 + 1)}.$$

22. $f(x) = \sqrt{1 + \sqrt{x}}$ — a chain inside a chain. Outer root, then inner root.

$$f'(x) = \frac{1}{2\sqrt{1 + \sqrt{x}}} \cdot \frac{d}{dx}(1 + \sqrt{x}) = \frac{1}{2\sqrt{1 + \sqrt{x}}} \cdot \frac{1}{2\sqrt{x}} = \boxed{\frac{1}{4\sqrt{x}\sqrt{1 + \sqrt{x}}}}.$$

23. Order of composition (chain rule each).

(a) $e^{\sin x}$: outer $e^{(\cdot)}$, inner $\sin x$. $\frac{d}{dx}e^{\sin x} = e^{\sin x} \cdot \cos x = \boxed{\cos x e^{\sin x}}.$

(b) $\sin(e^x)$: outer $\sin(\cdot)$, inner e^x . $\frac{d}{dx}\sin(e^x) = \cos(e^x) \cdot e^x = \boxed{e^x \cos(e^x)}.$

(c) $(\sin x)^2$: outer square, inner $\sin x$. $\frac{d}{dx}(\sin x)^2 = 2 \sin x \cdot \cos x = \boxed{2 \sin x \cos x}.$

Note (a) vs (b): swapping inner/outer gives entirely different derivatives — in (a) the inner-derivative factor is $\cos x$ and the outer keeps the exponential shape; in (b) the inner-derivative factor is e^x and the outer becomes $\cos$.

24. $f(x) = x e^x \sin x$ — product of three. Group as $x \cdot (e^x \sin x)$ and apply the product rule, using $(e^x \sin x)' = e^x \sin x + e^x \cos x$:

$$f'(x) = (1)(e^x \sin x) + x(e^x \sin x + e^x \cos x) = e^x \sin x + x e^x \sin x + x e^x \cos x.$$

$$\boxed{f'(x) = e^x[\sin x + x \sin x + x \cos x]} = e^x[(1 + x) \sin x + x \cos x].$$

(General pattern for a triple product uvw : $u'vw + uv'w + uvw'$ — here $u = x$, $v = e^x$, $w = \sin x$ gives the same thing.)

25. $f(x) = \frac{(x^2 + 1)^3}{x}$ — quotient with a chain on top. Top $u = (x^2 + 1)^3$ has $u' = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$; bottom $v = x$, $v' = 1$.

$$f'(x) = \frac{6x(x^2 + 1)^2 \cdot x - (x^2 + 1)^3 \cdot 1}{x^2} = \frac{6x^2(x^2 + 1)^2 - (x^2 + 1)^3}{x^2}.$$

Factor $(x^2 + 1)^2$ from the numerator:

$$f'(x) = \frac{(x^2 + 1)^2 [6x^2 - (x^2 + 1)]}{x^2} = \boxed{\frac{(x^2 + 1)^2 (5x^2 - 1)}{x^2}}.$$

(Check by rewriting $f = (x^2 + 1)^3 x^{-1}$ and using the product rule: $6x(x^2 + 1)^2 x^{-1} + (x^2 + 1)^3 (-x^{-2})$, which equals the same after combining over x^2 . ✓)

26. $f(x) = \frac{e^x}{x^2 + 1}$. Quotient rule, $u = e^x$, $v = x^2 + 1$:

$$f'(x) = \frac{e^x(x^2 + 1) - e^x(2x)}{(x^2 + 1)^2} = \boxed{\frac{e^x(x^2 - 2x + 1)}{(x^2 + 1)^2}} = \frac{e^x(x - 1)^2}{(x^2 + 1)^2}.$$

Horizontal tangent where $f'(x) = 0$: since $e^x > 0$ and $(x^2 + 1)^2 > 0$, we need $(x - 1)^2 = 0$, i.e. $\boxed{x = 1}$. (It is a single tangency point, not a sign change — the numerator is a perfect square.)

27. $f(x) = x^3 - 6x^2 + 9x$. Then $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = \boxed{3(x - 1)(x - 3)}$. Horizontal tangents where $f'(x) = 0$: $x = 1$ and $x = 3$. At $x = 1$: $f(1) = 1 - 6 + 9 = 4$, point $(1, 4)$. At $x = 3$: $f(3) = 27 - 54 + 27 = 0$, point $\boxed{(1, 4) \text{ and } (3, 0)}$.

28. $f(x) = x e^{-x}$. Product rule (e^{-x} via chain, derivative $-e^{-x}$):

$$f'(x) = (1)e^{-x} + x(-e^{-x}) = e^{-x}(1 - x).$$

Horizontal tangent where $f'(x) = 0$: $e^{-x} > 0$, so $1 - x = 0 \Rightarrow \boxed{x = 1}$. Second derivative: differentiate $f'(x) = e^{-x}(1 - x) = e^{-x} - x e^{-x}$:

$$f''(x) = -e^{-x} - [e^{-x} + x(-e^{-x})] = -e^{-x} - e^{-x} + x e^{-x} = e^{-x}(x - 2).$$

At $x = 1$: $f''(1) = e^{-1}(1 - 2) = \boxed{-e^{-1} = -\frac{1}{e} < 0}$ — concave down, so $x = 1$ is a local maximum (a Chapter 8 preview).

29. Inverse-function rule $g'(x) = \frac{1}{f'(g(x))}$.

$g(x) = x^{1/3}$ as the inverse of $f(x) = x^3$. Here $f'(x) = 3x^2$, so $f'(g(x)) = 3(x^{1/3})^2 = 3x^{2/3}$. Thus

$$g'(x) = \frac{1}{3x^{2/3}} = \boxed{\frac{1}{3}x^{-2/3}}.$$

Check against the power rule: $\frac{d}{dx}x^{1/3} = \frac{1}{3}x^{1/3-1} = \frac{1}{3}x^{-2/3}$. ✓ Same answer.

$g(x) = \sqrt{x}$ as the inverse of $f(x) = x^2$ (for $x \geq 0$). Here $f'(x) = 2x$, so $f'(g(x)) = 2\sqrt{x}$. Thus

$$g'(x) = \frac{1}{2\sqrt{x}} = \boxed{\frac{1}{2}x^{-1/2}}.$$

Check: $\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$. ✓

30. Hyperbolic functions, built from e^x (and e^{-x} via chain, $\frac{d}{dx}e^{-x} = -e^{-x}$).

$$\cosh x = \frac{e^x + e^{-x}}{2}:$$

$$\cosh' x = \frac{e^x + (-e^{-x})}{2} = \frac{e^x - e^{-x}}{2} = \boxed{\sinh x}.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}:$$

$$\sinh' x = \frac{e^x - (-e^{-x})}{2} = \frac{e^x + e^{-x}}{2} = \boxed{\cosh x}.$$

Note: each is the other's derivative with **no minus sign** — unlike $(\cos x)' = -\sin x$. The sign flip is absent because $\frac{d}{dx}e^{-x} = -e^{-x}$ contributes a minus that lines the two up positively, whereas the trig pair carries the minus through.

Concept & counterexample

31. True/false with justification.

(a) $(fg)' = f'g'$. **False.** The correct rule is $(fg)' = f'g + fg'$ — there is a missing cross-term. Counterexample: $f = g = x$ gives $(x \cdot x)' = (x^2)' = 2x$, but $f'g' = 1 \cdot 1 = 1 \neq 2x$.

(b) $\left(\frac{f}{g}\right)' = \frac{f'}{g'}$. **False.** The quotient rule is $\frac{f'g - fg'}{g^2}$. Counterexample: $f = x^2$, $g = x$ gives $\left(\frac{x^2}{x}\right)' = (x)' = 1$, but $\frac{f'}{g'} = \frac{2x}{1} = 2x \neq 1$.

(c) $(f \circ g)' = f' \cdot g'$. **False as written** — it omits that f' must be evaluated at $g(x)$. The chain rule is $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$, not $f'(x)g'(x)$. Counterexample: $f(u) = u^2$, $g(x) = x + 1$, so $(f \circ g)(x) = (x + 1)^2$ with derivative $2(x + 1)$. But $f'(x)g'(x) = (2x)(1) = 2x \neq 2(x + 1)$. The point of

evaluation is the whole content of the rule.

- (d) The derivative of e^{kx} is e^{kx} . **False** unless $k = 1$. By the chain rule, $\frac{d}{dx}e^{kx} = k e^{kx}$ — the constant k comes down. Only for $k = 1$ does it return itself.

32. Find the error. The student wrote $\frac{d}{dx}(x^2 \sin x) = 2x \cos x$ — they applied the **naïve (false) product rule** $(fg)' = f'g'$, multiplying the two derivatives $\frac{d}{dx}x^2 = 2x$ and $\frac{d}{dx}\sin x = \cos x$. The correct **product rule** $(fg)' = f'g + fg'$ gives

$$\frac{d}{dx}(x^2 \sin x) = \boxed{2x \sin x + x^2 \cos x}.$$

33. Find the error. The student wrote $\frac{d}{dx}(x^2+1)^5 = 5(x^2+1)^4$ — they differentiated the **outer** power but **forgot the chain-rule factor**, the derivative of the inside $(x^2 + 1)' = 2x$. Correct:

$$\frac{d}{dx}(x^2 + 1)^5 = 5(x^2 + 1)^4 \cdot 2x = \boxed{10x(x^2 + 1)^4}.$$

34. Construct f, g with $(fg)' = f'g'$ at a point. Take $f(x) = x$ and $g(x) = x$. Then the true derivative is $(fg)' = (x^2)' = 2x$, while $f'g' = 1$. These are equal where $2x = 1$, i.e. at $\boxed{x = \frac{1}{2}}$.

More systematically: $(fg)' = f'g + fg'$ equals $f'g'$ exactly when $f'g + fg' = f'g'$, a single equation that generically has isolated solutions x . (Another clean pick: $f(x) = g(x) = e^{2x}$ gives $(fg)' = (e^{4x})' = 4e^{4x}$ and $f'g' = (2e^{2x})^2 = 4e^{4x}$ — equal *everywhere*, by coincidence of these particular functions.) **Why you can't rely on it:** equality holds only at the specific x (or special functions) where the cross-terms happen to cancel into that form; for almost all functions and points the true derivative differs by exactly the missing cross-term $f'g + fg' - f'g'$.

35. Why $\frac{d}{dx}e^{\sin x}$ and $\frac{d}{dx}\sin(e^x)$ differ. The chain rule says: differentiate the outer at the inner, times the derivative of the inner.

- $e^{\sin x}$: outer $e^{(\cdot)}$ (derivative $e^{(\cdot)}$), inner $\sin x$ (derivative $\cos x$) $\Rightarrow e^{\sin x} \cos x$.
- $\sin(e^x)$: outer $\sin(\cdot)$ (derivative $\cos(\cdot)$), inner e^x (derivative e^x) $\Rightarrow \cos(e^x) e^x$.

Swapping inner and outer swaps **which function gets differentiated as the outer shell** and **which derivative becomes the multiplying inner factor**. The shells differ ($e^{\sin x}$ stays exponential-shaped; $\sin(e^x)$ becomes cosine-shaped), and the inner factors differ ($\cos x$ vs e^x). Composition is not commutative, and neither

is its derivative.

36. Is a nonzero self-derivative possible? **Yes** — it is e^x (and any constant multiple Ce^x), which we differentiated in this chapter: $\frac{d}{dx}e^x = e^x$. More generally $\frac{d}{dx}Ce^x = Ce^x$, so every constant multiple works. **Is it the only one (up to constant multiple)?** Yes. If $y' = y$, consider $\frac{d}{dx}(ye^{-x}) = y'e^{-x} + y(-e^{-x}) = (y' - y)e^{-x} = 0$, so ye^{-x} is constant, say C , giving $y = Ce^x$. Hence the only solutions are the constant multiples of e^x .

37. Sign of $\frac{d}{dx}e^{-x^2}$. By the chain rule, $\frac{d}{dx}e^{-x^2} = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$. Since $e^{-x^2} > 0$ always, the sign is the sign of $-2x$: - For $x > 0$: $-2x < 0$, so the derivative is **negative** — the curve is **falling**. - For $x < 0$: $-2x > 0$, so the derivative is **positive** — the curve is **rising**. - At $x = 0$: derivative is 0 — the **peak**.

So the bell curve e^{-x^2} rises on the left, peaks at the origin, and falls on the right — a single hump symmetric about $x = 0$, exactly the Gaussian/normal shape.

◆ Diamond

38. ◆ Product rule from the definition. Let $P(x) = f(x)g(x)$. Then

$$P'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h}.$$

Add and subtract $f(a+h)g(a)$ in the numerator:

$$\begin{aligned} & \frac{f(a+h)g(a+h) - f(a+h)g(a) + f(a+h)g(a) - f(a)g(a)}{h} \\ &= f(a+h) \cdot \frac{g(a+h) - g(a)}{h} + g(a) \cdot \frac{f(a+h) - f(a)}{h}. \end{aligned}$$

Now take $h \rightarrow 0$. The two difference quotients converge to $g'(a)$ and $f'(a)$. The factor $f(a+h) \rightarrow f(a)$ **because f is continuous at a** — and Chapter 3 guarantees this: differentiability implies continuity (if $f'(a)$ exists, then $f(a+h) - f(a) = \frac{f(a+h)-f(a)}{h} \cdot h \rightarrow f'(a) \cdot 0 = 0$). The factor $g(a)$ is constant. Therefore

$$P'(a) = f(a)g'(a) + g(a)f'(a) = \boxed{f'(a)g(a) + f(a)g'(a)}.$$

(Continuity of f is exactly what lets us replace $f(a+h)$ by $f(a)$ in the limit; without it the first term need not converge to $f(a)g'(a)$.)

39. ♦ Quotient rule via product + chain. First, $\frac{d}{dx}g^{-1}$. Write $g^{-1} = (g(x))^{-1}$; by the chain rule (power rule outer, g' inner),

$$\frac{d}{dx}g^{-1} = -1 \cdot (g)^{-2} \cdot g' = -\frac{g'}{g^2}.$$

Now write $\frac{f}{g} = f \cdot g^{-1}$ and apply the product rule:

$$\frac{d}{dx}(f \cdot g^{-1}) = f' \cdot g^{-1} + f \cdot \left(-\frac{g'}{g^2}\right) = \frac{f'}{g} - \frac{fg'}{g^2}.$$

Put over the common denominator g^2 (write $\frac{f'}{g} = \frac{f'g}{g^2}$):

$$\frac{d}{dx}\frac{f}{g} = \frac{f'g - fg'}{g^2}. \quad \boxed{\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}}$$

(Valid where $g \neq 0$.)

40. ♦ $\frac{d}{dx}\sin x = \cos x$ from the definition. Using $\sin(x+h) = \sin x \cos h + \cos x \sin h$,

$$\frac{\sin(x+h) - \sin x}{h} = \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.$$

Let $h \rightarrow 0$. By the two Chapter 2 limits, $\frac{\cos h - 1}{h} \rightarrow 0$ and $\frac{\sin h}{h} \rightarrow 1$. (Here $\sin x$, $\cos x$ are constants in h .) So the limit is $\sin x \cdot 0 + \cos x \cdot 1 = \boxed{\cos x}$.

For cosine, use $\cos(x+h) = \cos x \cos h - \sin x \sin h$:

$$\frac{\cos(x+h) - \cos x}{h} = \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \cos x \cdot \frac{\cos h - 1}{h} - \sin x \cdot \frac{\sin h}{h}.$$

As $h \rightarrow 0$: $\cos x \cdot 0 - \sin x \cdot 1 = \boxed{-\sin x}$.

41. ♦ Inverse-function rule, and $\frac{d}{dx}\ln x = \frac{1}{x}$. Suppose $g = f^{-1}$, so $f(g(x)) = x$ for all x in the domain. Differentiate both sides with respect to x . The right side: $\frac{d}{dx}x = 1$. The left side, by the chain rule: $f'(g(x)) \cdot g'(x)$. Equate:

$$f'(g(x)) \cdot g'(x) = 1 \quad \implies \quad \boxed{g'(x) = \frac{1}{f'(g(x))}}$$

(valid where $f'(g(x)) \neq 0$).

Now take $f(x) = e^x$, so $g(x) = \ln x$ is its inverse, and we are given $f'(x) = e^x$. Then $f'(g(x)) = e^{\ln x} = x$, so

$$\frac{d}{dx} \ln x = \frac{1}{f'(\ln x)} = \frac{1}{e^{\ln x}} = \boxed{\frac{1}{x}}.$$

This is the result Chapter 5 leans on.

42. ♦♦ Power rule for negative integers. Let n be a positive integer; we want $\frac{d}{dx} x^{-n}$. Write $x^{-n} = \frac{1}{x^n}$ and apply the quotient rule with $f = 1$ (so $f' = 0$) and $g = x^n$ (so $g' = nx^{n-1}$, the known positive-integer power rule):

$$\frac{d}{dx} \frac{1}{x^n} = \frac{(0)x^n - 1 \cdot nx^{n-1}}{(x^n)^2} = \frac{-nx^{n-1}}{x^{2n}} = -n x^{(n-1)-2n} = -n x^{-n-1}.$$

So $\boxed{\frac{d}{dx} x^{-n} = -n x^{-n-1}}$, which is precisely mx^{m-1} with $m = -n$ — the power rule extends to negative integers. (For $x \neq 0$.)

43. ♦♦ Power rule for rational exponents (implicit differentiation). Let $y = x^{p/q}$ with integers p, q ($q \neq 0$), and $x > 0$. Raise both sides to the q :

$$y^q = x^p.$$

Differentiate both sides with respect to x . Left side: chain rule on y^q gives $q y^{q-1} y'$. Right side: power rule gives $p x^{p-1}$. So

$$q y^{q-1} y' = p x^{p-1} \implies y' = \frac{p}{q} \cdot \frac{x^{p-1}}{y^{q-1}}.$$

Substitute $y = x^{p/q}$, so $y^{q-1} = x^{p(q-1)/q} = x^{p-p/q}$:

$$y' = \frac{p}{q} \cdot x^{(p-1)-(p-p/q)} = \frac{p}{q} x^{p/q-1}.$$

$$\boxed{\frac{d}{dx} x^{p/q} = \frac{p}{q} x^{p/q-1}}.$$

This closes the loan — the power rule holds for every rational exponent. (The trick of differentiating an unsolved equation in y is *implicit differentiation*.)

44. ♦♦ The chain rule, carefully – the $\Delta u = 0$ gap. The heuristic writes

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x},$$

which silently divides by Δu . **Why $\Delta u = 0$ can arise:** as x moves by a small Δx , the inner function $u = g(x)$ might not move at all – $\Delta u = g(x + \Delta x) - g(x)$ can be exactly 0 for some arbitrarily small Δx (for instance if g is locally constant, or, more subtly, if g oscillates so that g takes the value $g(x)$ again at nearby points). Whenever $\Delta u = 0$ the factored expression $\frac{\Delta y}{\Delta u}$ is undefined ($0/0$), so the algebra is invalid for those Δx .

How a rigorous proof gets around it. Replace the quotient by an equivalent *product* form that never divides by Δu . Define an auxiliary “error” function

$$E(k) = \begin{cases} \frac{f(u_0 + k) - f(u_0)}{k} - f'(u_0), & k \neq 0, \\ 0, & k = 0, \end{cases}$$

so that for **all** k (including $k = 0$),

$$f(u_0 + k) - f(u_0) = [f'(u_0) + E(k)] k,$$

and $E(k) \rightarrow 0$ as $k \rightarrow 0$ because f is differentiable at u_0 . Put $k = \Delta u = g(x + \Delta x) - g(x)$ and $u_0 = g(x)$:

$$\Delta y = [f'(u_0) + E(\Delta u)] \Delta u.$$

Divide by Δx (which is allowed – $\Delta x \neq 0$):

$$\frac{\Delta y}{\Delta x} = [f'(g(x)) + E(\Delta u)] \cdot \frac{\Delta u}{\Delta x}.$$

This identity holds **even when $\Delta u = 0$** (both sides are then handled correctly: the bracket multiplies $\Delta u/\Delta x$, no division by Δu occurs). As $\Delta x \rightarrow 0$: $\frac{\Delta u}{\Delta x} \rightarrow g'(x)$, and since g is continuous $\Delta u \rightarrow 0$, hence $E(\Delta u) \rightarrow 0$. The limit is $f'(g(x)) \cdot g'(x)$, the chain rule. The gap the heuristic hides is exactly the illegal division by Δu ; the fix is the error-term identity, which is a product and so survives $\Delta u = 0$.

Teach it

45. (*Chain rule via gear ratios, 3–4 sentences.*) Imagine three meshed gears: x drives u , and u drives y . If turning x spins u three times as fast ($\frac{du}{dx} = 3$), and turning u spins y twice as fast ($\frac{dy}{du} = 2$), then turning x spins y six times as fast — the ratios **multiply**: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 2 \cdot 3 = 6$. That is all the chain rule says: a small nudge in x travels through the chain, each stage scaling it by that stage's rate, so the total scaling is the product of the rates. To differentiate $f(g(x))$, find how fast the outer responds to its input, $f'(g(x))$, and multiply by how fast the inner responds to x , namely $g'(x)$.

46. (*Debug the classmate.*) One-line settler: take $f = g = x$. Then $(fg)' = (x \cdot x)' = (x^2)' = 2x$, but $f'g' = 1 \cdot 1 = 1$, and $2x \neq 1$. So derivatives do **not** distribute over multiplication. The correct rule is the **product rule**:

$$(fg)' = f'g + fg'.$$

47. (*Worked solution for $\sqrt{x^2 + 1}$, narrated so it transfers to $\sqrt{x^3 + 1}$.)*

We differentiate $y = \sqrt{x^2 + 1}$.

- **Identify the composition.** This is an *outer* function applied to an *inner* function. Outer: $f(u) = \sqrt{u} = u^{1/2}$ (the square root). Inner: $u = g(x) = x^2 + 1$ (what's under the root).
- **Differentiate the outer, inside left alone.** $f'(u) = \frac{1}{2}u^{-1/2} = \frac{1}{2\sqrt{u}}$, so $f'(g(x)) = \frac{1}{2\sqrt{x^2 + 1}}$.
- **Differentiate the inner.** $g'(x) = \frac{d}{dx}(x^2 + 1) = 2x$.
- **Multiply (chain rule).** $y' = f'(g(x)) \cdot g'(x) = \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \boxed{\frac{x}{\sqrt{x^2 + 1}}}$.

Redo for $\sqrt{x^3 + 1}$: same outer $\sqrt{u}$ (so $\frac{1}{2\sqrt{x^3+1}}$), new inner $x^3 + 1$ with derivative $3x^2$; multiply to get $\frac{3x^2}{2\sqrt{x^3 + 1}}$.

48. (*Design a problem — chain inside a product, full key.*)

Problem. Differentiate $f(x) = x^2 e^{3x}$.

Solution key. This needs the **product rule** (two factors x^2 and e^{3x}), and the

second factor needs the **chain rule**. - Let $u = x^2$, $v = e^{3x}$. Then $u' = 2x$. - Chain rule on v : outer $e^{(\cdot)}$, inner $3x$ with derivative 3, so $v' = e^{3x} \cdot 3 = 3e^{3x}$. - Product rule: $f'(x) = u'v + uv' = 2x e^{3x} + x^2 (3e^{3x})$. - Factor: $f'(x) = e^{3x}(2x + 3x^2) = \boxed{x e^{3x}(2 + 3x)}$.

(Check at $x = 0$: $f'(0) = 0$; indeed $f = x^2 e^{3x}$ has a double zero at the origin, so its slope there is 0. ✓)

49. (Why the trig derivatives need radians.) The clean formula $\frac{d}{dx} \sin x = \cos x$ rests entirely on the limit $\frac{\sin h}{h} \rightarrow 1$ as $h \rightarrow 0$. That limit is **true only when h is measured in radians** — geometrically, on the unit circle a small central angle of h radians cuts off an arc of length exactly h , so the chord (related to $\sin h$) and the arc h become equal in the limit, giving the ratio 1. If you measured the angle in degrees, the arc length would be $\frac{\pi}{180}h$, the limit would be $\frac{\pi}{180}$ instead of 1, and you'd be stuck with $\frac{d}{dx} \sin x^\circ = \frac{\pi}{180} \cos x^\circ$ — an ugly constant smeared through every trig derivative. Radians are *defined* so arc length equals angle, which is precisely what makes $\frac{\sin h}{h} \rightarrow 1$ and keeps the calculus clean; mathematicians measure angles in radians for exactly this reason.

Looking ahead

50. Differentiate a^x using the hint $a^x = e^{(\ln a)x}$. The exponent is $(\ln a)x$, a constant $\ln a$ times x . By the chain rule (outer $e^{(\cdot)}$, inner $(\ln a)x$ with derivative $\ln a$):

$$\frac{d}{dx} a^x = \frac{d}{dx} e^{(\ln a)x} = e^{(\ln a)x} \cdot \ln a = \boxed{(\ln a) a^x}.$$

So differentiating a general exponential multiplies it by the constant $\ln a$. (Chapter 5 confirms this; note $a = e$ gives $\ln e = 1$, recovering $(e^x)' = e^x$.)

51. Derivative of 2^x . By the result of problem 50 (or directly $2^x = e^{(\ln 2)x}$):

$$\frac{d}{dx} 2^x = (\ln 2) 2^x.$$

The constant multiple is $\boxed{\ln 2 \approx 0.693}$, **not** 1. Why not 1? Because for the base-2 exponential the difference-quotient limit is $\frac{2^h - 1}{h} \rightarrow \ln 2$, not 1. The number e is *defined* as the unique base for which that limit is exactly 1; any other base $a \neq e$ inherits the factor $\ln a$, which equals 1 only when $a = e$. Since $2 < e$, we get $\ln 2 < 1$.

52. The 100th derivative of $\sin x$. The derivatives cycle with period 4:

$$\sin x \xrightarrow{' } \cos x \xrightarrow{' } -\sin x \xrightarrow{' } -\cos x \xrightarrow{' } \sin x \text{ (back to start).}$$

The result depends only on $100 \bmod 4$. Since $100 = 4 \cdot 25$ with remainder 0, the 100th derivative is the same as the 0th — the function itself:

$$\boxed{\frac{d^{100}}{dx^{100}} \sin x = \sin x.}$$

53. Hyperbolic second derivatives. From problem 30, $\cosh' = \sinh$ and $\sinh' = \cosh$. Differentiating again:

$$\cosh'' = (\sinh)' = \cosh, \quad \sinh'' = (\cosh)' = \sinh.$$

So each hyperbolic function satisfies $\boxed{y'' = y}$ — it is its own second derivative. By contrast, from problem 19, $\sin'' = -\sin$ and $\cos'' = -\cos$, so **sine and cosine** satisfy $\boxed{y'' = -y}$. That contrast — $y'' = y$ for $\cosh, \sinh$ versus $y'' = -y$ for $\sin, \cos$ — is the seed of the exponential–trig duality (and the sign flip $i^2 = -1$) behind Euler’s formula in Chapter 7.

Chapter 5

The Function That Is Its Own Slope

Fluency

1. Differentiate. (Throughout, $\frac{d}{dx}e^{kx} = k e^{kx}$ by the chain rule.)

(a) $\frac{d}{dx}e^{5x} = 5e^{5x}$ (inner $5x$, inner derivative 5). $5e^{5x}$

(b) $\frac{d}{dx}e^{-x} = -e^{-x}$ (inner $-x$, inner derivative -1). $-e^{-x}$

(c) $\frac{d}{dx}3e^{2x} = 3 \cdot 2e^{2x} = 6e^{2x}$. $6e^{2x}$

(d) $\frac{d}{dx}(e^x + x^3) = e^x + 3x^2$. $e^x + 3x^2$

(e) $\frac{d}{dx}e^{x/2} = \frac{1}{2}e^{x/2}$ (inner $\frac{x}{2}$, inner derivative $\frac{1}{2}$). $\frac{1}{2}e^{x/2}$

(f) $\frac{d}{dx}7e^{-0.1x} = 7 \cdot (-0.1)e^{-0.1x} = -0.7e^{-0.1x}$. $-0.7e^{-0.1x}$

2. Differentiate. (Use $\frac{d}{dx}\ln x = \frac{1}{x}$, plus the chain rule and log laws.)

(a) $\frac{d}{dx}\ln x = \frac{1}{x}$. $\frac{1}{x}$

(b) $\ln(3x) = \ln 3 + \ln x$, and $\ln 3$ is a constant, so $\frac{d}{dx}\ln(3x) = \frac{1}{x}$. (Chain-rule check: $\frac{1}{3x} \cdot 3 = \frac{1}{x}$.) $\frac{1}{x}$

(c) $\frac{d}{dx}5\ln x = \frac{5}{x}$. $\frac{5}{x}$

(d) $\frac{d}{dx}(\ln x + x^2) = \frac{1}{x} + 2x$. $\frac{1}{x} + 2x$

3. Rewrite using log laws ($\ln(AB) = \ln A + \ln B$, $\ln \frac{A}{B} = \ln A - \ln B$, $\ln(A^p) = p \ln A$).

(a) $\ln 6 = \ln(2 \cdot 3) = \ln 2 + \ln 3$. $\ln 2 + \ln 3$

(b) $\ln \frac{x}{5} = \ln x - \ln 5$. $\ln x - \ln 5$

(c) $\ln(x^3) = 3 \ln x$. $3 \ln x$

(d) $\ln(2x^2) = \ln 2 + \ln(x^2) = \ln 2 + 2 \ln x$. $\ln 2 + 2 \ln x$

4. Evaluate. (Use $\ln 1 = 0$, $\ln e = 1$, $\ln(e^a) = a$, $e^{\ln a} = a$.)

(a) $\ln 1 = 0$ (because $e^0 = 1$). 0

(b) $\ln e = 1$ (because $e^1 = e$). 1

(c) $\ln e^4 = 4$. 4

(d) $e^{\ln 7} = 7$ ($\ln$ and e^x are inverses). 7

(e) $\ln e^{-2} = -2$. -2

5. $N(t) = 200 e^{0.4t}$.

$N(0) = 200 e^0 = 200 \cdot 1 = \mathbf{200}$.

$N(5) = 200 e^{0.4 \cdot 5} = 200 e^2 = 200(7.389 \dots) \approx \mathbf{1478}$.

6. Half-life 4 h, start 80 mg. Each 4 hours the amount halves:

- After 4 h (one half-life): $80 \rightarrow \mathbf{40}$ mg.
- After 8 h (two half-lives): $40 \rightarrow \mathbf{20}$ mg.
- After 12 h (three half-lives): $20 \rightarrow \mathbf{10}$ mg.

Practice

7. Differentiate, naming outer/inner functions.

(a) e^{3x^2} : outer $e^{(\cdot)}$, inner $u = 3x^2$ with $u' = 6x$. So $\frac{d}{dx} e^{3x^2} = e^{3x^2} \cdot 6x = \mathbf{6x e^{3x^2}}$.

(b) $\ln(x^2 + 1)$: outer $\ln(\cdot)$, inner $u = x^2 + 1$ with $u' = 2x$. So $\frac{d}{dx} \ln(x^2 + 1) = \frac{1}{x^2 + 1} \cdot 2x = \frac{\mathbf{2x}}{x^2 + 1}$.

(c) e^{-x^2} : outer $e^{(\cdot)}$, inner $u = -x^2$ with $u' = -2x$. So $\frac{d}{dx} e^{-x^2} = e^{-x^2} \cdot (-2x) = \mathbf{-2x e^{-x^2}}$.

(d) $\ln(\sqrt{x})$: first simplify with a log law, $\ln(\sqrt{x}) = \ln(x^{1/2}) = \frac{1}{2} \ln x$. Then

$$\frac{d}{dx} \frac{1}{2} \ln x = \frac{1}{2x}. \quad (\text{Chain-rule check: outer } \ln, \text{ inner } u = x^{1/2}, u' = \frac{1}{2}x^{-1/2}; \frac{1}{\sqrt{x}} \cdot \frac{1}{2}x^{-1/2} = \frac{1}{2x}.)$$

8. $N(t) = N_0 e^{kt}$. Differentiate with the chain rule (inner kt , inner derivative k):

$$\frac{dN}{dt} = N_0 \cdot k e^{kt} = k(N_0 e^{kt}) = kN.$$

So $N(t)$ satisfies $\frac{dN}{dt} = kN$. **In one sentence:** $N_0 = N(0)$ is the initial amount (the value at $t = 0$), and k is the proportional rate of change per unit time — positive k means growth, negative k means decay.

9. $\lambda = 0.05$ per year.

$$\text{Half-life: } t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{0.05} \approx \mathbf{13.9} \text{ years.}$$

$$\text{Fraction after 30 years: } \frac{N(30)}{N_0} = e^{-\lambda \cdot 30} = e^{-0.05 \cdot 30} = e^{-1.5} \approx \mathbf{0.223} \text{ (about 22\%).}$$

Sanity check: 30 years is $30/13.9 \approx 2.16$ half-lives, and $2^{-2.16} \approx 0.223$ — agrees.

10. $A(t) = 1000 e^{0.05t}$. Doubling means $A(t) = 2000$:

$$1000 e^{0.05t} = 2000 \Rightarrow e^{0.05t} = 2 \Rightarrow 0.05t = \ln 2 \Rightarrow t = \frac{\ln 2}{0.05} = \frac{0.693}{0.05} \approx \mathbf{13.9} \text{ years.}$$

Why “ $70 \div \text{rate\%}$ ” works: doubling time is $t = \frac{\ln 2}{\lambda}$. Since $\ln 2 \approx 0.693 \approx 0.70$, writing the rate as a percent $r = \frac{p}{100}$ gives $t = \frac{0.70}{p/100} = \frac{70}{p}$. So at $p\%$ the money doubles in about $70/p$ years — the “Rule of 70.”

11. Convert half-life to λ via $\lambda = \frac{\ln 2}{t_{1/2}}$.

$$\text{Drug A } (t_{1/2} = 2 \text{ h}): \lambda_A = \frac{\ln 2}{2} \approx \mathbf{0.347} \text{ per h, so } N_A(t) = N_0 e^{-0.347t}.$$

$$\text{Drug B } (t_{1/2} = 6 \text{ h}): \lambda_B = \frac{\ln 2}{6} \approx \mathbf{0.116} \text{ per h, so } N_B(t) = N_0 e^{-0.116t}.$$

$$\text{Time to fall to 10\% } (N/N_0 = 0.1): \text{ solve } e^{-\lambda t} = 0.1 \Rightarrow t = \frac{\ln 10}{\lambda}.$$

- A: $t = \frac{\ln 10}{\lambda_A} = \frac{2 \ln 10}{\ln 2} = 2 \log_2 10 \approx \mathbf{6.6} \text{ h.}$
- B: $t = \frac{\ln 10}{\lambda_B} = \frac{6 \ln 10}{\ln 2} = 6 \log_2 10 \approx \mathbf{19.9} \text{ h.}$

(Each is $\log_2 10 \approx 3.32$ half-lives — sensible, since 10% is between $1/8$ and $1/16$ of

the dose.)

12. $\text{pH} = -\log_{10}[\text{H}^+]$. If $[\text{H}^+]_1 = 100 [\text{H}^+]_2$:

$$\text{pH}_1 - \text{pH}_2 = -\log_{10}(100 [\text{H}^+]_2) + \log_{10}[\text{H}^+]_2 = -\log_{10} 100 = -2.$$

The pH values differ by 2 (the more concentrated solution is 2 pH units *lower*). A factor of 100 in concentration is just $\log_{10} 100 = 2$ units on the scale — that is the whole point of a log scale.

13. $h(x) = e^{2x} + \ln(5x)$. Note $\ln(5x) = \ln 5 + \ln x$, so its derivative is $\frac{1}{x}$. Thus

$$h'(x) = 2e^{2x} + \frac{1}{x}.$$

At $x = 1$: $h'(1) = 2e^2 + 1 = 2(7.389 \dots) + 1 \approx \mathbf{15.78}$. $h'(x) = 2e^{2x} + \frac{1}{x}$, $h'(1) = 2e^2 + 1 \approx 15.78$

Concept & counterexample

14. State true/false and justify.

- (a) **False.** The “power rule” $\frac{d}{dx}x^n = nx^{n-1}$ applies when the *base* is the variable and the exponent is constant. Here the variable is in the *exponent*, so you cannot write $\frac{d}{dx}e^x = xe^{x-1}$. The correct rule is $\frac{d}{dx}e^x = e^x$.
- (b) **False.** $\ln$ turns *products* into sums, not sums: $\ln(AB) = \ln A + \ln B$, but $\ln(A + B) \neq \ln A + \ln B$ in general. Counterexample: $A = B = 1$ gives $\ln(1 + 1) = \ln 2 \approx 0.693$, while $\ln 1 + \ln 1 = 0$.
- (c) **True.** $\frac{d}{dx}e^x = e^x$, so at every point the slope equals the height of the curve. That self-equality is exactly why e was chosen.
- (d) **True.** The slope of a^x at $x = 0$ is $M(a) = \ln a$. For $a = 2$, $M(2) = \ln 2 \approx 0.693 < 1$. So 2^x does grow ($\ln 2 > 0$), but it leaves the y -axis less steeply than slope 1.

15. e is the *natural* base because it is the unique base for which $\frac{d}{dx}a^x = a^x$ — i.e. e^x is its own derivative, leaving the y -axis at slope exactly 1. Every other base a carries an awkward constant factor $\ln a$ when differentiated ($\frac{d}{dx}a^x = (\ln a)a^x$); only base e makes that factor 1, so calculus with e is friction-free. (10 is natural only to our ten fingers.)

16. Only e^x satisfies $f' = f$ (its derivative equals itself). The other two do **not**:

$$\frac{d}{dx}2^x = (\ln 2)2^x \approx 0.693 \cdot 2^x, \quad \frac{d}{dx}3^x = (\ln 3)3^x \approx 1.099 \cdot 3^x.$$

Each derivative is the function scaled by $\ln a$ ($\ln 2$ for 2^x , $\ln 3$ for 3^x), neither equal to 1.

17. **Find the error.** The student applied $\frac{d}{dx} \ln x = \frac{1}{x}$ as if the input were x , but the input is x^2 — they **skipped the chain rule** (they forgot to multiply by the derivative of the inside, $2x$). Correct derivative two ways:

- *Chain rule:* outer $\ln$, inner $u = x^2$, $u' = 2x$: $\frac{d}{dx} \ln(x^2) = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$.
- *Log law first:* $\ln(x^2) = 2 \ln x$, so $\frac{d}{dx} \ln(x^2) = 2 \cdot \frac{1}{x} = \frac{2}{x}$.

Both give $\boxed{\frac{2}{x}}$ (valid for $x \neq 0$).

18. **Is it possible? No** — a first-order decay never reaches exactly zero in finite time. From $N(t) = N_0 e^{-\lambda t}$ with $N_0 > 0$ and $\lambda > 0$: the exponential $e^{-\lambda t}$ is *positive for every real t* (it is never 0). To get $N = 0$ you would need $e^{-\lambda t} = 0$, which has no solution; $e^{-\lambda t} \rightarrow 0$ only as $t \rightarrow \infty$. **What this says:** mathematically the drug is “completely gone” only in the limit — never at a finite time. In practice we use the half-life and declare it negligible once it has dropped below some threshold (a few percent), but strictly it only *approaches* zero.

19. The three loans, in plain words, with where each is repaid (the book repays all three in Chapter 10):

1. We assumed the slope-at-the-axis limit $M(a) = \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$ actually *exists* — that a^x has a genuine slope there. (*Repaid in Chapter 10.*)
2. We assumed $M(a)$ moves smoothly/continuously from below 1 (at $a = 2$) up through above 1 (at $a = 3$), so some base gives slope exactly 1. (*Repaid in Chapter 10.*)
3. We assumed the “slope-1 base” e is the **same** number as Chapter 1’s limit $\left(1 + \frac{1}{n}\right)^n$. (*Repaid in Chapter 10.*)

◆ Diamond

20. ◆ Prove $\frac{d}{dx} \ln x = \frac{1}{x}$. Let $y = \ln x$. By definition of the natural log as the inverse of e^x , this means

$$e^y = x.$$

Differentiate both sides with respect to x . The right side: $\frac{d}{dx} x = 1$. The left side needs the chain rule, since y is a function of x :

$$e^y \cdot \frac{dy}{dx} = 1.$$

Solve and substitute $e^y = x$:

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}. \quad \blacksquare$$

(Valid for $x > 0$, the domain of $\ln$.)

21. ◆ Prove $M(a) = \ln a$. Write every base in terms of e . Since $a = e^{\ln a}$,

$$a^x = (e^{\ln a})^x = e^{(\ln a)x}.$$

Differentiate with the chain rule, taking the constant $k = \ln a$:

$$\frac{d}{dx} a^x = \frac{d}{dx} e^{(\ln a)x} = (\ln a) e^{(\ln a)x} = (\ln a) a^x.$$

Comparing with the boxed rule $\frac{d}{dx} a^x = M(a) a^x$ gives $\boxed{M(a) = \ln a}$, and the derivative formula $\frac{d}{dx} a^x = (\ln a) a^x$. Check at $a = e$: $M(e) = \ln e = 1$, so $\frac{d}{dx} e^x = 1 \cdot e^x = e^x$ — exactly the condition that defined e . ■

22. ◆ Derive the half-life, and prove memorylessness. Start from $N(t) = N_0 e^{-\lambda t}$ ($\lambda > 0$). The half-life $t_{1/2}$ is the time when N has fallen to $\frac{1}{2} N_0$:

$$N_0 e^{-\lambda t_{1/2}} = \frac{1}{2} N_0.$$

Cancel N_0 (it is nonzero): $e^{-\lambda t_{1/2}} = \frac{1}{2}$. Take $\ln$ of both sides — justified because $\ln$ is the inverse of e^x , so $\ln(e^{-\lambda t_{1/2}}) = -\lambda t_{1/2}$:

$$-\lambda t_{1/2} = \ln \frac{1}{2} = -\ln 2 \implies \boxed{t_{1/2} = \frac{\ln 2}{\lambda}}.$$

Memoryless property. Suppose at some arbitrary time t_0 the amount is $N(t_0) = A$ (any level). After a further time $t_{1/2}$,

$$N(t_0 + t_{1/2}) = N_0 e^{-\lambda(t_0 + t_{1/2})} = N_0 e^{-\lambda t_0} \cdot e^{-\lambda t_{1/2}} = A \cdot e^{-\lambda t_{1/2}} = A \cdot \frac{1}{2} = \frac{1}{2}A.$$

So from *any* starting level A , it takes the same time $t_{1/2} = \frac{\ln 2}{\lambda}$ to halve — the elapsed time, not the absolute amount, is what matters. ■

23. ♦ Why a base with $M(a) = 1$ exists between 2 and 4. By Problem 21, $M(a) = \ln a$. Evaluate at the endpoints:

$$M(4) = \ln 4 = 2 \ln 2 \approx 2(0.693) = 1.386 > 1, \quad M(2) = \ln 2 \approx 0.693 < 1.$$

So $M(2) < 1 < M(4)$. **Assuming $M(a) = \ln a$ is continuous in a** (this is one of the chapter's loans — that M varies smoothly/continuously with the base), a continuous quantity that is below 1 at $a = 2$ and above 1 at $a = 4$ must equal 1 somewhere in between, by the Intermediate Value Theorem (Chapter 2). That in-between base is e , defined by $M(e) = \ln e = 1$. (The same argument works with $a = 3$ in place of $a = 4$, since $M(3) = \ln 3 \approx 1.099 > 1$.) ■

24. ♦ Prove $\frac{d}{dx} \ln(kx) = \frac{1}{x}$ for $k > 0$. Two ways.

Log law: $\ln(kx) = \ln k + \ln x$. Here $\ln k$ is a constant, so

$$\frac{d}{dx} \ln(kx) = \frac{d}{dx} (\ln k + \ln x) = 0 + \frac{1}{x} = \frac{1}{x}.$$

Chain rule: outer $\ln$, inner $u = kx$, $u' = k$:

$$\frac{d}{dx} \ln(kx) = \frac{1}{kx} \cdot k = \frac{1}{x}.$$

Both give $\boxed{\frac{1}{x}}$. **Why the constant vanishes:** multiplying the input by k adds the *constant* $\ln k$ to $\ln x$, which slides the graph of $\ln$ straight up (or down) by a fixed amount. A vertical shift changes no slopes, so the derivative is unchanged. ■

25. ♦♦ Uniqueness of the growth/decay solution. Suppose $f'(x) = k f(x)$ for all x , with $f(0) = N_0$. Define $g(x) = f(x) e^{-kx}$. Differentiate with the product rule (and chain rule on e^{-kx}):

$$g'(x) = f'(x) e^{-kx} + f(x) \cdot (-k) e^{-kx} = e^{-kx} [f'(x) - k f(x)].$$

By hypothesis $f'(x) - k f(x) = 0$, so $g'(x) = e^{-kx} \cdot 0 = 0$ for all x . A function with zero derivative everywhere is constant (a consequence of the Mean Value Theorem, Chapter 6), so $g(x) = g(0)$ for all x . Evaluate the constant: $g(0) = f(0)e^0 = N_0$. Therefore

$$f(x)e^{-kx} = N_0 \implies f(x) = N_0 e^{kx}.$$

This shows $N_0 e^{kx}$ is *the* (unique) solution, not merely *a* solution. ■

26. ♦♦ Differentiate x^x ($x > 0$) via $e/\ln$. The trick: never differentiate “variable-to-a-variable” directly. Route through e and $\ln$ using $u^v = e^{v \ln u}$. Here $u = v = x$, so

$$x^x = e^{x \ln x}.$$

Differentiate with the chain rule; the inner function is $w(x) = x \ln x$, whose derivative (product rule) is

$$w'(x) = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1.$$

Therefore

$$\frac{d}{dx} x^x = e^{x \ln x} \cdot (\ln x + 1) = x^x (\ln x + 1).$$

$$\frac{d}{dx} x^x = x^x (\ln x + 1)$$

Check (numeric): at $x = 2$, $x^x(\ln x + 1) = 4(\ln 2 + 1) \approx 4(1.693) = 6.77$, matching a direct numerical derivative of x^x at 2. ■

Teach it

27. (Model answer, friend-to-friend.) Ask: what function changes at a rate proportional to *itself* — bigger means faster-growing, in exact step with its own size? Every exponential a^x already *almost* does this: its derivative is the same curve scaled by a fixed number $M(a) = \ln a$. So we just pick the one base where that number is exactly 1 — and that base is $e \approx 2.718$. With that choice $\frac{d}{dx} e^x = e^x$: at every height, the slope equals the height, so e^x literally is its own slope.

28. (Worked solution; method generalizes to any half-life T .)

Set up the model. Decay is first-order, so $N(t) = N_0 e^{-\lambda t}$, where N_0 is the starting

dose and λ the elimination rate constant.

Find λ from the half-life. Given $t_{1/2} = 3$ h, use $t_{1/2} = \frac{\ln 2}{\lambda}$, so

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3} \approx 0.231 \text{ per hour.}$$

Solve for “5% remains.” We want $N/N_0 = 0.05$:

$$e^{-\lambda t} = 0.05.$$

Take $\ln$ of both sides (undoing the exponential) and solve for t :

$$-\lambda t = \ln 0.05 = -\ln 20 \implies t = \frac{\ln 20}{\lambda} = \frac{\ln 20}{\ln 2/3} = 3 \frac{\ln 20}{\ln 2} = 3 \log_2 20.$$

Numerically $t = \frac{\ln 20}{0.231} = \frac{3.00}{0.231} \approx \mathbf{13.0}$ hours.

Redo for any half-life T : same steps give $t = \frac{\ln 20}{(\ln 2)/T} = T \log_2 20 \approx 4.32 T$. (Sanity check: 5% is just past four halvings, since $2^{-4} = 6.25\%$ and $2^{-4.32} \approx 5\%$.)

29. Debug. The classmate’s slogan “ e to anything is its own derivative” is the error: $\frac{d}{dx}e^u = e^u$ holds only when the exponent is *literally* x . Here the exponent is $2x$, so they **skipped the chain rule** (multiply by the derivative of the inside, 2). Correct derivative:

$$\frac{d}{dx}e^{2x} = e^{2x} \cdot \frac{d}{dx}(2x) = 2e^{2x}.$$

$2e^{2x}$, **not** e^{2x} .

30. Design a problem. (*Problem + key.*)

Problem. You invest \$1000 in an account with continuous compounding. What annual rate r makes the balance double in exactly 10 years?

Answer key. Continuous compounding gives $A(t) = 1000 e^{rt}$. “Doubles in 10 years” means $A(10) = 2000$:

$$1000 e^{10r} = 2000 \implies e^{10r} = 2 \implies 10r = \ln 2 \implies r = \frac{\ln 2}{10} \approx 0.0693.$$

The required rate is $r = \frac{\ln 2}{10} \approx 6.93\%$. (Consistent with the Rule of 70: $70/6.93 \approx 10$ years. Verify: $1000 e^{0.0693 \cdot 10} = 1000 e^{0.693} = 1000 \cdot 2 = 2000$. ✓)

31. (Model answer.) Mathematicians prefer $\ln$ (base e) because its calculus is clean: $\frac{d}{dx} \ln x = \frac{1}{x}$ exactly, with no stray constant. By contrast $\frac{d}{dx} \log_{10} x = \frac{1}{x \ln 10}$ — every derivative and integral drags along an extra factor of $\ln 10 \approx 2.303$. Likewise $\frac{d}{dx} e^x = e^x$ but $\frac{d}{dx} 10^x = (\ln 10)10^x$. Base 10 is friendlier for reading off powers of ten; base e is friendlier for *doing calculus*, which is what we are doing.

Looking ahead

32. Match the value and derivatives of e^x at $x = 0$. Since e^x is its own derivative, every derivative equals e^x , and at $x = 0$ each equals $e^0 = 1$: the value is 1, the slope is 1, the second derivative (curvature) is 1, the third is 1, Building the polynomial whose successive derivatives at 0 are all 1:

- constant term (value 1): 1
- linear term (slope 1): x — since $\frac{d}{dx} x = 1$
- quadratic term (second derivative 1): $\frac{x^2}{2}$ — since $\frac{d^2}{dx^2} \frac{x^2}{2} = 1$
- cubic term (third derivative 1): $\frac{x^3}{6}$ — since $\frac{d^3}{dx^3} \frac{x^3}{6} = 1$

So the first four terms are

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \quad \left(\text{i.e. } \sum \frac{x^n}{n!} \right).$$

33. From a calculator, $e^{0.1} = 1.10517 \dots$ Compare the two approximations:

- $1 + 0.1 = 1.1$, error $\approx 1.10517 - 1.1 = 0.00517$.
- $1 + 0.1 + \frac{0.1^2}{2} = 1 + 0.1 + 0.005 = 1.105$, error $\approx 1.10517 - 1.105 = 0.00017$.

The second is much closer — its error (≈ 0.00017) is about 30× smaller than the first (≈ 0.005). Adding the quadratic term $\frac{x^2}{2}$ sharpens the approximation; this is a Taylor polynomial improving as you keep terms (Chapter 6).

34. Reading $\ln x = \int_1^x \frac{1}{t} dt$ as the (signed) area under $\frac{1}{t}$ from 1 to x :

- $\ln 1$ should be **0**: the integral from 1 to 1 spans zero width, enclosing no area, so $\int_1^1 \frac{1}{t} dt = 0$. (Matches $\ln 1 = 0$.)
- $\ln x$ should **increase** as x grows (for $x > 1$): the integrand $\frac{1}{t}$ is positive on $[1, x]$, so extending the upper limit to a larger x adds more positive area underneath — the accumulated area can only grow. (And for $0 < x < 1$

the integral runs backward from 1, giving negative area, so $\ln x < 0$ there — consistent with $\ln$ being negative below 1.)

Chapter 10 makes this construction airtight and discharges the chapter's three loans.

Chapter 6

The Mean Value Theorem

Fluency

1. Verify Rolle's hypotheses, then find every guaranteed c .

- (a) $f(x) = x^2 - 4x$ on $[0, 4]$. It is a polynomial, hence continuous on $[0, 4]$ and differentiable on $(0, 4)$. Endpoints: $f(0) = 0$ and $f(4) = 16 - 16 = 0$, so $f(0) = f(4)$. Rolle applies. Set $f'(x) = 2x - 4 = 0$, giving $x = 2$.

$c = 2$ (and $2 \in (0, 4)$; check $f'(2) = 0$). ✓

- (b) $f(x) = x^3 - x$ on $[-1, 1]$. Polynomial, so continuous and differentiable everywhere; $f(-1) = -1 + 1 = 0$ and $f(1) = 1 - 1 = 0$, so $f(-1) = f(1)$. Rolle applies. Set $f'(x) = 3x^2 - 1 = 0$, so $x^2 = \frac{1}{3}$, $x = \pm \frac{1}{\sqrt{3}}$.

$c = \pm \frac{1}{\sqrt{3}} = \pm \frac{\sqrt{3}}{3} \approx \pm 0.577$ (both lie in $(-1, 1)$). ✓

2. Find every c the MVT guarantees.

- (a) $f(x) = x^2$ on $[1, 3]$. Average slope $= \frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$. Set $f'(c) = 2c = 4$, so $c = 2$.

$c = 2$ (in $(1, 3)$; check $f'(2) = 4$). ✓

- (b) $f(x) = x^3$ on $[0, 3]$. Average slope $= \frac{27 - 0}{3 - 0} = 9$. Set $f'(c) = 3c^2 = 9$, so $c^2 = 3$, $c = \sqrt{3}$ (reject $-\sqrt{3}$, not in the interval).

$c = \sqrt{3} \approx 1.732$ (in $(0, 3)$; check $3(\sqrt{3})^2 = 9$). ✓

3. Name the single failing hypothesis.

- (a) $f(x) = |x|$ on $[-1, 1]$: continuous on $[-1, 1]$, but **not differentiable at $x = 0$** (a corner), an interior point. The differentiability-on- (a, b) hypothesis fails.

(b) $f(x) = \frac{1}{x}$ on $[-1, 1]$: **not continuous on** $[-1, 1]$ — it isn't even defined at $x = 0$ (infinite discontinuity). The continuity hypothesis fails (and the function isn't defined on the whole interval).

(c) $f(x) = \lfloor x \rfloor$ on $[0, 2]$: **not continuous** — it jumps at $x = 1$ and $x = 2$. The continuity-on- $[a, b]$ hypothesis fails.

4. $f'(x) = (x - 1)(x - 3)$. This upward parabola in x is positive outside its roots and negative between them.

Increasing on $(-\infty, 1)$ **and on** $(3, \infty)$ (where $f' > 0$); **decreasing on** $(1, 3)$ (where $f' < 0$). By Consequence 2.

5. $s(t) = t^2$ on $[0, 3]$. Average speed $= \frac{s(3) - s(0)}{3 - 0} = \frac{9 - 0}{3} = 3$ mph. The MVT (here, just $s'(t) = 2t$) gives the matching instant: $s'(t) = 2t = 3 \Rightarrow t = 1.5$.

Average speed = 3 mph, achieved exactly at $t = 1.5$ hours (in $(0, 3)$). ✓

6. $|f'(x)| \leq 3$ everywhere and $f(0) = 5$. By Consequence 3 on $[0, 4]$, $|f(4) - f(0)| \leq 3(4 - 0) = 12$, so $5 - 12 \leq f(4) \leq 5 + 12$.

Largest possible $f(4) = 17$; smallest possible $f(4) = -7$. (Both are attained by the straight lines $f(x) = 5 \pm 3x$, whose slopes are ± 3 .)

Practice

7. $f(x) = x^3 - x$ on $[0, 2]$ (polynomial — MVT hypotheses hold). Average slope $= \frac{f(2) - f(0)}{2 - 0} = \frac{(8 - 2) - 0}{2} = 3$. Set $f'(c) = 3c^2 - 1 = 3$, so $3c^2 = 4$, $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}}$ (reject the negative root).

$$c = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3} \approx 1.155 \text{ (in } (0, 2)\text{); check } 3c^2 - 1 = 4 - 1 = 3\text{). } \checkmark$$

8. Show $f(x) = x^5 + 2x + 1$ has **exactly one** real root.

Existence (IVT). f is a polynomial, hence continuous. $f(-1) = -1 - 2 + 1 = -2 < 0$ and $f(0) = 1 > 0$. Since f changes sign on $[-1, 0]$, the Intermediate Value Theorem gives a root in $(-1, 0)$.

Uniqueness (Consequence 2). $f'(x) = 5x^4 + 2$. Since $x^4 \geq 0$, we have $f'(x) \geq 2 > 0$ for all x , so f is strictly increasing on $\mathbb{R}$. A strictly increasing function takes each value at most once, so it can cross 0 at most once.

f has exactly one real root (at least one by IVT, at most one by strict monotonicity).

9. $f'(x) = g'(x) = 2x$ for all x , with $f(0) = 3$, $g(0) = 5$.

Since $f' = g'$ everywhere, by the corollary to Consequence 1 ($f - g$ has derivative $f' - g' = 0$, so it is constant). Evaluate the constant at 0: $f(0) - g(0) = 3 - 5 = -2$.

$$f(x) - g(x) = -2 \text{ for all } x.$$

Now identify each: any function with derivative $2x$ has the form $x^2 + C$ (again Consequence 1, comparing to x^2). Matching the given values $f(0) = 3$ and $g(0) = 5$:

$$f(x) = x^2 + 3 \text{ and } g(x) = x^2 + 5 \text{ (and indeed } f - g = -2\text{)}. \checkmark$$

10. Prove $|e^{-a} - e^{-b}| \leq |a - b|$ for all $a, b \geq 0$.

If $a = b$ both sides are 0. Otherwise take $f(x) = e^{-x}$, continuous and differentiable everywhere. Apply the MVT on the interval with endpoints a and b : there is a c strictly between them with

$$\frac{e^{-a} - e^{-b}}{a - b} = f'(c) = -e^{-c}.$$

Now $c > 0$ (it lies between $a, b \geq 0$, and the case $a = b$ is excluded), so $|f'(c)| = e^{-c} \leq e^0 = 1$. Therefore

$$|e^{-a} - e^{-b}| = |f'(c)| |a - b| \leq 1 \cdot |a - b| = |a - b|. \blacksquare$$

$|e^{-a} - e^{-b}| \leq |a - b|$ for all $a, b \geq 0$. (This is Consequence 3 with $M = 1$: e^{-x} is 1-Lipschitz on $[0, \infty)$.)

11. Prove $x > \ln(1 + x)$ for all $x > 0$.

Let $h(x) = x - \ln(1 + x)$ on $[0, \infty)$ (defined and differentiable for $x > -1$). Then $h(0) = 0 - \ln 1 = 0$, and

$$h'(x) = 1 - \frac{1}{1+x} = \frac{(1+x) - 1}{1+x} = \frac{x}{1+x}.$$

For $x > 0$ both numerator and denominator are positive, so $h'(x) > 0$. By Consequence 2, h is strictly increasing on $[0, \infty)$, so for any $x > 0$ we have $h(x) > h(0) = 0$, i.e. $x - \ln(1 + x) > 0$.

$x > \ln(1 + x)$ for all $x > 0$. ■

12. f differentiable with $f(1) = 3$ and $f'(x) \geq 2$ on $[1, 5]$. Prove $f(5) \geq 11$.

Apply the MVT on $[1, 5]$: there is a $c \in (1, 5)$ with

$$f(5) - f(1) = f'(c)(5 - 1) = 4f'(c).$$

Since $f'(c) \geq 2$, we get $f(5) - f(1) \geq 4 \cdot 2 = 8$, so $f(5) \geq f(1) + 8 = 3 + 8 = 11$.

$f(5) \geq 11$. ■ (A one-sided Consequence 3: a slope floor forces a value floor.)

Concept & counterexample

13. (True/false with justification.)

- (a) **True.** This is exactly the Mean Value Theorem: with those hypotheses, some interior c has $f'(c) = \frac{f(b) - f(a)}{b - a}$, the average slope.
- (b) **False.** $f'(c) = 0$ at an interior point says nothing about the endpoints. Example: $f(x) = x^3$ on $[-1, 2]$ has $f'(0) = 0$, yet $f(-1) = -1 \neq 8 = f(2)$. (Rolle is a one-way implication; its conclusion does not force equal endpoints.)
- (c) **True.** This is Consequence 2: $f' > 0$ on (a, b) implies f is (strictly) increasing there, since for $x < y$ the MVT gives $f(y) - f(x) = f'(c)(y - x) > 0$.
- (d) **False.** The MVT is an *existence* theorem: it guarantees *some* c exists but gives no formula or procedure to locate it. (You find c by solving $f'(c) =$ average slope yourself, when you can.)

14. **Find the error.** The student applied Rolle to $f(x) = |x|$ on $[-1, 1]$ even though f fails the **differentiability hypothesis**: $|x|$ has a corner at the interior point $x = 0$, where $f'(0)$ does not exist. Rolle requires differentiability at *every* interior point. With a hypothesis violated, the conclusion ($f'(c) = 0$ somewhere) need not hold — and indeed it doesn't, since $f'(x) = \pm 1$ wherever it exists.

Fix: Rolle does not apply; the missing slope at $x = 0$ is exactly the turning point a smooth curve would have used, so no flat tangent is required.

15. **Construct** a continuous f on $[0, 1]$ with $f(0) = f(1)$ but $f'(c) = 0$ at no

interior point.

Take the tent function $f(x) = 1 - |2x - 1|$, i.e. $f(x) = 2x$ on $[0, \frac{1}{2}]$ and $f(x) = 2 - 2x$ on $[\frac{1}{2}, 1]$. It is continuous, and $f(0) = 0 = f(1)$. But its slope is $+2$ on $(0, \frac{1}{2})$ and -2 on $(\frac{1}{2}, 1)$, and it is **not differentiable at the peak** $x = \frac{1}{2}$, so f' is never 0.

Broken hypothesis: differentiability on $(0, 1)$ (a corner at $\frac{1}{2}$).

That is allowed to ruin Rolle's conclusion because Rolle *requires* differentiability everywhere inside; the single missing derivative is precisely where a smooth curve would have flattened.

16. Construct a function increasing on an interval yet with $f' = 0$ at a point inside it.

Take $f(x) = x^3$ on $[-1, 1]$. For any $x < y$, $x^3 < y^3$, so f is strictly increasing on the whole interval. Yet $f'(x) = 3x^2$ gives $f'(0) = 0$.

So f **increasing does not require $f' > 0$ everywhere** — the converse of Consequence 2 fails. (Consequence 2 says “ $f' > 0 \Rightarrow$ increasing,” not the reverse; an increasing function may have isolated points where $f' = 0$.)

17. Is it possible? A differentiable f with $f(a) = f(b)$ but $f'(x) \neq 0$ at *every* point of (a, b) ?

No. If f is differentiable on (a, b) (hence continuous on $[a, b]$) and $f(a) = f(b)$, then Rolle's theorem *guarantees* at least one interior c with $f'(c) = 0$. So “ f' never zero on (a, b) ” directly contradicts Rolle.

Impossible — Rolle forbids it exactly when f is differentiable throughout. (The only way to avoid a zero slope is to break a hypothesis, as in Problems 14–15.)

18. “ $f' = g'$ everywhere” does not mean “ $f = g$,” because knowing two functions have identical slopes everywhere fixes only their *difference*, not their *height*: $f - g$ has derivative 0, so by Consequence 1 it is a constant, but that constant can be anything.

It means exactly $f(x) = g(x) + C$ **for some constant C** . One-line example: $f(x) = x^2 + 7$ and $g(x) = x^2$ both have derivative $2x$, yet differ by 7.

◆ Diamond

19. ◆ Prove Fermat's lemma. *Claim:* if f has an interior local maximum at c and $f'(c)$ exists, then $f'(c) = 0$.

Since c is a local maximum, there is a neighborhood in which $f(c + h) \leq f(c)$ for all small h , so the numerator $f(c + h) - f(c) \leq 0$. Examine the difference quotient $\frac{f(c + h) - f(c)}{h}$ from each side:

- **Right ($h > 0$):** numerator ≤ 0 , denominator > 0 , so the quotient ≤ 0 .
Letting $h \rightarrow 0^+$, $f'(c) = \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h} \leq 0$.
- **Left ($h < 0$):** numerator ≤ 0 , denominator < 0 , so the quotient ≥ 0 . Letting $h \rightarrow 0^-$, $f'(c) = \lim_{h \rightarrow 0^-} \frac{f(c + h) - f(c)}{h} \geq 0$.

Because $f'(c)$ exists, both one-sided limits equal $f'(c)$. A number that is simultaneously ≤ 0 and ≥ 0 must be 0.

$f'(c) = 0$. ■ (For a local minimum, reverse all inequalities; the same squeeze gives $f'(c) = 0$.)

20. ◆ Prove Rolle's theorem from the EVT and Fermat's lemma.

Hypotheses: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$.

By the **Extreme Value Theorem**, the continuous f attains a maximum value M and a minimum value m on the closed interval $[a, b]$. Two cases.

- **Both extremes occur only at the endpoints.** Then the largest and smallest values of f are among $\{f(a), f(b)\}$, which are equal, so $M = m$. A function whose max equals its min is **constant** on $[a, b]$. Then $f'(x) = 0$ at every interior point, and any $c \in (a, b)$ works.
- **At least one extreme occurs at an interior point $c \in (a, b)$.** That point is a local max or local min, and $f'(c)$ exists (differentiability on (a, b)), so by **Fermat's lemma** $f'(c) = 0$.

In every case some $c \in (a, b)$ has $f'(c) = 0$. ■

21. ◆ Prove the MVT from Rolle by the tilt trick.

Hypotheses: f continuous on $[a, b]$, differentiable on (a, b) . Define the auxiliary

function (the gap between f and its secant line):

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right].$$

Verify Rolle's hypotheses for g . The bracketed term is linear (continuous, differentiable everywhere), so g inherits continuity on $[a, b]$ and differentiability on (a, b) from f . At the endpoints:

$$g(a) = f(a) - f(a) = 0, \quad g(b) = f(b) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (b - a) \right] = f(b) - f(b) = 0.$$

So $g(a) = g(b) = 0$, and Rolle applies to g : there exists $c \in (a, b)$ with $g'(c) = 0$.

Finish. Differentiating,

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

so $g'(c) = 0$ becomes $f'(c) - \frac{f(b) - f(a)}{b - a} = 0$, i.e.

$$f'(c) = \frac{f(b) - f(a)}{b - a}. \quad \blacksquare$$

22. ♦ Prove Consequence 1 and its corollary.

Claim: if $f'(x) = 0$ at every point of an interval I , then f is constant on I .

Pick any two points $x < y$ in I . On $[x, y]$, f is differentiable (so continuous), so by the **MVT** there is a $c \in (x, y)$ with

$$f(y) - f(x) = f'(c)(y - x).$$

By hypothesis $f'(c) = 0$, hence $f(y) - f(x) = 0$, i.e. $f(y) = f(x)$. Since x, y were arbitrary, f takes one common value throughout I — it is constant. ■

Corollary: if $f'(x) = g'(x)$ everywhere on I , then $f - g$ is constant. Let $\varphi = f - g$. Then $\varphi'(x) = f'(x) - g'(x) = 0$ on I , so by the result just proved φ is constant; that is, $f(x) - g(x) = C$ for some constant C . ■

MVT is invoked at “ $f(y) - f(x) = f'(c)(y - x)$ ” — the single bridge from “zero slope” to “no change in value.”

23. ♦♦ Prove Consequence 2 ($f' > 0$ on an interval $\Rightarrow f$ strictly increasing).

Suppose $f'(x) > 0$ at every point of an interval I . Take any $x < y$ in I . On $[x, y]$, f is differentiable (hence continuous), so the **MVT** gives a $c \in (x, y)$ with

$$f(y) - f(x) = f'(c)(y - x).$$

Here $y - x > 0$, and $f'(c) > 0$ by hypothesis, so the right-hand side is a product of two positives:

$$f(y) - f(x) > 0 \quad \implies \quad f(y) > f(x).$$

Thus $x < y$ forces $f(x) < f(y)$: f is strictly increasing on I . ■

(The decreasing case is identical: $f'(c) < 0$ makes $f(y) - f(x) < 0$, so f is strictly decreasing.)

24. ♦♦♦ Prove Cauchy's generalized MVT.

Claim: with f, g continuous on $[a, b]$ and differentiable on (a, b) , some $c \in (a, b)$ satisfies $[f(b) - f(a)] g'(c) = [g(b) - g(a)] f'(c)$.

Define

$$h(x) = [f(b) - f(a)] g(x) - [g(b) - g(a)] f(x).$$

As a linear combination of f and g , h is continuous on $[a, b]$ and differentiable on (a, b) .

Verify $h(a) = h(b)$.

$$h(a) = [f(b) - f(a)] g(a) - [g(b) - g(a)] f(a) = f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) = f(b)g(a) - g(b)f(a)$$

$$h(b) = [f(b) - f(a)] g(b) - [g(b) - g(a)] f(b) = f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) = g(a)f(b) - g(b)f(b)$$

Both equal $f(b)g(a) - f(a)g(b)$, so $h(a) = h(b)$.

Apply Rolle. There is a $c \in (a, b)$ with $h'(c) = 0$. Since

$$h'(x) = [f(b) - f(a)] g'(x) - [g(b) - g(a)] f'(x),$$

setting $h'(c) = 0$ yields

$$[f(b) - f(a)] g'(c) = [g(b) - g(a)] f'(c). \quad \blacksquare$$

25. ♦♦ Prove $e^x \geq 1 + x$ for all real x , with equality only at $x = 0$.

Let $h(x) = e^x - 1 - x$. Then $h(0) = 1 - 1 - 0 = 0$, and $h'(x) = e^x - 1$. Since e^x is strictly increasing and $e^0 = 1$:

- For $x > 0$: $e^x > 1$, so $h'(x) > 0$. By Consequence 2, h is strictly increasing on $[0, \infty)$, hence $h(x) > h(0) = 0$ for $x > 0$.
- For $x < 0$: $e^x < 1$, so $h'(x) < 0$. By Consequence 2, h is strictly decreasing on $(-\infty, 0]$, so going *left* from 0 the value rises: $h(x) > h(0) = 0$ for $x < 0$.

Thus $h(x) > 0$ for every $x \neq 0$, and $h(0) = 0$. Equivalently $e^x \geq 1 + x$, with equality only at $x = 0$.

$e^x \geq 1 + x$ **for all x , equality iff $x = 0$.** ■ Since $1 + x$ is the tangent to e^x at 0, this says e^x never dips below that tangent line — a first taste of convexity.

Teach it

26. (Model answer — MVT via the speedometer, including what it does not give.)

Imagine a road trip where you cover the distance at an *average* speed of 60 mph. The Mean Value Theorem promises that at some single instant during the trip your speedometer read *exactly* 60 — your instantaneous speed had to equal your average speed at least once. More generally, for any smooth function the average rate of change over an interval is actually achieved as an instantaneous rate (a tangent parallel to the secant) somewhere strictly inside. What the theorem does **not** tell you is *when* that instant occurred — it guarantees such a moment exists but gives no clue where to find it; you'd have to solve for it yourself.

27. Debug your classmate. $f(x) = x^4 - 2x^2$, with $f(-\sqrt{2}) = f(\sqrt{2}) = 0$.

The misreading: Rolle promises **at least one** interior point with zero slope, never “exactly one.” Rolle is an existence statement, not a count.

The actual count: $f'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 4x(x - 1)(x + 1)$, which is zero at $x = -1, 0, 1$. All three lie in $(-\sqrt{2}, \sqrt{2})$.

Rolle guarantees ≥ 1 such point; this f actually has three ($x = -1, 0, 1$). The classmate confused “at least one” with “exactly one.”

28. (Textbook-quality worked solution — exactly one real root of $x^5 + 2x + 1$.)

Claim. $f(x) = x^5 + 2x + 1$ has exactly one real root.

The phrase “exactly one” hides two separate jobs: show a root *exists*, and show no *second* root can. Each half uses a different tool, and the pair is a pattern worth memorizing.

Existence (Intermediate Value Theorem). f is a polynomial, so it is continuous on all of $\mathbb{R}$. Evaluate at two convenient points: $f(-1) = (-1)^5 + 2(-1) + 1 = -2 < 0$ and $f(0) = 1 > 0$. A continuous function that is negative at one point and positive at another must hit every value between — in particular 0 — so by the IVT there is a root $r \in (-1, 0)$.

Uniqueness (monotonicity, Consequence 2). Differentiate: $f'(x) = 5x^4 + 2$. Since $x^4 \geq 0$ always, $f'(x) \geq 2 > 0$ for every x . By Consequence 2 of the MVT, f is strictly increasing on the whole line. A strictly increasing function never repeats a value, so it can equal 0 at most once.

Conclusion. At least one root (IVT) and at most one (strict increase) together give **exactly one real root**. ■

Reusable pattern: IVT manufactures a root from a sign change; a derivative of constant sign rules out any second crossing.

29. **Design a problem.** (Invent a cubic and interval where Rolle applies; give the key.)

Problem. Let $f(x) = x^3 - 3x$ on $[-\sqrt{3}, \sqrt{3}]$. Verify Rolle’s hypotheses and find every c it guarantees.

Solution key. f is a polynomial, so it is continuous on $[-\sqrt{3}, \sqrt{3}]$ and differentiable on the interior. Endpoints: $f(-\sqrt{3}) = (-\sqrt{3})^3 - 3(-\sqrt{3}) = -3\sqrt{3} + 3\sqrt{3} = 0$ and $f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0$, so $f(-\sqrt{3}) = f(\sqrt{3})$. Rolle applies. Set $f'(x) = 3x^2 - 3 = 0$, so $x^2 = 1$, $x = \pm 1$.

Guaranteed c = ± 1 , and both lie in $(-\sqrt{3}, \sqrt{3}) \approx (-1.732, 1.732)$. Check: $f'(\pm 1) = 3 - 3 = 0$. ✓

30. (Model answer — IVT vs. MVT.)

The Intermediate Value Theorem is a promise about **values**: a func-

tion continuous on $[a, b]$ must take every height between $f(a)$ and $f(b)$ somewhere in between — it cannot skip a value. The Mean Value Theorem is a promise about **slopes**: a function differentiable on (a, b) must, at some interior point, have an instantaneous rate of change equal to its average rate of change over $[a, b]$. One constrains the heights a continuous curve must pass through; the other constrains the slopes a smooth curve must achieve.

Looking ahead

31. The MVT is the order-1 statement $f(b) = f(a) + f'(c)(b-a)$. Chapter 7's order-2 version is $f(b) = f(a) + f'(a)(b-a) + \frac{1}{2}f''(c)(b-a)^2$. The pattern adds one more term, raising the power of $(b-a)$ by one and dividing by the next factorial. So the order-3 version should read:

$$f(b) = f(a) + f'(a)(b-a) + \frac{1}{2!}f''(a)(b-a)^2 + \frac{1}{3!}f'''(c)(b-a)^3.$$

The new term carries **power** $(b-a)^3$ and **factorial** $3! = 6$ (so coefficient $\frac{1}{6}$), with the deepest derivative evaluated at an unknown interior c . (This is the Taylor expansion with Lagrange remainder.)

32. At a point where f switches from increasing to decreasing, f' changes from positive to negative, so (where f' exists and is continuous) it must pass through $f' = 0$ at that point. The graph rises then falls, so:

$f' = 0$ **there, and the point is a local maximum** (a peak). This is the First Derivative Test of Chapter 8: $a + \rightarrow -$ sign change in f' marks a local max; $a - \rightarrow +$ change marks a local min.

33. A convex function lies above all its tangent lines. Lying above the tangent means the curve bends upward everywhere — its slope f' is itself increasing. The slope of the slope is f'' , and “ f' increasing” is exactly “ $f'' \geq 0$ ” (by Consequence 2 applied to f'). So:

$f'' \geq 0$ **everywhere** for a convex function (strictly $f'' > 0$ for strict convexity). Curvature has a constant sign: the graph never bends back down, which is why gradient descent on a convex loss meets no false floors. (For $f = e^x$ in Problem 25, $f'' = e^x > 0$ confirms it.)

Chapter 7

Building a Function From Its Derivatives

Fluency

1. For e^{bx} at 0: $f^{(k)}(0) = b^k$, so $c_k = b^k/k!$ and $T_2(x) = 1 + bx + \frac{b^2}{2}x^2$.

(a) $b = 1$: $T_2(x) = 1 + x + \frac{x^2}{2}$

(b) $b = 2$: $T_2(x) = 1 + 2x + 2x^2$

(c) $b = -1$: $T_2(x) = 1 - x + \frac{x^2}{2}$

(d) $b = \frac{1}{3}$: $T_2(x) = 1 + \frac{x}{3} + \frac{x^2}{18}$

2. Start from $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ and $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$.

(a) $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120}$

(b) $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24}$

(c) Substitute $2x$ into $\sin$: $2x - \frac{(2x)^3}{6} + \frac{(2x)^5}{120} = 2x - \frac{4x^3}{3} + \frac{4x^5}{15}$

(d) Substitute $\frac{x}{2}$ into $\cos$: $1 - \frac{(x/2)^2}{2} + \frac{(x/2)^4}{24} = 1 - \frac{x^2}{8} + \frac{x^4}{384}$

3. All are geometric. $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \dots$.

(a) $u = x$: $1 + x + x^2 + x^3$; coefficients all 1 ($c_k = 1$).

(b) $u = -x$: $1 - x + x^2 - x^3$; coefficients $(-1)^k$.

(c) $u = 2x$: $1 + 2x + 4x^2 + 8x^3$; coefficients 2^k .

4. Compute $f(0)$, $f'(0)$, $f''(0)$; then $T_2 = f(0) + f'(0)x + \frac{f''(0)}{2}x^2$.

$$(a) \sqrt{1+x}: f(0) = 1, f' = \frac{1}{2}(1+x)^{-1/2} \Rightarrow f'(0) = \frac{1}{2}, f'' = -\frac{1}{4}(1+x)^{-3/2} \Rightarrow f''(0) = -\frac{1}{4}. \quad \boxed{1 + \frac{x}{2} - \frac{x^2}{8}}$$

$$(b) (1+x)^{1/3}: f(0) = 1, f' = \frac{1}{3}(1+x)^{-2/3} \Rightarrow \frac{1}{3}, f'' = -\frac{2}{9}(1+x)^{-5/3} \Rightarrow -\frac{2}{9}. \quad \boxed{1 + \frac{x}{3} - \frac{x^2}{9}}$$

$$(c) \ln(1+x): f(0) = 0, f' = \frac{1}{1+x} \Rightarrow 1, f'' = -\frac{1}{(1+x)^2} \Rightarrow -1. \quad \boxed{x - \frac{x^2}{2}}$$

$$(d) \ln(1-x): f(0) = 0, f' = -\frac{1}{1-x} \Rightarrow -1, f'' = -\frac{1}{(1-x)^2} \Rightarrow -1. \quad \boxed{-x - \frac{x^2}{2}}$$

5. Reuse known series.

$$(a) \frac{1}{1+x^2}: \text{put } u = -x^2 \text{ into } \frac{1}{1-u}: 1 + (-x^2) + (-x^2)^2 + \dots = \boxed{1 - x^2 + x^4}$$

$$(b) e^{x^2}: \text{put } x^2 \text{ into } e^u: 1 + x^2 + \frac{x^4}{2!} = \boxed{1 + x^2 + \frac{x^4}{2}}$$

$$(c) x e^x: \text{multiply } x \text{ by } 1 + x + \frac{x^2}{2} + \dots: \boxed{x + x^2 + \frac{x^3}{2}}$$

6.

$$(a) e^x: f^{(k)}(0) = 1 \text{ for all } k, \text{ so } f^{(0)} = f' = f'' = f''' = 1, \text{ and } c_k = \frac{1}{k!}: c_0 = 1, c_1 = 1, c_2 = \frac{1}{2}, c_3 = \frac{1}{6}.$$

$$(b) \cos x: \text{derivatives at } 0 \text{ are } f(0) = 1, f'(0) = 0, f''(0) = -1, f'''(0) = 0; \text{ coefficients } c_0 = 1, c_1 = 0, c_2 = -\frac{1}{2}, c_3 = 0.$$

7. (Calculator values rounded.)

$$(a) e^{0.2} \approx 1 + 0.2 + \frac{0.04}{2} + \frac{0.008}{6} = 1.22133. \text{ True } 1.22140; \text{ off by } \approx 7 \times 10^{-5}.$$

$$(b) \cos(0.1) \approx 1 - \frac{0.01}{2} = 0.99500. \text{ True } 0.995004; \text{ off by } \approx 4 \times 10^{-6}.$$

$$(c) \sin(0.3) \approx 0.3 - \frac{0.027}{6} = 0.29550. \text{ True } 0.295520; \text{ off by } \approx 2 \times 10^{-5}.$$

$$(d) \sqrt{1.1} \text{ with } x = 0.1 \text{ in } 1 + \frac{x}{2} - \frac{x^2}{8}: 1 + 0.05 - 0.00125 = 1.04875. \text{ True } 1.048809; \text{ off by } \approx 6 \times 10^{-5}.$$

8. Use $(x-a)$ powers: $T_2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$.

$$(a) e^x \text{ at } a = 1: f(1) = f'(1) = f''(1) = e. \quad \boxed{e \left[1 + (x-1) + \frac{(x-1)^2}{2} \right]}$$

$$(b) \sqrt{x} \text{ at } a = 4: f(4) = 2, f' = \frac{1}{2}x^{-1/2} \Rightarrow \frac{1}{4}, f'' = -\frac{1}{4}x^{-3/2} \Rightarrow -\frac{1}{4} \cdot \frac{1}{8} = -\frac{1}{32}. \quad \boxed{2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2}$$

Practice

9. e^{-x} : $f^{(k)}(0) = (-1)^k$, so $c_k = \frac{(-1)^k}{k!}$ and

$$e^{-x} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

Check: replacing $x \rightarrow -x$ in $e^x = \sum x^k/k!$ flips the sign on every odd power — exactly the alternating signs above. 3

10. $f(x) = \frac{1}{1-x}$ has $f^{(k)}(x) = \frac{k!}{(1-x)^{k+1}}$, so $f^{(k)}(0) = k!$ and $c_k = \frac{k!}{k!} = 1$. Hence

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots,$$

the geometric series. The equality is valid exactly for $|x| < 1$ (where the series converges); for $|x| \geq 1$ the right side diverges.

11. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$; first three nonzero terms $x - \frac{x^2}{2} + \frac{x^3}{3}$. One line:

the leading term is x , so $\frac{\ln(1+t)}{t} = \frac{t-t^2/2+\dots}{t} = 1 - \frac{t}{2} + \dots \rightarrow 1$ as $t \rightarrow 0$.

12. $T_2(0.1) = 1 + 0.1 + \frac{0.01}{2} = 1.105$. With $|f'''| = |e^x| \leq 1.2$ on $[0, 0.1]$:

$$|R_2(0.1)| \leq \frac{1.2(0.1)^3}{3!} = \frac{1.2 \times 0.001}{6} = 0.0002.$$

True value $e^{0.1} = 1.105171$, so the actual error is $|1.105171 - 1.105| = 0.000171 < 0.0002$. Inside the bound. 3

13. Estimate: $1 - \frac{(0.2)^2}{2} = 1 - 0.02 = 0.98$. Error bounded by the next term $\frac{x^4}{24} = \frac{(0.2)^4}{24} = \frac{0.0016}{24} \approx 6.7 \times 10^{-5}$. (True $\cos 0.2 = 0.980067$, actual error $\approx 6.6 \times 10^{-5}$ — within the bound.) The error jumps by **two** powers because $\cos$ is even: only even powers appear, so the first omitted term after x^2 is x^4 , not x^3 . The odd term is absent, so the leading error is $O(x^4)$.

14. Antidifferentiate $1 - x^2 + x^4 - \dots$ term by term and fix the constant by $\arctan 0 = 0$:

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots, \quad \text{first three: } \boxed{x - \frac{x^3}{3} + \frac{x^5}{5}}$$

At $x = 1$: $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ (Leibniz's series).

15. Binomial $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \dots$.

$(1+x)^{1/2}$ ($\alpha = \frac{1}{2}$): $1 + \frac{1}{2}x + \frac{(1/2)(-1/2)}{2}x^2 = \boxed{1 + \frac{x}{2} - \frac{x^2}{8}}$ $(1+x)^{-1}$ ($\alpha = -1$):
 $1 + (-1)x + \frac{(-1)(-2)}{2}x^2 = \boxed{1 - x + x^2}$ Check: $\frac{1}{1+x} = 1 - x + x^2 - \dots$ is the geometric series with $u = -x$. 3

16. Multiply $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$ by $\cos x = 1 - \frac{x^2}{2} + \dots$: $-x^0$: $1 - x^1$: $1 - x^2$:
 $\frac{1}{2} + 1 \cdot (-\frac{1}{2}) = 0$ $-x^3$: $\frac{1}{6} + 1 \cdot (-\frac{1}{2}) = -\frac{1}{3}$

First three nonzero terms: $\boxed{1 + x - \frac{x^3}{3}}$. The x^2 term drops out.

17. Substitute $-x^2$ into $e^u = 1 + u + \frac{u^2}{2} + \dots$:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \dots, \quad \text{first three: } \boxed{1 - x^2 + \frac{x^4}{2}}$$

Only even powers appear, so it is **even** ($f(-x) = f(x)$). It is largest at $x = 0$, where $e^0 = 1$ (the exponent $-x^2 \leq 0$ is maximized at $x = 0$).

18. (Notebook.) Plotting $\sin x$ with its 1-, 3-, 5-, 7-term polynomials on $[-2\pi, 2\pi]$: the 1-term line x matches only the central rising stretch through the origin; the 3-term $x - \frac{x^3}{6}$ follows the first hump on each side before diving off; 5 terms hold through roughly the second hump; 7 terms track out toward $|x| \approx 2\pi$, capturing several humps before peeling away. Each added pair of terms buys roughly one more hump of fidelity; all eventually shoot to $\pm\infty$ since a polynomial cannot stay bounded like $\sin$.

Concept & counterexample

19. (a) **True.** By construction $T_n^{(j)}(a) = f^{(j)}(a)$ for $j = 0, \dots, n$ — that is exactly how the coefficients $c_k = f^{(k)}(a)/k!$ were chosen. (b) **False.** Larger n improves accuracy *near* a , but outside the interval of convergence it can be worse. E.g. the series for $\frac{1}{1-x}$ diverges for $|x| > 1$, so higher n makes $T_n(2)$ blow up, not improve. (c) **True.** A degree- d polynomial has $f^{(k)} \equiv 0$ for $k > d$, so the Taylor series terminates and reconstructs the polynomial exactly (it equals itself). (d) **False** in general. Two functions can share all Taylor coefficients at a yet differ — see problem 33 (e^{-1/x^2} vs. the zero function, both with all-zero series at 0).

20. The mistake is using the series outside its **interval of convergence**. The

identity $\frac{1}{1-x} = 1 + x + x^2 + \dots$ holds only for $|x| < 1$; at $x = 2$ (> 1) the series diverges, so writing “ $-1 = 1 + 2 + 4 + \dots$ ” equates a finite number to a divergent sum. The function $\frac{1}{1-x} = -1$ is fine at $x = 2$; its *series representation* simply isn’t valid there.

21. $\sin$ is **odd**: $\sin(-x) = -\sin x$. A power series equals its function only if it has the same parity, and an odd function’s series can contain only odd powers (even powers would survive the sign flip and break oddness). Concretely, the even derivatives of $\sin$ at 0 are all 0, killing every even coefficient. $\cos$ is **even**: $\cos(-x) = \cos x$, so only even powers survive; its odd derivatives vanish at 0.

22. Degree-1 approximation $1 + 3x$ means $f(0) = 1$, $f'(0) = 3$. One example: $f(x) = 1 + 3x$ itself. A different one: $g(x) = 1 + 3x + x^2$ (or e^{3x} , since $e^{3x} = 1 + 3x + \frac{9x^2}{2} + \dots$). All such functions share the same value and slope at 0 — they pass through $(0, 1)$ with slope 3 — and differ in their second-and-higher derivatives at 0, i.e. in how they curve away from the common tangent line.

23. The factorial appears because differentiating $(x - a)^k$ exactly k times peels off the factors $k, k - 1, \dots, 1$ in turn, leaving the constant $k!$. So $P^{(k)}(a) = k! c_k$, and dividing by $k!$ to solve for c_k is what plants the $1/k!$ in each coefficient.

24. Yes to both. Tangent-line error is $R_1 = \frac{1}{2}f''(c)(x - a)^2 = O((x - a)^2)$ whenever f is twice differentiable near a — e.g. any smooth f like e^x . Error shrinking only like $|x - a|$ happens when $f'(a)$ exists but f is *not* twice differentiable (or the tangent doesn’t match slope) — e.g. a function with a kink in its derivative. The separating property: a genuine matched tangent line at a twice-differentiable point gives second-order $((x - a)^2)$ error; lose that smoothness and you drop to first-order.

25. The series should start failing at $x = 1$ (radius of convergence 1). Reason: $\ln(1 + x)$ misbehaves at $x = -1$, where $1 + x = 0$ and $\ln \rightarrow -\infty$. A power series centered at 0 converges in a symmetric interval reaching out to the nearest bad point — distance 1 from the center — so it can converge no farther than $|x| < 1$; thus it must fail by $|x| = 1$. (At $x = 1$ it happens to converge to $\ln 2$, at $x = -1$ it diverges.)

◆ Diamond

26. Let $P(x) = \sum_{j=0}^n c_j(x-a)^j$. Differentiate k times. For a single term $(x-a)^j$: - if $j < k$, the k -th derivative is 0 (you differentiate a degree- j term more than j times); - if $j > k$, the k -th derivative still has a factor $(x-a)^{j-k}$, which vanishes at $x = a$; - if $j = k$, $\frac{d^k}{dx^k}(x-a)^k = k!$ (a constant).

So $P^{(k)}(a) = c_k \cdot k!$, the only surviving term. Setting $P^{(k)}(a) = f^{(k)}(a)$ gives $c_k k! =$

$$f^{(k)}(a), \text{ i.e. } \boxed{c_k = \frac{f^{(k)}(a)}{k!}}. \blacksquare$$

27. Let $P(x) = c_0 + c_1(x-a) + c_2(x-a)^2$ be any parabola matching $f(a), f'(a), f''(a)$. Evaluating at a : $P(a) = c_0 = f(a)$; $P'(a) = c_1 = f'(a)$; $P''(a) = 2c_2 = f''(a) \Rightarrow c_2 = \frac{f''(a)}{2}$. The three conditions determine c_0, c_1, c_2 **uniquely**, so there is exactly one such parabola, namely T_2 . $\blacksquare$

28. Taylor's remainder: $f(x) = T_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ for some c between a and x . - $n = 0$: $T_0(x) = f(a)$, so $f(x) = f(a) + \frac{f'(c)}{1!}(x-a) = f(a) + f'(c)(x-a)$. Rearranged, $\frac{f(x)-f(a)}{x-a} = f'(c)$ - the **Mean Value Theorem**. - $n = 1$: $T_1(x) = f(a) + f'(a)(x-a)$, so $f(x) = f(a) + f'(a)(x-a) + \frac{f''(c)}{2!}(x-a)^2 = f(a) + f'(a)(x-a) + \frac{1}{2}f''(c)(x-a)^2$ - the **second-order MVT** of Chapter 6. $\blacksquare$

29. From the series, $\cos x = \sum_{m \geq 0} \frac{(-1)^m}{(2m)!} x^{2m}$ has only even powers, and $x^{2m} = (-x)^{2m}$, so $\cos(-x) = \cos x$ - **even**. From $\sin x = \sum_{m \geq 0} \frac{(-1)^m}{(2m+1)!} x^{2m+1}$, every power is odd and $(-x)^{2m+1} = -x^{2m+1}$, so $\sin(-x) = -\sin x$ - **odd**. (Term-by-term sign behavior under $x \rightarrow -x$ is justified by the series' convergence everywhere, problem 32.) $\blacksquare$

30. Start from $e^u = \sum_{k \geq 0} \frac{u^k}{k!}$ and put $u = ix$. The powers of i cycle: $i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, \dots$ Group by parity of k :

$$e^{ix} = \sum_{m \geq 0} \frac{(ix)^{2m}}{(2m)!} + \sum_{m \geq 0} \frac{(ix)^{2m+1}}{(2m+1)!}.$$

Even part: $(ix)^{2m} = (i^2)^m x^{2m} = (-1)^m x^{2m}$, giving $\sum \frac{(-1)^m x^{2m}}{(2m)!} = \cos x$. Odd part: $(ix)^{2m+1} = i(-1)^m x^{2m+1}$, giving $i \sum \frac{(-1)^m x^{2m+1}}{(2m+1)!} = i \sin x$. Hence

$$\boxed{e^{ix} = \cos x + i \sin x}.$$

Set $x = \pi$: $\cos \pi = -1$, $\sin \pi = 0$, so $e^{i\pi} = -1$. ■

31. Fix x and choose M so that $f(x) = T_n(x) + M(x-a)^{n+1}$ exactly. Define

$$g(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k - M(x-t)^{n+1}.$$

Endpoints. At $t = x$, only the $k = 0$ term survives ($f(x)$) and the last term is 0: $g(x) = f(x) - f(x) - 0 = 0$. At $t = a$, the sum is $T_n(x)$, so $g(a) = f(x) - T_n(x) - M(x-a)^{n+1} = 0$ by the choice of M . Thus $g(a) = g(x) = 0$, and **Rolle's theorem** gives c strictly between with $g'(c) = 0$.

Derivative telescopes. Differentiate the sum in t using the product rule on each term:

$$\frac{d}{dt} \frac{f^{(k)}(t)}{k!} (x-t)^k = \frac{f^{(k+1)}(t)}{k!} (x-t)^k - \frac{f^{(k)}(t)}{(k-1)!} (x-t)^{k-1}.$$

Summing over $k = 0, \dots, n$, consecutive pieces cancel and only the last forward term remains:

$$\frac{d}{dt} \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k = \frac{f^{(n+1)}(t)}{n!} (x-t)^n.$$

So $g'(t) = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n + M(n+1)(x-t)^n$. Setting $g'(c) = 0$ and dividing by $(x-c)^n \neq 0$:

$$M(n+1) = \frac{f^{(n+1)}(c)}{n!} \implies M = \frac{f^{(n+1)}(c)}{(n+1)!},$$

which is the Lagrange remainder $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$. ■

32. Every derivative of $\sin x$ is $\pm \sin x$ or $\pm \cos x$, so $|f^{(n+1)}(t)| \leq 1$ for all t . With $a = 0$,

$$|R_n(x)| = \frac{|f^{(n+1)}(c)|}{(n+1)!} |x|^{n+1} \leq \frac{|x|^{n+1}}{(n+1)!}.$$

Fix x . As $n \rightarrow \infty$, $\frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$ (factorial growth beats any fixed power), so $R_n(x) \rightarrow 0$ and $T_n(x) \rightarrow \sin x$ for **every** x . This is cleaner than the e^x case because the bound $|f^{(n+1)}| \leq 1$ is a *uniform constant* — no $e^{|x|}$ factor needs carrying — so the estimate holds with no extra work for all real x . ■

33. All derivatives vanish at 0: $f^{(k)}(0) = 0$ for every k . Hence its Taylor series at

0 is

$$\sum_{k=0}^{\infty} \frac{0}{k!} x^k = 0 + 0 + 0 + \cdots = 0,$$

the identically-zero series, which converges (to 0) for every x . But $f(x) = e^{-1/x^2} > 0$ for all $x \neq 0$, so the series does **not** equal f anywhere except $x = 0$. This refutes 19(d): f and the zero function share every Taylor coefficient at 0 yet are different functions. A function need not equal its Taylor series even when that series converges — having matching derivatives to all orders is not enough; the remainder must also vanish (here it does not).

Teach it

34. (Model answer.) “Near a point, a curve looks almost straight, and the derivative is the slope of that straight-line stand-in — the tangent line is the best you can do with a line, because it matches the curve’s height and its slope right there. Taylor’s idea is to keep going: also match the curvature with an x^2 term, then the next bend with an x^3 term, and so on. Each new term matches one more derivative, so the polynomial hugs the curve over a wider stretch. The k -th coefficient is $f^{(k)}(a)/k!$ — derivative over factorial — and adding terms trades a line for a parabola, then a cubic, getting as close as you like near a .”

35. Missing ingredient: the factorial denominators. Every derivative of e^x equals e^x , which at 0 gives 1 — but the Taylor coefficient is $f^{(k)}(0)/k! = 1/k!$, not 1. The classmate forgot to divide by $k!$. Correct series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

One-line reason for $\frac{1}{k!}$: differentiating $(x)^k$ exactly k times yields $k!$, so to match $f^{(k)}(0)$ the coefficient must carry a compensating $1/k!$.

36. (Worked solution, redoable for ln 1.2.) Use $\ln(1+x)$ at 0 with $x = 0.1$. - **Polynomial.** $T_2(x) = x - \frac{x^2}{2}$, so $T_2(0.1) = 0.1 - \frac{0.01}{2} = 0.095$. - **Remainder.** $f'''(x) = \frac{2}{(1+x)^3}$. On $[0, 0.1]$, $(1+x)^3 \geq 1$, so $|f'''| \leq 2$. Then

$$|R_2(0.1)| \leq \frac{2(0.1)^3}{3!} = \frac{2 \times 0.001}{6} \approx 3.3 \times 10^{-4}.$$

- **Conclusion.** $\ln 1.1 \approx 0.095$ with error $< 3.4 \times 10^{-4}$. (True $\ln 1.1 = 0.0953102$, actual error $\approx 3.1 \times 10^{-4}$ — inside the bound.) - **For $\ln 1.2$:** repeat with $x = 0.2$. $T_2(0.2) = 0.2 - 0.02 = 0.18$; on $[0, 0.2]$, $|f'''| \leq 2$, so $|R_2| \leq \frac{2(0.2)^3}{6} = \frac{0.016}{6} \approx 2.7 \times 10^{-3}$. (True $\ln 1.2 = 0.182322$.)

37. (Design + key.) Problem. Find the degree-2 Taylor polynomial of $f(x) = \tan x$ at $a = 0$. **Key.** $f(0) = \tan 0 = 0$. $f'(x) = \sec^2 x$, $f'(0) = 1$. $f''(x) = 2 \sec^2 x \tan x$, $f''(0) = 0$. So

$$T_2(x) = 0 + 1 \cdot x + \frac{0}{2}x^2 = x.$$

(The first curvature shows only at order 3: $\tan x = x + \frac{x^3}{3} + \dots$, a good sanity check that the x^2 term is genuinely 0 because $\tan$ is odd.)

38. (Model answer.) “You already know three series: e^x , $\cos x$, $\sin x$. The single move is to feed ix — an imaginary input — into the e^x series. Then the powers of i cycle through $1, i, -1, -i$, which sorts the terms into two piles: the real pile is exactly the cosine series, and the imaginary pile is exactly the sine series. So $e^{ix} = \cos x + i \sin x$. It’s surprising because exponential growth and circular oscillation look like opposites, yet one substitution shows they’re the same series wearing different clothes — and at $x = \pi$ it collapses to $e^{i\pi} = -1$, tying $e, \pi, i, 1$ together.”

39. (Model answer.) A Taylor **polynomial** T_n is a finite sum (degrees 0 through n); it is only an approximation, and it always comes with an exact remainder $R_n = f - T_n$ measuring the gap. A Taylor **series** is the infinite limit, $\lim_{n \rightarrow \infty} T_n$. The series equals the function only at x where (i) it converges and (ii) the remainder $R_n(x) \rightarrow 0$. So T_n is honest arithmetic you can always write down with an error bar; the series is a *claim of equality* that holds only on its interval of convergence — and even convergence isn’t enough without $R_n \rightarrow 0$ (problem 33).

Looking ahead

40. At a critical point the approximation is $f(x) \approx f(a) + \frac{1}{2}f''(a)(x-a)^2$, a parabola through $(a, f(a))$. - $f''(a) > 0$: opens **upward**, so $f(x) > f(a)$ nearby — a **minimum**. - $f''(a) < 0$: opens **downward**, so $f(x) < f(a)$ nearby — a **maximum**.

This is the second-derivative test, read straight off the leading Taylor term.

41. $R_1 = \frac{1}{2}f''(c)(x-a)^2$. Since $(x-a)^2 \geq 0$, the sign of R_1 matches the sign

of $f''(c)$. So $R_1 \geq 0$ for all x exactly when $f'' \geq 0$ throughout. Then $f(x) = T_1(x) + R_1 \geq T_1(x)$: the curve never dips below any tangent line — the function is **convex** (Chapter 8).

42. Newton replaces f by its tangent line T_1 at the current guess x_n and solves $T_1 = 0$. The discrepancy between f and T_1 is the T_1 remainder, $R_1 = \frac{1}{2}f''(c)(x-a)^2$ — quadratic in the distance from x_n to the true root. So the new error inherits that (old error)² factor: roughly $|e_{n+1}| \approx \frac{|f''|}{2|f'|} e_n^2$. Squaring a small error makes it much smaller — which is why the number of correct digits roughly **doubles** each Newton step (quadratic convergence).

43. From problem 17, $e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \dots$ — and $e^{-x^2/2} = 1 - \frac{x^2}{2} + \frac{x^4}{8} - \dots$ similarly has **only even powers**, hence is **even** (symmetric about the y -axis). Its single maximum is at $x = 0$, where the exponent $-x^2/2$ is largest (equals 0) and the value is 1; it decays on both sides. So the bell curve of statistics is, structurally, just the exponential series with $-x^2/2$ substituted in — symmetry and a single central peak both fall out of the series immediately.

Chapter 8

Maxima, Minima, and the Shape of a Function

Fluency

1. Critical points are where $f' = 0$ (none of these have f' undefined anywhere).

(a) $f(x) = x^2 - 6x + 1$: $f'(x) = 2x - 6 = 0 \Rightarrow \boxed{x = 3}$.

(b) $f(x) = x^3 - 3x$: $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \Rightarrow \boxed{x = -1, x = 1}$.

(c) $f(x) = x^4 - 2x^2$: $f'(x) = 4x^3 - 4x = 4x(x - 1)(x + 1) = 0 \Rightarrow \boxed{x = -1, x = 0, x = 1}$.

2. Second-derivative test: $f'' > 0 \Rightarrow \text{min}$, $f'' < 0 \Rightarrow \text{max}$.

(a) $f''(x) = 2 > 0$ everywhere. At $x = 3$: $f''(3) = 2 > 0 \Rightarrow \boxed{\text{local min at } (3, -8)}$.

(b) $f''(x) = 6x$. At $x = -1$: $f''(-1) = -6 < 0 \Rightarrow \boxed{\text{local max at } (-1, 2)}$. At $x = 1$: $f''(1) = 6 > 0 \Rightarrow \boxed{\text{local min at } (1, -2)}$.

(c) $f''(x) = 12x^2 - 4$. At $x = -1$: $f''(-1) = 8 > 0 \Rightarrow \boxed{\text{local min at } (-1, -1)}$. At $x = 0$: $f''(0) = -4 < 0 \Rightarrow \boxed{\text{local max at } (0, 0)}$. At $x = 1$: $f''(1) = 8 > 0 \Rightarrow \boxed{\text{local min at } (1, -1)}$.

3. Concavity from the sign of f'' ; inflection where f'' changes sign.

(a) $f(x) = x^3$: $f''(x) = 6x$. $f'' < 0$ for $x < 0$ (concave **down**), $f'' > 0$ for $x > 0$ (concave **up**), sign changes at 0. $\boxed{\text{concave down on } (-\infty, 0), \text{ up on } (0, \infty); \text{ inflection at } (0, 0)}$

(b) $f(x) = x^3 - 3x^2$: $f''(x) = 6x - 6 = 6(x - 1)$. $f'' < 0$ for $x < 1$ (concave **down**), $f'' > 0$ for $x > 1$ (concave **up**). At $x = 1$, $f(1) = 1 - 3 = -2$. $\boxed{\text{concave down on } (-\infty, 1), \text{ up on } (1, \infty); \text{ inflection at } (1, -2)}$.

4. $f(x) = \sin x + \cos x$ on $[0, 2\pi]$. $f'(x) = \cos x - \sin x = 0 \Rightarrow \tan x = 1 \Rightarrow x =$

$\frac{\pi}{4}, \frac{5\pi}{4}$. Second derivative $f''(x) = -\sin x - \cos x = -(\sin x + \cos x) = -f(x)$. At $x = \frac{\pi}{4}$: $f = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$, and $f'' = -\sqrt{2} < 0 \Rightarrow \text{max}$. At $x = \frac{5\pi}{4}$: $f = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$, and $f'' = \sqrt{2} > 0 \Rightarrow \text{min}$. $\text{local max } (\frac{\pi}{4}, \sqrt{2}); \quad \text{local min } (\frac{5\pi}{4}, -\sqrt{2})$.

5. Convex $\Leftrightarrow f'' \geq 0$ everywhere; concave $\Leftrightarrow f'' \leq 0$ everywhere.

(a) x^2 : $f'' = 2 > 0 \Rightarrow$ **convex**.

(b) e^x : $f'' = e^x > 0 \Rightarrow$ **convex**.

(c) $\ln x$ ($x > 0$): $f'' = -1/x^2 < 0 \Rightarrow$ **concave**.

(d) $-x^2$: $f'' = -2 < 0 \Rightarrow$ **concave**.

(e) x^3 : $f'' = 6x$, which is < 0 for $x < 0$ and > 0 for $x > 0$, so it changes sign $\Rightarrow$ **neither** on all of $\mathbb{R}$.

6. $f(x) = x^2 - 4x + 3$ on $[0, 3]$. $f'(x) = 2x - 4 = 0 \Rightarrow x = 2 \in (0, 3)$. Evaluate at candidates $\{0, 2, 3\}$: $f(0) = 3$, $f(2) = 4 - 8 + 3 = -1$, $f(3) = 9 - 12 + 3 = 0$. $\text{global max} = 3 \text{ at } x = 0; \quad \text{global min} = -1 \text{ at } x = 2$.

Practice

7. $f(x) = x^3 - 3x + 1$ on $[-2, 2]$. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1) = 0 \Rightarrow x = \pm 1$, both in $(-2, 2)$. Evaluate at $\{-2, -1, 1, 2\}$: $f(-2) = -8 + 6 + 1 = -1$, $f(-1) = -1 + 3 + 1 = 3$, $f(1) = 1 - 3 + 1 = -1$, $f(2) = 8 - 6 + 1 = 3$. $\text{global max} = 3 \text{ (at } x = -1 \text{ and } x = 2); \quad \text{global min} = -1 \text{ (at } x = -2 \text{ and } x = 1)$. (Both extreme values are attained twice — once at an interior critical point and once at an endpoint.)

8. Fencing 200 m against a river; no fence on the river side. Let x be the length of each of the two sides perpendicular to the river and y the side parallel to it. Fencing used: $2x + y = 200$, so $y = 200 - 2x$. Area:

$$A(x) = xy = x(200 - 2x) = 200x - 2x^2.$$

$A'(x) = 200 - 4x = 0 \Rightarrow x = 50$. Confirm a max: $A''(x) = -4 < 0$ (concave down), so this critical point is the maximum. Then $y = 200 - 2(50) = 100$. $50 \text{ m} \times 100 \text{ m (the 100 m side parallel to the river), area} = 5000 \text{ m}^2$.

9. Closest point on $y = 2x + 3$ to the origin. Minimize squared distance $D^2(x) = x^2 + (2x + 3)^2 = 5x^2 + 12x + 9$. $\frac{d}{dx}D^2 = 10x + 12 = 0 \Rightarrow$

$x = -\frac{6}{5}$. Confirm a min: $\frac{d^2}{dx^2}D^2 = 10 > 0$. Then $y = 2(-\frac{6}{5}) + 3 = \frac{3}{5}$.

closest point $(-\frac{6}{5}, \frac{3}{5})$, distance $= \sqrt{\frac{36}{25} + \frac{9}{25}} = \frac{3\sqrt{5}}{5}$. (Check: the line from origin to this point has slope $\frac{3/5}{-6/5} = -\frac{1}{2}$, perpendicular to the line's slope 2 — as it must be.)

10. Box, square base x , **open top**, volume 32 cm^3 . Volume $x^2h = 32 \Rightarrow h = \frac{32}{x^2}$. Material = base + four sides (no top):

$$S(x) = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}.$$

$S'(x) = 2x - \frac{128}{x^2} = 0 \Rightarrow 2x^3 = 128 \Rightarrow x^3 = 64 \Rightarrow x = 4$. Confirm a min: $S''(x) = 2 + \frac{256}{x^3} > 0$ for all $x > 0$ (so S is convex). Then $h = \frac{32}{16} = 2$.

base $4 \times 4 \text{ cm}$, height 2 cm ; material $= 16 + 32 = 48 \text{ cm}^2$.

11. $f(x) = xe^{-x}$ for $x \geq 0$. By the product rule $f'(x) = e^{-x} - xe^{-x} = (1-x)e^{-x} = 0 \Rightarrow x = 1$ (since $e^{-x} \neq 0$). *Two confirmations it is a maximum:* **First-derivative test:** $e^{-x} > 0$ always, so the sign of f' matches $(1-x)$: positive for $x < 1$, negative for $x > 1$ — slope goes $+$ $\rightarrow$ $-$, a local max. **Second-derivative test:** $f''(x) = (x-2)e^{-x}$, so $f''(1) = (-1)e^{-1} = -e^{-1} < 0$, confirming a max. The value is $f(1) = e^{-1}$. Since $f(0) = 0$, $f > 0$ on $(0, \infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$, this lone interior peak is the **global** maximum on $[0, \infty)$. maximum at $x = 1$, value $\frac{1}{e}$.

12. $f(x) = \sin x(1 + \cos x)$ on $[0, \pi]$. Expand $f = \sin x + \sin x \cos x$. Then

$$f'(x) = \cos x + (\cos^2 x - \sin^2 x) = \cos x + \cos 2x.$$

Using $\cos 2x = 2\cos^2 x - 1$: $f'(x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1) = 0$. So $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$, or $\cos x = -1 \Rightarrow x = \pi$. Evaluate the candidates plus the endpoint $x = 0$: $f(0) = 0$, $f(\pi) = 0$, and $f(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}(1 + \frac{1}{2}) = \frac{\sqrt{3}}{2} \cdot \frac{3}{2} = \frac{3\sqrt{3}}{4} \approx 1.30$.

largest at $x = \frac{\pi}{3}$, value $\frac{3\sqrt{3}}{4}$.

Concept & counterexample

13. (a) False. A critical point can be neither: $f(x) = x^3$ has $f'(0) = 0$ but 0 is neither a max nor a min (the slope is positive on both sides). Fermat's condition is necessary, not sufficient. **(b) False.** $f''(a) = 0$ is necessary for an inflection but not sufficient — f'' must actually *change sign*. Example: $f(x) = x^4$ has $f''(0) = 0$, yet $f''(x) = 12x^2 \geq 0$ never changes sign, so 0 is **not** an inflection point (it's a minimum, concave up on both sides). **(c) False.** A convex function cannot have two distinct local minima. If it had local minima at $a \neq b$, the chord joining $(a, f(a))$ and $(b, f(b))$ would lie on or above the graph (Face 1), forcing the graph between them to dip below the chord; combined with both being local valley floors this is impossible for a single bowl. (Equivalently: by the keystone theorem any interior point with $f' = 0$ is *the* global min, so there can be only one minimizing value.) **(d) False.** $f'(a) = 0$ and $f''(a) = 0$ is exactly the inconclusive case — it could be an extremum. Example: $f(x) = x^4$ has $f'(0) = f''(0) = 0$ yet $x = 0$ is a (global) minimum. So the point may well be an extremum; you just can't tell from these two facts alone.

14. The error: the second-derivative test is *not* a verdict of “neither” when $f''(a) = 0$ — it is **inconclusive**. With $f'(0) = 0$ and $f''(0) = 0$ the test simply fails to decide; the student wrongly read “inconclusive” as “neither.” **What's true:** fall back to the first-derivative test (or higher Taylor terms). For $f(x) = x^4$, $f'(x) = 4x^3$ is negative for $x < 0$ and positive for $x > 0$ — slope goes $- \rightarrow +$, so $x = 0$ is in fact a **local (indeed global) minimum**, with $f(0) = 0$.

15. Take any function with a shallow dip and a deeper one. A clean choice: $f(x) = x^4 - 2x^2$ (or equivalently a two-well shape). Better still, an asymmetric one whose floors differ in height — e.g.

$$f(x) = x^4 - 3x^2 + x.$$

Its derivative $f'(x) = 4x^3 - 6x + 1$ has two negative-to-positive crossings (two local minima), and evaluating shows the two valley floors sit at **different** depths — the higher one is a *local* minimum that is **not** the global minimum. That higher valley floor is exactly the “false floor” convexity forbids.

$f(x) = x^4 - 3x^2 + x$ has a local min that is not global.

16. A critical point that is neither max nor min: $f(x) = x^3$ at $a = 0$. $f'(0) = 0$,

but $f'(x) = 3x^2 > 0$ on both sides, so it's a saddle-flat point — neither. x^3 at 0. A point with $f''(a) = 0$ that nonetheless is a minimum: $f(x) = x^4$ at $a = 0$. Here $f'(0) = 0$ and $f''(0) = 0$ (so the second-derivative test is inconclusive), yet $f'(x) = 4x^3$ goes $- \rightarrow +$, making 0 a genuine minimum. x^4 at 0. Together: the first shows “ $f' = 0$, no sign change” really can fail to be an extremum, and the second shows “ $f'' = 0$ ” really can hide a true extremum — each inconclusive case earns its name.

17. Not possible (for a non-constant convex f) — there is no interior local maximum. Argue from the faces: - *Face 3 (slopes increasing)*: if f' is increasing, it can cross zero at most once and only ever from $-$ to $+$, which is a **minimum** profile, never $+$ $\rightarrow -$ (the max profile). - *Face 2 (tangent below graph)*: at an interior point a the graph lies **on or above** its tangent line. A local maximum requires the graph to lie *below* a (horizontal-or-not) supporting line near a — the opposite — so no interior point can be a strict local max. (The only loophole is a constant or affine stretch, where the “max” is also a min — a flat tie, not a genuine peak.)

No interior local maximum for a convex function.

18. For a **convex** loss, every point with zero slope is the single global minimum (the keystone theorem: tangent-below with a horizontal tangent forces $f(x) \geq f(a)$ everywhere). So gradient descent, which only walks downhill until the slope vanishes, must arrive at that one true bottom — there is no other valley to fall into. For a **non-convex** loss like the two-valley curve, there are *several* points with zero slope at *different* depths; descent halts at whichever valley is downhill from the start, which may be the shallower one. That is precisely why, in the two-valley picture, where the ball started decided which floor it reached — convexity removes the second floor and with it the dependence on the start.

◆ Diamond

19. Fermat's theorem. *Claim:* if f has a local extremum at an interior point c , then $f'(c) = 0$ or $f'(c)$ does not exist. *Proof.* Suppose $f'(c)$ **exists**; we show it is 0. This is precisely **Fermat's lemma** (Chapter 6): say c is a local maximum, so $f(c+h) \leq f(c)$ for all small h . For $h > 0$ the difference quotient $\frac{f(c+h) - f(c)}{h} \leq 0$, so the right-hand limit gives $f'(c) \leq 0$. For $h < 0$ the quotient $\frac{f(c+h) - f(c)}{h} \geq 0$ (negative over negative), so the left-hand limit gives $f'(c) \geq 0$. Since $f'(c)$ exists,

both one-sided limits equal it, forcing $0 \leq f'(c) \leq 0$, i.e. $f'(c) = 0$. (For a local minimum the inequalities flip but the squeeze is identical.) ■ *Why the “or”:* the lemma only delivers $f'(c) = 0$ **when the derivative exists**. The hypothesis “interior local extremum” does not guarantee differentiability – e.g. $f(x) = |x|$ has a minimum at 0 where $f'(0)$ **does not exist**. So the honest conclusion must allow the second alternative; otherwise it would miss every corner extremum.

20. First-derivative test (local max). *Claim:* if f is continuous at c , $f'(x) > 0$ on $(c - \delta, c)$ and $f'(x) < 0$ on $(c, c + \delta)$, then c is a local maximum. *Proof.* On the left interval $(c - \delta, c)$, $f' > 0$, so by Chapter 6’s monotonicity consequence (positive derivative $\Rightarrow$ strictly increasing) f is increasing there; hence for any x with $c - \delta < x < c$, $f(x) < f(c)$ (taking $x \uparrow c$ and using continuity, $f(x) \leq f(c)$, with strictness from increase). On the right interval $(c, c + \delta)$, $f' < 0$, so f is decreasing; hence for any x with $c < x < c + \delta$, $f(x) < f(c)$. Thus $f(x) \leq f(c)$ for all x in $(c - \delta, c + \delta)$, which is exactly a local maximum at c . ■

21. Second-derivative test. *Claim:* if $f'(a) = 0$ and $f''(a) > 0$, then a is a local minimum. *Proof (via the Taylor quadratic, Chapter 7).* Taylor’s theorem with remainder gives, for x near a ,

$$f(x) = f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2 + R(x),$$

where $R(x) = o((x - a)^2)$, i.e. $\frac{R(x)}{(x - a)^2} \rightarrow 0$ as $x \rightarrow a$. Since $f'(a) = 0$,

$$f(x) - f(a) = \frac{1}{2}f''(a)(x - a)^2 + R(x) = (x - a)^2 \left[\frac{1}{2}f''(a) + \frac{R(x)}{(x - a)^2} \right].$$

As $x \rightarrow a$ the bracket tends to $\frac{1}{2}f''(a) > 0$, so there is a $\delta > 0$ on which the bracket stays positive. For $0 < |x - a| < \delta$ we then have $(x - a)^2 > 0$ times a positive bracket, giving $f(x) - f(a) > 0$, i.e. $f(x) > f(a)$. Hence a is a (strict) local minimum. ■ (The quadratic term $\frac{1}{2}f''(a)(x - a)^2$ **dominates** the remainder because the remainder is of smaller order than $(x - a)^2$.)

22. Face 4 $\Rightarrow$ Face 2 (tangent below graph). *Claim:* if $f'' \geq 0$ on an interval, then for every a the graph lies on or above the tangent at a : $f(x) \geq f(a) + f'(a)(x - a)$. *Proof.* By Taylor with Lagrange remainder, for each x there is a c strictly

between a and x with

$$f(x) = f(a) + f'(a)(x - a) + \frac{1}{2}f''(c)(x - a)^2.$$

The gap from the tangent line is therefore

$$f(x) - [f(a) + f'(a)(x - a)] = \frac{1}{2}f''(c)(x - a)^2 \geq 0,$$

since $f''(c) \geq 0$ and $(x - a)^2 \geq 0$. So the graph never dips below the tangent. ■
Specialize to e^x : take $f(x) = e^x$, which has $f''(x) = e^x > 0$ everywhere, and use the tangent at $a = 0$, where $f(0) = 1$ and $f'(0) = 1$. The tangent line is $1 + x$, so the result reads

$$e^x \geq 1 + x \quad \text{for all } x,$$

re-proving Chapter 6's inequality as the statement " e^x is convex."

23. Keystone theorem. *Claim:* if f is convex and $f'(a) = 0$, then a is the global minimum. *Proof.* Convexity gives Face 2: the graph lies on or above the tangent line at a ,

$$f(x) \geq f(a) + f'(a)(x - a) \quad \text{for all } x.$$

But $f'(a) = 0$, so the tangent line is the **horizontal** line $y = f(a)$. The inequality collapses to

$$f(x) \geq f(a) \quad \text{for every } x,$$

which says $f(a)$ is the least value f attains anywhere — a is the **global** minimum. ■ (There are no false floors: a single zero-slope point already pins down the bottom of the entire function.)

24. Face 2 $\Rightarrow$ Face 1 (tangent below $\Rightarrow$ chord above). *Claim:* if the graph lies above each tangent (equivalently $f'' \geq 0$, so f' is increasing), then for any $p < q$ and any $x \in [p, q]$,

$$f(x) \leq L(x) := f(p) + \frac{f(q) - f(p)}{q - p}(x - p),$$

i.e. the chord from $(p, f(p))$ to $(q, f(q))$ lies on or above the graph. *Proof (clean route via f' increasing).* Write $x = p + t(q - p)$ with $t \in [0, 1]$; the claim $f(x) \leq L(x)$ is equivalent to

$$f(p + t(q - p)) \leq (1 - t)f(p) + t f(q).$$

Define $g(t) = (1-t)f(p) + tf(q) - f(p+t(q-p))$ on $[0, 1]$. Then $g(0) = g(1) = 0$, and

$$g'(t) = f(q) - f(p) - (q-p)f'(p+t(q-p)).$$

Because f' is **increasing** (Face 3, from $f'' \geq 0$), the term $f'(p+t(q-p))$ increases in t , so $g'(t)$ is **decreasing** in t . A function with decreasing derivative on $[0, 1]$ that vanishes at both endpoints is ≥ 0 in between: indeed by the Mean Value Theorem there is t^* with $g'(t^*) = \frac{g(1) - g(0)}{1} = 0$; since g' is decreasing, $g' > 0$ on $[0, t^*)$ and $g' < 0$ on $(t^*, 1]$, so g rises from $g(0) = 0$ then falls back to $g(1) = 0$, staying ≥ 0 throughout. Hence $g(t) \geq 0$, i.e. $f(x) \leq L(x)$ for all $x \in [p, q]$: the chord lies on or above the graph. ■

Teach it

25. Picture a smooth bowl: convexity means that wherever you stretch a straight string between two points of the curve, the string never cuts below the curve — the bowl bulges down in one clean piece, with no bumps or secondary dips inside it. Because there's only ever the one downward bulge, there's only one lowest point, and any spot where the ground goes flat *is* that lowest point. So a convex bowl has no “false floors” — no shallow ledge where a rolling ball could stall thinking it's reached the bottom when a deeper bottom lies elsewhere; in a convex bowl the first flat spot you find is the true bottom of the world.

26. What they've actually found is a **critical point** — a *candidate* for an extremum, the place Fermat's theorem says extrema can hide, but not yet a verdict (e.g. x^3 has $f'(0) = 0$ and it's neither). Before calling it a **global** minimum they still must (1) **classify** it — confirm it's even a *local* min, via $f'' > 0$ there or a $- \rightarrow +$ sign change of f' — and (2) **rule out lower values elsewhere** — either show f is convex (so this lone critical point is automatically global) or, on a closed interval, compare f here against the other critical points and the **endpoints**.

27. Worked solution — rectangle of maximum area with perimeter 40.
Setup. Let the rectangle have side lengths x and y . The perimeter constraint is $2x + 2y = 40$, so $x + y = 20$, giving $y = 20 - x$. (Geometrically we need $0 < x < 20$.) *Single-variable function.* Area is $A = xy$. Substitute the constraint to get one variable:

$$A(x) = x(20 - x) = 20x - x^2.$$

Optimize. $A'(x) = 20 - 2x = 0 \Rightarrow x = 10$. *Confirm.* $A''(x) = -2 < 0$ for all x , so A is concave down and $x = 10$ is a maximum (not a min). Then $y = 20 - 10 = 10$. *Answer.* The maximum-area rectangle is a **square**, 10×10 , with area $A = 100$. *General recipe (any perimeter P):* $x + y = P/2$, $A(x) = x(P/2 - x)$, $A' = P/2 - 2x = 0 \Rightarrow x = P/4$, so $y = P/4$ too — a square of side $P/4$, area $P^2/16$. The maximum-area rectangle of fixed perimeter is **always a square**.

28. Design a problem (with full key). *Problem.* An open-top rectangular tank with a **square base** must hold 4000 cm^3 of water. Material costs the same per cm^2 everywhere. What base length and height use the least material, and how much material is that? *Key.* Let the base side be x and height h . Volume: $x^2h = 4000 \Rightarrow h = \frac{4000}{x^2}$. Material (base + four sides, open top):

$$S(x) = x^2 + 4xh = x^2 + 4x \cdot \frac{4000}{x^2} = x^2 + \frac{16000}{x}.$$

Optimize: $S'(x) = 2x - \frac{16000}{x^2} = 0 \Rightarrow 2x^3 = 16000 \Rightarrow x^3 = 8000 \Rightarrow x = 20$. f''

check: $S''(x) = 2 + \frac{32000}{x^3} > 0$ for all $x > 0$, so S is convex and $x = 20$ is the **global** minimum. Then $h = \frac{4000}{400} = 10$. *Answer.* Base $20 \times 20 \text{ cm}$, height 10 cm ; material $= 400 + \frac{16000}{20} = 400 + 800 = 1200 \text{ cm}^2$.

29. A **local** extremum is best only *in its own neighborhood* — a hilltop or valley floor that beats nearby points but may be dwarfed by a taller hill or undercut by a deeper valley somewhere else on the domain. A **global** extremum is best *over the entire domain* — the single highest or lowest value anywhere. The two can disagree (a function can have many local minima at different depths), and the gap between them is the whole danger in optimization. **Convexity removes the gap:** for a convex function the graph never rises above its chords / never falls below its tangents, so there is only one valley — every local minimum *is* the global minimum, and the cautious “local” and the ambitious “global” become the same point.

Looking ahead

30. Yes — from any start, gradient descent on $f(x) = x^2$ reaches the global minimum $x = 0$. Here $f''(x) = 2 > 0$ everywhere, so f is convex, and the only point with $f'(x) = 2x = 0$ is $x = 0$. By the keystone theorem that lone critical point is the global minimum, and convexity means there are **no other valleys** to

fall into: every downhill step (the update $x \leftarrow x - \eta f'(x) = x - 2\eta x = (1 - 2\eta)x$, which for a reasonable step shrinks $|x|$) drives x toward 0. There is nowhere to get stuck because there is nothing to get stuck in.

31. Which valley wins is decided by the starting point — specifically, by which side of the central ridge (the local *maximum* of $f(x) = x^4 - 3x^2 + x$, where $f' = 4x^3 - 6x + 1 = 0$ at the middle root) the start lands on. Gradient descent only ever steps downhill, so it slides into whichever of the two valleys it begins above and halts at that floor, with no mechanism to climb back over the ridge to check the other (possibly deeper) valley. This is exactly the Chapter 9 lab: the function is **non-convex** (two separate dips at different depths), the keystone theorem does not apply, and so “where you start” determines the outcome — sometimes the global minimum, sometimes the shallower false floor.

32. In many input directions the single second derivative f'' is replaced by the **Hessian** — the *matrix* of all second partial derivatives $[\partial^2 f / \partial x_i \partial x_j]$, recording curvature in every direction and how the directions couple. “Convex” should mean *curving upward in every direction at once*: the multivariable bowl, where the analogue of “ $f'' \geq 0$ ” is that this Hessian matrix is **positive semidefinite** (no direction in which the curvature is negative), equivalently that the graph still lies above all its tangent planes and every chord lies above the surface. As in one dimension, such a bowl has a single global minimum and gradient descent cannot get lost — which is why convexity, now a statement about a matrix of curvatures, remains the dividing line in the multivariable optimization where machine learning actually lives.

Chapter 9

When You Can't Solve It, Chase It

Computational Lab

1. Learning-rate sweep. Run gradient descent on $f(x) = x^2$ from $x_0 = 1$. The update is $x_{n+1} = (1 - 2\eta)x_n$, so $x_n = (1 - 2\eta)^n$ and the entire behavior is fixed by the factor $r = 1 - 2\eta$.

η	$r = 1 - 2\eta$	$ r $	first few x_n	behavior
0.1	0.8	0.8	1, 0.8, 0.64, 0.512, ...	smooth glide ✓
0.3	0.4	0.4	1, 0.4, 0.16, 0.064, ...	smooth glide ✓
0.5	0	0	1, 0, 0, ...	one-step (best case) ✓
0.7	-0.4	0.4	1, -0.4, 0.16, -0.064, ...	chattering spiral ✓
0.9	-0.8	0.8	1, -0.8, 0.64, -0.512, ...	chattering spiral ✓
1.0	-1	1	1, -1, 1, -1, ...	stuck bounce ✗
1.1	-1.2	1.2	1, -1.2, 1.44, -1.728, ...	explosion ✗
1.5	-2	2	1, -2, 4, -8, ...	explosion ✗

Sorting into four behaviors: - *Smooth (monotone) glide* - $0 < r < 1$, i.e. $0 <$

$\eta < 0.5$: positive factor, x_n decreases to 0 without changing sign ($\eta = 0.1, 0.3$). - *Shrinking spiral* — $-1 < r < 0$, i.e. $0.5 < \eta < 1$: negative factor, x_n alternates sign but shrinks ($\eta = 0.7, 0.9$). - *Stuck bounce* — $r = -1$, i.e. $\eta = 1$: x_n flips between ± 1 forever ($\eta = 1.0$). - *Explosion* — $|r| > 1$, i.e. $\eta > 1$: $|x_n|$ grows without bound ($\eta = 1.1, 1.5$).

Boundary check. Convergence requires $|r| = |1 - 2\eta| < 1 \iff -1 < 1 - 2\eta < 1 \iff 0 < \eta < 1$. Every $\eta < 1$ in the table converges; $\eta = 1$ is exactly the critical bounce; every $\eta > 1$ diverges. The prediction $|1 - 2\eta| < 1$ is confirmed exactly.

2. Hunt the fastest rate. With $x_n = (1 - 2\eta)^n$ and $x_0 = 1$, reaching $|x_n| < 10^{-6}$ needs $|1 - 2\eta|^n < 10^{-6}$, i.e.

$$n > \frac{-6 \ln 10}{\ln |1 - 2\eta|}.$$

The right side is *smallest* when $|1 - 2\eta|$ is smallest, and $|1 - 2\eta|$ hits its minimum value 0 at $\eta = \frac{1}{2}$.

η	$ 1 - 2\eta $	steps to 10^{-6}
0.1	0.8	62
0.3	0.4	16
0.5	0	1
0.7	0.4	16
0.9	0.8	62
1.0	1	never
1.1	1.2	never

The experimental winner is $\eta = 0.5$, hitting the minimum in a *single step* ($x_1 = 0$ exactly). This matches the predicted optimum $\eta = \frac{1}{2}$. Notice the symmetry about $\eta = \frac{1}{2}$: rates equidistant from $\frac{1}{2}$ (e.g. 0.3 and 0.7) take the same number of steps, because they give factors of equal magnitude.

3. Newton, fast and broken. For $\sqrt{2}$ iterate $x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$ from $x_0 = 1.5$ (true value $\sqrt{2} = 1.41421356237 \dots$):

n	x_n	error $ x_n - \sqrt{2} $	correct digits
0	1.5	8.6×10^{-2}	—
1	1.41666666...	2.5×10^{-3}	≈ 2
2	1.41421568...	2.1×10^{-6}	≈ 5
3	1.41421356237...	1.6×10^{-12}	≈ 11

The error exponent roughly doubles each step ($-2 \rightarrow -3 \rightarrow -6 \rightarrow -12$): correct digits double — quadratic convergence.

For $f(x) = x^{1/3}$ from $x_0 = 1$, the rule simplifies to $x_{n+1} = -2x_n$ (see Problem 18):

n	0	1	2	3	4	5
x_n	1	-2	4	-8	16	-32

It flips sign and doubles every step — flies apart, never approaching the root at 0. **Same method, opposite outcomes: speed and reliability are different promises.**

4. A landscape with two valleys. $f(x) = x^4 - 3x^2 + x$, so $f'(x) = 4x^3 - 6x + 1$. Setting $f' = 0$ gives three critical points:

$$x \approx -1.30084 \text{ (min, } f \approx -3.514), \quad x \approx 0.16994 \text{ (max, } f \approx 0.084), \quad x \approx 1.13090 \text{ (min, } f \approx -1.07)$$

(Check $f'' = 12x^2 - 6$: positive at the two outer points — minima; negative at the middle — a maximum.) The left valley is **deeper** (the global minimum); the right valley is a shallower **local** minimum.

Running $x_{n+1} = x_n - \eta f'(x_n)$ with a small η (say 0.01):

start x_0	rolls to	which minimum
-2.0	≈ -1.30084	deep (global)
-0.5	≈ -1.30084	deep (global)
0.5	≈ 1.13090	shallow (local)
2.0	≈ 1.13090	shallow (local)

The maximum at 0.16994 is the watershed: starts to its left fall into the deep valley, starts to its right into the shallow one. **Gradient descent finds a local minimum — which one depends entirely on where you begin, not on which is best.** That is the central headache of training.

5. The race. Take the quadratic $f(x) = x^2 - 4x + 7$ (minimizer $x = 2$, $f'' = 2$).

Gradient descent, good rate $\eta = 0.25$ (so $1 - \eta f'' = 1 - 0.5 = 0.5$), from $x_0 = 0$: $x_n - 2 = 0.5^n(0 - 2)$, so to reach $|x_n - 2| < 10^{-6}$ needs $0.5^n \cdot 2 < 10^{-6}$, i.e. $n \geq 21$ steps.

Newton on the same f : $x_{n+1} = x_n - \frac{2x_n - 4}{2} = 2$ — it lands on the exact minimizer in **one step** from any start (Problem 21). Newton wins by a landslide (1 vs ~ 21).

A setup where Newton fails and gradient descent doesn't: point Newton at the **root** problem $f(x) = x^{1/3}$ from $x_0 = 1$ — it diverges (1, -2 , 4 , -8 , ..., Problem 18). To minimize the bowl $g(x) = x^2$ instead, gradient descent with any $\eta \in (0, 1)$ converges happily. (Likewise, Newton from a near-flat slope, e.g. $f(x) = \arctan x$ from a large x_0 , overshoots and diverges while a small-step descent on a convex loss stays safe.) **Newton is the fastest and the most fragile; gradient descent is slower but does not blow up from a flat spot.**

6. (Stretch) A first real optimizer. Momentum: $x_{n+1} = x_n - \eta f'(x_n) + \beta(x_n - x_{n-1})$, $\beta = 0.9$, on the stretched bowl $f(x) = 10x^2$ (so $f'(x) = 20x$, $f'' = 20$). Use $x_{-1} = x_0 = 1$.

Plain gradient descent needs $\eta < \frac{2}{f''} = 0.1$. Take $\eta = 0.04$: factor $1 - \eta f'' = 1 - 0.8 = 0.2$, so $x_n = 0.2^n$ — converges, but the tight cap on η (because curvature $a = 20$ is large) forces small steps.

With momentum ($\eta = 0.04$, $\beta = 0.9$), the first iterates from $x_0 = 1$:

n	plain GD x_n	momentum x_n
0	1	1
1	0.2	0.2
2	0.04	-0.68
3	0.008	-0.928

n	plain GD x_n	momentum x_n
4	0.0016	-0.40880
5	0.00032	0.38552

Momentum overshoots hard — it carries velocity straight through the bottom — and with $\beta = 0.9$ this much accumulated velocity sends it into a slow, ringing oscillation. The momentum recurrence here is the second-order linear rule $x_{n+1} = (1 + \beta - \eta a)x_n - \beta x_{n-1}$ with $a = f'' = 20$, i.e. $x_{n+1} = 1.2x_n - 0.9x_{n-1}$; its characteristic roots are complex with modulus $\sqrt{\beta} = \sqrt{0.9} \approx 0.949$, so the envelope decays by only about 0.949 *per step* — far **slower** than plain descent's 0.2^n for this aggressive β . That is the cautionary half of the lesson: **too much β turns the helpful overshoot into a sluggish, oscillating crawl** (push it further and the roots leave the unit circle and it diverges outright). Momentum *can* beat plain descent — but only with β tuned to the curvature; the gain is not automatic. (The well-tuned regime, where momentum genuinely accelerates the long, gently-curved direction, is exactly what real optimizers chase.) **This is a baby version of the momentum/Adam optimizers that train real networks.**

By hand

7. $f(x) = x^2 - 5$, so $f'(x) = 2x$. Newton's rule:

$$x_{n+1} = x_n - \frac{x_n^2 - 5}{2x_n} = \frac{x_n}{2} + \frac{5}{2x_n} = \frac{1}{2} \left(x_n + \frac{5}{x_n} \right).$$

From $x_0 = 2$:

n	x_n
0	2
1	$\frac{1}{2}(2 + \frac{5}{2}) = 2.25$
2	$\frac{1}{2}(2.25 + \frac{5}{2.25}) = 2.236111 \dots$

$\sqrt{5} = 2.2360679 \dots$, so after two steps $x_2 = 2.23611$ is already correct to four decimals. **Answer:** $x_1 = 2.25$, $x_2 \approx 2.23611$.

8. $f(x) = x^2$, $f'(x) = 2x$, update $x_{n+1} = x_n - 0.1(2x_n) = 0.8x_n$, from $x_0 = 1$:

$$x_1 = 0.8, \quad x_2 = 0.64, \quad x_3 = 0.512.$$

(Geometric, ratio 0.8, gliding toward 0.) **Answer:** $x_1 = 0.8$, $x_2 = 0.64$, $x_3 = 0.512$.

9. $f(x) = x^2 - 4x + 7$, so $f'(x) = 2x - 4$. Exact minimizer: $f' = 0 \Rightarrow 2x - 4 = 0 \Rightarrow x = 2$ (and $f'' = 2 > 0$, a minimum). Gradient descent $x_{n+1} = x_n - 0.1(2x_n - 4)$ from $x_0 = 0$:

$$x_1 = 0 - 0.1(2 \cdot 0 - 4) = 0 - 0.1(-4) = 0.4,$$

$$x_2 = 0.4 - 0.1(2 \cdot 0.4 - 4) = 0.4 - 0.1(-3.2) = 0.72.$$

The values $0 \rightarrow 0.4 \rightarrow 0.72$ are increasing toward the minimizer 2 — correct direction (we started left of the bottom, so the step is to the right). **Answer:** minimizer $x = 2$; $x_1 = 0.4$, $x_2 = 0.72$, heading the right way.

10. Behavior is set by $r = 1 - 2\eta$ (converges iff $|r| < 1$; sign of r tells monotone vs. oscillating):

η	$r = 1 - 2\eta$	classification
0.2	0.6	monotone glide ($0 < r < 1$)
0.5	0	one-step (lands on minimum immediately)
0.8	-0.6	oscillating glide ($-1 < r < 0$)
1.2	-1.4	explosion ($ r > 1$)

Concept & counterexample

11. (a) **False.** A smaller learning rate is *safer* (you stay inside the convergence range $0 < \eta < \frac{2}{f''}$), but not *better*: too small and you crawl, taking many more steps. “Safe” $\neq$ “fast.” There is an optimal $\eta = 1/f''$, and going below it costs speed.

(b) True. Whenever $\eta > \frac{1}{f''}$ (factor $1 - \eta f'' < 0$), each step lands on the *other* side of the minimum — it overshoots. On x^2 with $\eta = 0.7$, x jumps from $+1$ to -0.4 , crossing the bottom. As long as $|1 - \eta f''| < 1$ the overshoots shrink and it still converges (the spiral).

(c) False. Newton is faster *near a well-behaved root*, but it can be slower or fail entirely elsewhere — at a near-flat slope it overshoots wildly, and on $f(x) = x^{1/3}$ it diverges while gentle gradient descent on a convex loss converges. “Faster” is a conditional promise, not a universal one.

(d) False. Steps shrinking does *not* guarantee approach to the minimum. Counterexample: the harmonic-style step sizes, or simply a step rule whose increments sum to a finite total — you can shrink steps yet halt short of the bottom. (Concretely, if a method’s moves were $0.5, 0.25, 0.125, \dots$ summing to 1, it stops at most 1 away from where it began.) Shrinking steps are necessary near a true minimum but not sufficient for reaching it; you also need the steps pointed correctly and not summing short.

12. Predict, then check. For $f(x) = x^2$, $\eta = 0.5$: the factor is $1 - 2(0.5) = 0$, so $x_1 = 0 \cdot x_0 = 0$ — gradient descent reaches the exact minimum in **one step**, landing at $x_1 = 0$ from any start. (Check: $x_1 = 1 - 0.5 \cdot 2(1) = 0$. ✓)

For $\eta = 1$: factor $1 - 2(1) = -1$, so $x_n = (-1)^n x_0$ — **it bounces between $+x_0$ and $-x_0$ forever, never settling at the minimum.**

13. When it breaks. (a) Newton diverges: $f(x) = x^{1/3}$ from any $x_0 \neq 0$. The rule becomes $x_{n+1} = -2x_n$, so the guess flips sign and doubles each step. *Cause:* the slope $f'(x) = \frac{1}{3}x^{-2/3}$ blows up at the root, making the tangent line throw the next guess outward (a *structural* failure of the function/start). A flatter version: $f(x) = \arctan x$ from a large x_0 , where the near-flat tangent flings x_{n+1} far past the root.

(b) Gradient descent diverges: $f(x) = x^2$ with $\eta = 1.5$ from $x_0 = 1$ gives $1, -2, 4, -8, \dots$ — it explodes. *Cause:* the learning rate exceeds the safe range $\eta < \frac{2}{f''} = 1$, so each step overshoots farther than the last.

Which is fixable by one number? The gradient-descent failure is — just lower η below $\frac{2}{f''}$ and it converges. The Newton failure on $x^{1/3}$ is *not* fixable by tuning

one number: there is no step-size dial in Newton's formula, and the trouble lives in the function's misbehaving slope at the root, not in a parameter you chose.

14. The two-valley trap (on paper). Sketch a W-shaped (double-well) curve whose **left valley is shallow** and **right valley is deep**, separated by a hump (a local maximum). Mark the hump's location x_{peak} . - A start *just left of the hump* slides **left into the shallow (worse) minimum**. - A start *just right of the hump* slides **right into the deep (better) minimum**.

In one sentence: gradient descent finds the nearest *downhill* basin, not the best one overall — so “gradient descent finds the best answer” is false; it finds the answer your starting point can reach.

15. An asymmetry worth seeing. On $f(x) = x^2$ the factor is $r = 1 - 2\eta$. - *Too small* an η (near 0): r is just below 1, so $|r| < 1$ and $x_n = r^n x_0 \rightarrow 0$ — you **still converge**, only slowly. Smallness costs *time*, never correctness. - *Too large* an η (above 1): $r < -1$, so $|r| > 1$ and $x_n = r^n x_0$ **grows without bound** — you converge to nothing; the method fails outright.

The two are not symmetric because the controlling quantity is $|r|$ versus the hard wall at 1. Shrinking η pushes $|r|$ toward 1 *from below* (always safe); growing η pushes $|r|$ past 1 (catastrophe). One direction merely approaches the boundary of success; the other crosses it. Slow is recoverable — wait longer; divergence is not — there is no answer to wait for.

◆ Diamond

16. ◆ Start from the tangent-line approximation at x_n :

$$f(x) \approx f(x_n) + f'(x_n)(x - x_n).$$

We want the next guess x_{n+1} to be where this *line* equals zero. Set the right side to 0 with $x = x_{n+1}$:

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n).$$

Provided $f'(x_n) \neq 0$, solve:

$$f'(x_n)(x_{n+1} - x_n) = -f(x_n) \implies x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \implies \boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}.$$

This is Newton's method, *derived* (not memorized) from “near a point, a curve looks like its tangent.” ■

17. ♦ For $f(x) = x^2 - A$, $f'(x) = 2x$:

$$x_{n+1} = x_n - \frac{x_n^2 - A}{2x_n} = x_n - \frac{x_n}{2} + \frac{A}{2x_n} = \frac{x_n}{2} + \frac{A}{2x_n} = \frac{1}{2} \left(x_n + \frac{A}{x_n} \right),$$

the Babylonian rule. ■ For $\sqrt{7}$ take $A = 7$, $x_0 = 3$:

n	x_n
0	3
1	$\frac{1}{2} \left(3 + \frac{7}{3} \right) = 2.666667$
2	$\frac{1}{2} \left(2.666667 + \frac{7}{2.666667} \right) = 2.645833$
3	$\frac{1}{2} \left(2.645833 + \frac{7}{2.645833} \right) = 2.645751$

$\sqrt{7} = 2.6457513\dots$, so to five decimals $\boxed{\sqrt{7} \approx 2.64575}$ (stable by x_3).

18. ♦ The cube-root disaster. $f(x) = x^{1/3}$, $f'(x) = \frac{1}{3}x^{-2/3}$. Then

$$x_{n+1} = x_n - \frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = x_n - 3x_n^{1/3} \cdot x_n^{2/3} = x_n - 3x_n^{1/3+2/3} = x_n - 3x_n = -2x_n.$$

So $x_{n+1} = -2x_n$, giving $x_n = (-2)^n x_0$. For any $x_0 \neq 0$, $|x_n| = 2^n |x_0| \rightarrow \infty$ — divergence from **every** nonzero start (the only root, $x = 0$, is never reached). *Geometrically*: the Newton step is $-\frac{f(x_n)}{f'(x_n)} = -\frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = -3x_n$ — always proportional to x_n itself, never to the *distance to the root* in a way that shrinks it. The trouble is the slope: as $x_n \rightarrow 0$ the slope $f'(x) = \frac{1}{3}x^{-2/3} \rightarrow +\infty$, so the tangent steepens toward **vertical** at the root. But the function value $x_n^{1/3}$ shrinks *slower* than the

slope grows, so their ratio $f/f' = 3x_n$ stays a full three steps wide. Newton subtracts that whole $3x_n$, overshooting the root and landing at $-2x_n$ — the opposite side, twice as far out — every step. The steep tangent overcorrects rather than homing in. ■

19. ♦ The learning-rate range, in general. $f(x) = \frac{a}{2}(x - m)^2$ with $a > 0$, so $f'(x) = a(x - m)$. Gradient descent:

$$x_{n+1} = x_n - \eta a(x_n - m).$$

Subtract m from both sides:

$$x_{n+1} - m = (x_n - m) - \eta a(x_n - m) = (1 - \eta a)(x_n - m).$$

Iterating,

$$\boxed{x_n - m = (1 - \eta a)^n (x_0 - m)}.$$

This $\rightarrow 0$ (i.e. $x_n \rightarrow m$) iff $|1 - \eta a| < 1$:

$$-1 < 1 - \eta a < 1 \iff 0 < \eta a < 2 \iff \boxed{0 < \eta < \frac{2}{a}}.$$

The **optimal** η kills the factor: $1 - \eta a = 0 \Rightarrow \boxed{\eta = \frac{1}{a}}$, which sends $x_1 = m$ exactly in one step. (Since $a = f''$, this is $\eta = 1/f''$.)

The practical lesson: the safe range is $0 < \eta < \frac{2}{a}$, whose width is inversely proportional to the curvature a . *As the curvature a grows, the safe learning rate must shrink* (like $1/a$): a sharply curved valley demands a tiny step or gradient descent overshoots and explodes. This is why steep loss landscapes force small learning rates.

20. ♦♦ Why Newton's error squares. Let $e_n = x_n - \sqrt{A}$ for $x_{n+1} = \frac{1}{2}\left(x_n + \frac{A}{x_n}\right)$. Then

$$e_{n+1} = x_{n+1} - \sqrt{A} = \frac{1}{2}\left(x_n + \frac{A}{x_n}\right) - \sqrt{A} = \frac{x_n^2 + A - 2\sqrt{A}x_n}{2x_n} = \frac{(x_n - \sqrt{A})^2}{2x_n}.$$

The numerator is a perfect square because $x_n^2 - 2\sqrt{A}x_n + A = (x_n - \sqrt{A})^2$. Hence

the exact identity

$$e_{n+1} = \frac{e_n^2}{2x_n}.$$

Near the root $x_n \rightarrow \sqrt{A}$, so the factor $\frac{1}{2x_n} \rightarrow \frac{1}{2\sqrt{A}}$ is essentially constant, giving $e_{n+1} \approx C e_n^2$ with $C = \frac{1}{2\sqrt{A}}$. The error **squares** each step: if $e_n \sim 10^{-k}$ then $e_{n+1} \sim 10^{-2k}$ — the number of correct digits roughly *doubles* every iteration. That is precisely quadratic convergence, matching the $2 \rightarrow 5 \rightarrow 11$ digits in §“Watching it work.” ■

(Numerical check, $A = 2$, $x_0 = 1.5$: $e_0 = 8.58 \times 10^{-2}$, predicted $e_1 = e_0^2/(2x_0) = 2.45 \times 10^{-3}$ — matches the actual $x_1 - \sqrt{2}$ exactly; then $e_2 = 2.12 \times 10^{-6}$, $e_3 = 1.59 \times 10^{-12}$.)

21. ♦♦ Newton reaches a quadratic’s minimum in one step. For $f(x) = ax^2 + bx + c$ with $a > 0$: $f'(x) = 2ax + b$, $f''(x) = 2a$. The Newton-for-minima step:

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = x_0 - \frac{2ax_0 + b}{2a} = x_0 - x_0 - \frac{b}{2a} = -\frac{b}{2a}.$$

The x_0 terms cancel, so $x_1 = -\frac{b}{2a}$ — the exact minimizer (where $f' = 0$) — **regardless of the starting point x_0** . ■

Why it had to happen (Chapter 6/7): Newton’s method works by replacing f with its second-order Taylor model and jumping to that model’s minimum. A quadratic is its own second-order Taylor polynomial — the model is not an approximation, it is exact everywhere. So the model’s minimum is the true minimum, and one step lands on it precisely. There is no leftover error to iterate away.

22. ♦♦ The fixed-point view that unifies everything. Both methods are $x_{n+1} = g(x_n)$, and a limit L satisfies $g(L) = L$. Convergence holds when $|g'(L)| < 1$, and smaller $|g'(L)|$ means faster.

(a) Gradient descent: $g(x) = x - \eta f'(x)$, so $g'(x) = 1 - \eta f''(x)$. At a minimum L the curvature is $f''(L) = a > 0$, hence

$$g'(L) = 1 - \eta a.$$

The criterion $|g'(L)| < 1$ becomes $|1 - \eta a| < 1 \iff 0 < \eta < \frac{2}{a}$ — *exactly* the range

from Problem 19. The fastest case is $g'(L) = 0$, i.e. $\eta = 1/a$, the optimum (one-step convergence).

(b) Newton: $g(x) = x - \frac{f(x)}{f'(x)}$. Differentiate with the quotient rule:

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{(f'(x))^2} = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = \frac{f(x)f''(x)}{(f'(x))^2}.$$

At a **simple root** L we have $f(L) = 0$ (and $f'(L) \neq 0$), so the numerator vanishes:

$$g'(L) = \frac{0 \cdot f''(L)}{(f'(L))^2} = 0.$$

Explanation of both phenomena from one criterion. For gradient descent $g'(L) = 1 - \eta a$ is some nonzero number you can only *push toward* 0 by tuning η — and if you push η too far it exceeds $|g'| < 1$ and diverges; this is the learning-rate range. For Newton $g'(L) = 0$ *automatically*, with no tuning — a vanishing derivative means error not just shrinks but shrinks faster than any fixed ratio (it squares, cf. Problem 20). The single fact “ $|g'(L)| < 1$ converges, smaller is faster” explains both the delicate learning-rate window *and* Newton’s blazing self-tuned speed: gradient descent is a generic contraction ($|g'|$ between 0 and 1 if you’re lucky), while Newton is the special contraction that drives g' all the way to 0. ■

Teach it

23. Model answer. Picture a ball on a U-shaped valley in the dark; the learning rate η is how big a step the ball takes for each unit of steepness it feels under its feet. Too **small** an η and the ball inches downhill, technically arriving but taking forever. Too **big** an η and the ball leaps clear past the bottom and lands higher up the far wall — and if η is large enough each leap overshoots more than the last, so the ball climbs out of the valley entirely and flies off. The craft is choosing η large enough to make real progress but small enough that the steps still shrink toward the bottom.

24. The one true sentence: Newton needs the curvature f'' , which for a model with

N parameters is an $N \times N$ table of N^2 entries that is impossible to store or invert at the billion-parameter scale of real networks — so we accept gradient descent's slower, fiddly learning rate because it needs only the N -number slope, which we *can* compute.

25. Worked solution: Newton's method for $\sqrt{10}$ to four decimals.

Setup. Finding $\sqrt{10}$ means solving $x^2 = 10$, i.e. finding the root of $f(x) = x^2 - 10$. Its derivative is $f'(x) = 2x$.

The rule. Substitute into $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$:

$$x_{n+1} = x_n - \frac{x_n^2 - 10}{2x_n} = \frac{1}{2} \left(x_n + \frac{10}{x_n} \right).$$

(General square-root recipe: to find $\sqrt{A}$, iterate $x_{n+1} = \frac{1}{2}(x_n + A/x_n)$.)

Iterate from $x_0 = 3$ (since $3^2 = 9$ is close to 10):

n	x_n
0	3
1	$\frac{1}{2}(3 + \frac{10}{3}) = 3.166667$
2	$\frac{1}{2}(3.166667 + \frac{10}{3.166667}) = 3.162281$
3	$\frac{1}{2}(3.162281 + \frac{10}{3.162281}) = 3.162278$

Stop when the digits stop changing: x_2 and x_3 agree to four decimals. **Answer:** $\sqrt{10} \approx 3.1623$ (true value 3.16227766...). A peer can reuse this for any $\sqrt{A}$ by swapping 10 for A and starting near a known nearby square.

26. Design a problem. *Problem:* “Run gradient descent on $f(x) = 3x^2 - 12x + 5$ with learning rate $\eta = \frac{1}{6}$ from $x_0 = 10$. Show it reaches the minimum in exactly one step.”

Solution key. Here $f'(x) = 6x - 12$ and $f''(x) = 6$, so the true minimizer is $f' = 0 \Rightarrow x = 2$. One gradient step:

$$x_1 = x_0 - \eta f'(x_0) = 10 - \frac{1}{6}(6 \cdot 10 - 12) = 10 - \frac{1}{6}(48) = 10 - 8 = 2.$$

It lands exactly on the minimizer $x = 2$ in one step. *Why η works:* one-step convergence on a quadratic happens precisely when $\eta = \frac{1}{f''}$ (Problem 19's optimum), since then the factor $1 - \eta f'' = 0$. Here $f'' = 6$, so $\eta = \frac{1}{6}$ — and indeed it lands in one step from *any* x_0 , not just 10. (Builder's recipe: pick any quadratic $ax^2 + bx + c$ and set $\eta = \frac{1}{2a}$.)

27. Model answer. The link is that **a minimum is a place where the slope is zero**. Gradient descent hunts for the smallest value of f ; but “smallest” happens exactly where $f'(x) = 0$, which is a *root-finding* question — about the function f' . Newton's method is the root-finder, so pointing Newton at f' (replacing f by f' , hence f' by f'') gives $x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$, a minimum-finder. Both updates have the same shape, $x_{n+1} = x_n - (\text{step}) \cdot f'(x_n)$: they step against the slope. The *only* difference is the step size — gradient descent uses a fixed η you choose, Newton uses $1/f''$ it computes from the curvature. The same machine — “step against the slope” — solves both jobs; you only change what it points at and how it sizes its step.

Looking ahead

28. When x becomes a vector $\mathbf{x}$ and f' becomes the **gradient** ∇f (a vector pointing in the steepest *uphill* direction), the rule becomes

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \eta \nabla f(\mathbf{x}_n).$$

What changes: x and the derivative are now vectors with many components, so “downhill” is a direction in many-dimensional space, not just left/right; the subtraction is vector subtraction (each parameter updated by its own slope component). **What stays exactly the same:** the *form* of the rule — “new position = old position minus learning-rate times slope” — and the role of η as the single dial controlling step size, with the same too-small-crawls / too-large-explodes tension. The ball-in-a-valley picture is unchanged; the ball just has more directions to roll.

29. In one dimension there is a single curvature f'' and a single safe range $0 <$

$\eta < \frac{2}{f''}$. In many dimensions the curvature differs by direction: a valley can be steeply curved across (large effective a) and gently curved along (small effective a). The safe range for *each* direction is $0 < \eta < \frac{2}{a_{\text{that direction}}}$. A single number η must serve all directions at once, so if it is small enough to be safe in the **steep** direction (small $\frac{2}{a}$, demanding small η) it is far below the optimum $\frac{1}{a}$ for the **gentle** direction — too big to stay stable in one place yet too small to make progress in another *simultaneously*. No single η can be right for both. This mismatch is exactly why advanced optimizers (Adam, RMSProp, etc.) adapt the rate **per direction**.

30. A model with one million parameters has $N = 10^6$, so Newton's curvature table is $N \times N = (10^6)^2 = 10^{12}$ — **a trillion entries**. *Why that ends Newton at real scale:* you cannot even store a trillion numbers for the table, let alone do the (far-more-expensive-than-division) work of inverting it every step — and a trillion is small next to today's billion-parameter models, where the table would have a *billion-billion* (10^{18}) entries. This reconnects to the chapter's punchline: Newton's self-tuning second-order step is physically impossible at scale, so the field trades it for first-order gradient descent — which needs only the N -number slope (10^6 numbers, not 10^{12}) — and pays for that affordability by babysitting a hand-tuned learning rate.

Chapter 10

Adding Up the Infinite

Fluency

1. Reverse power rule: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$. (Check each by differentiating back.)

(a) $\int x^4 dx = \frac{x^5}{5} + C$. Check: $\frac{d}{dx} \frac{x^5}{5} = x^4$. $\boxed{\frac{x^5}{5} + C}$

(b) $\int x^7 dx = \frac{x^8}{8} + C$. Check: $\frac{d}{dx} \frac{x^8}{8} = x^7$. $\boxed{\frac{x^8}{8} + C}$

(c) $\int x^{-2} dx = \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$. Check: $\frac{d}{dx}(-x^{-1}) = x^{-2}$. $\boxed{-\frac{1}{x} + C}$

(d) $\int \sqrt{x} dx = \int x^{1/2} dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$. Check: $\frac{d}{dx} \frac{2}{3}x^{3/2} = \frac{2}{3} \cdot \frac{3}{2}x^{1/2} = x^{1/2}$. $\boxed{\frac{2}{3}x^{3/2} + C}$

(e) $\int 1 dx = \int x^0 dx = x + C$. Check: $\frac{d}{dx}x = 1$. $\boxed{x + C}$

2. Linearity: integrate term by term, constants pull out, one C at the end.

(a) $\int (3x^2 - 2x + 5) dx = 3 \cdot \frac{x^3}{3} - 2 \cdot \frac{x^2}{2} + 5x + C = x^3 - x^2 + 5x + C$. Check: $\frac{d}{dx}(x^3 - x^2 + 5x) = 3x^2 - 2x + 5$. $\boxed{x^3 - x^2 + 5x + C}$

(b) $\int (6x^5 + 4) dx = 6 \cdot \frac{x^6}{6} + 4x + C = x^6 + 4x + C$. Check: $\frac{d}{dx}(x^6 + 4x) = 6x^5 + 4$. $\boxed{x^6 + 4x + C}$

(c) $\int \left(\frac{1}{x^2} + x \right) dx = \int (x^{-2} + x) dx = \frac{x^{-1}}{-1} + \frac{x^2}{2} + C = -\frac{1}{x} + \frac{x^2}{2} + C$. Check:

$$\frac{d}{dx}\left(-x^{-1} + \frac{x^2}{2}\right) = x^{-2} + x. \quad \boxed{-\frac{1}{x} + \frac{x^2}{2} + C}$$

3. Straight from the standard library.

(a) $\int \cos x \, dx = \sin x + C$. Check: $\frac{d}{dx} \sin x = \cos x$. $\boxed{\sin x + C}$

(b) $\int \sin x \, dx = -\cos x + C$. Check: $\frac{d}{dx}(-\cos x) = \sin x$. $\boxed{-\cos x + C}$

(c) $\int e^x \, dx = e^x + C$. Check: $\frac{d}{dx} e^x = e^x$. $\boxed{e^x + C}$

(d) $\int \sec^2 x \, dx = \tan x + C$. Check: $\frac{d}{dx} \tan x = \sec^2 x$. $\boxed{\tan x + C}$

(e) $\int \frac{1}{x} \, dx = \ln|x| + C$ (the $n = -1$ exception). Check: $\frac{d}{dx} \ln|x| = \frac{1}{x}$. $\boxed{\ln|x| + C}$

4. Mixed library, term by term.

(a) $\int (2 \cos x - 3 \sin x) \, dx = 2 \sin x - 3(-\cos x) + C = 2 \sin x + 3 \cos x + C$. Check: $\frac{d}{dx}(2 \sin x + 3 \cos x) = 2 \cos x - 3 \sin x$. $\boxed{2 \sin x + 3 \cos x + C}$

(b) $\int (e^x + x^2) \, dx = e^x + \frac{x^3}{3} + C$. Check: $\frac{d}{dx}(e^x + \frac{x^3}{3}) = e^x + x^2$. $\boxed{e^x + \frac{x^3}{3} + C}$

(c) $\int (4 \sec^2 x - \frac{5}{x}) \, dx = 4 \tan x - 5 \ln|x| + C$. Check: $\frac{d}{dx}(4 \tan x - 5 \ln|x|) = 4 \sec^2 x - \frac{5}{x}$. $\boxed{4 \tan x - 5 \ln|x| + C}$

5. Rewrite into power form first, then integrate.

(a) Split the fraction: $\frac{x^2 - 1}{x} = x - \frac{1}{x}$. So $\int \left(x - \frac{1}{x}\right) dx = \frac{x^2}{2} - \ln|x| + C$. Check: $\frac{d}{dx}\left(\frac{x^2}{2} - \ln|x|\right) = x - \frac{1}{x} = \frac{x^2 - 1}{x}$. $\boxed{\frac{x^2}{2} - \ln|x| + C}$

(b) One power: $x^2 \sqrt{x} = x^2 \cdot x^{1/2} = x^{5/2}$. So $\int x^{5/2} \, dx = \frac{x^{7/2}}{7/2} + C = \frac{2}{7} x^{7/2} + C$.

Check: $\frac{d}{dx} \frac{2}{7} x^{7/2} = \frac{2}{7} \cdot \frac{7}{2} x^{5/2} = x^{5/2}$. $\boxed{\frac{2}{7} x^{7/2} + C}$

(c) Expand: $(x + 1)^2 = x^2 + 2x + 1$. So $\int (x^2 + 2x + 1) \, dx = \frac{x^3}{3} + x^2 + x + C$.

Check: $\frac{d}{dx}\left(\frac{x^3}{3} + x^2 + x\right) = x^2 + 2x + 1$. $\boxed{\frac{x^3}{3} + x^2 + x + C}$

(Equivalently $\frac{1}{3}(x + 1)^3 + C$; the two differ by the constant $\frac{1}{3}$.)

6. Simple substitution — the inside's derivative is already present.

(a) $u = x^2 + 1$, $du = 2x \, dx$. $\int 2x(x^2 + 1)^5 \, dx = \int u^5 \, du = \frac{u^6}{6} + C = \frac{(x^2 + 1)^6}{6} + C$.

Check (chain rule): $\frac{d}{dx} \frac{(x^2+1)^6}{6} = (x^2+1)^5 \cdot 2x \cdot \boxed{\frac{(x^2+1)^6}{6} + C}$

(b) $u = x^3+1, du = 3x^2 dx. \int 3x^2(x^3+1)^4 dx = \int u^4 du = \frac{u^5}{5} + C = \frac{(x^3+1)^5}{5} + C.$

Check: $\frac{d}{dx} \frac{(x^3+1)^5}{5} = (x^3+1)^4 \cdot 3x^2 \cdot \boxed{\frac{(x^3+1)^5}{5} + C}$

(c) $u = \sin x, du = \cos x dx. \int \cos x e^{\sin x} dx = \int e^u du = e^u + C = e^{\sin x} + C.$

Check: $\frac{d}{dx} e^{\sin x} = e^{\sin x} \cos x \cdot \boxed{e^{\sin x} + C}$

(d) $u = x^4+2, du = 4x^3 dx. \int 4x^3(x^4+2)^6 dx = \int u^6 du = \frac{u^7}{7} + C = \frac{(x^4+2)^7}{7} + C.$

Check: $\frac{d}{dx} \frac{(x^4+2)^7}{7} = (x^4+2)^6 \cdot 4x^3 \cdot \boxed{\frac{(x^4+2)^7}{7} + C}$

(e) $u = \sin x, du = \cos x dx. \int \sin^4 x \cos x dx = \int u^4 du = \frac{u^5}{5} + C = \frac{\sin^5 x}{5} + C.$

Check: $\frac{d}{dx} \frac{\sin^5 x}{5} = \sin^4 x \cos x \cdot \boxed{\frac{\sin^5 x}{5} + C}$

7. Substitution needing a constant factor, or landing on a log.

(a) $u = x^2, du = 2x dx, \text{ so } x dx = \frac{1}{2} du. \int x e^{x^2} dx = \int e^u \cdot \frac{1}{2} du = \frac{1}{2} e^u + C = \frac{1}{2} e^{x^2} + C. \text{ Check: } \frac{d}{dx} \frac{1}{2} e^{x^2} = \frac{1}{2} e^{x^2} \cdot 2x = x e^{x^2} \cdot \boxed{\frac{1}{2} e^{x^2} + C}$

(b) $u = x^2 + 1, du = 2x dx, \text{ so } x dx = \frac{1}{2} du. \int x(x^2+1)^3 dx = \int u^3 \cdot \frac{1}{2} du = \frac{1}{2} \cdot \frac{u^4}{4} + C = \frac{(x^2+1)^4}{8} + C. \text{ Check: } \frac{d}{dx} \frac{(x^2+1)^4}{8} = \frac{4(x^2+1)^3 \cdot 2x}{8} = x(x^2+1)^3. \boxed{\frac{(x^2+1)^4}{8} + C}$

(c) $u = 3x, du = 3 dx, \text{ so } dx = \frac{1}{3} du. \int \sin(3x) dx = \int \sin u \cdot \frac{1}{3} du = -\frac{1}{3} \cos u + C = -\frac{1}{3} \cos(3x) + C. \text{ Check: } \frac{d}{dx} \left(-\frac{1}{3} \cos 3x \right) = -\frac{1}{3} (-\sin 3x) \cdot 3 = \sin 3x. \boxed{-\frac{1}{3} \cos(3x) + C}$

(d) $u = 5x, du = 5 dx, \text{ so } dx = \frac{1}{5} du. \int \cos(5x) dx = \int \cos u \cdot \frac{1}{5} du = \frac{1}{5} \sin u + C = \frac{1}{5} \sin(5x) + C. \text{ Check: } \frac{d}{dx} \frac{1}{5} \sin 5x = \frac{1}{5} \cos 5x \cdot 5 = \cos 5x. \boxed{\frac{1}{5} \sin(5x) + C}$

(e) Logarithm pattern: top is (a constant times) the derivative of the bottom.

$u = x^3 + 1, du = 3x^2 dx, \text{ so } x^2 dx = \frac{1}{3} du. \int \frac{x^2}{x^3+1} dx = \int \frac{1}{u} \cdot \frac{1}{3} du =$

$$\frac{1}{3} \ln |u| + C = \frac{1}{3} \ln |x^3 + 1| + C. \text{ Check: } \frac{d}{dx} \frac{1}{3} \ln |x^3 + 1| = \frac{1}{3} \cdot \frac{3x^2}{x^3 + 1} = \frac{x^2}{x^3 + 1}.$$

$$\boxed{\frac{1}{3} \ln |x^3 + 1| + C}$$

8. Right Riemann sum for $f(x) = x^2$ on $[0, 4]$, $n = 4$, $\Delta x = 1$. Right endpoints are $x = 1, 2, 3, 4$:

$$\sum_{i=1}^4 f(x_i) \Delta x = (1^2 + 2^2 + 3^2 + 4^2) \cdot 1 = 1 + 4 + 9 + 16 = 30.$$

$$\boxed{\text{Estimate} = 30.}$$

Since $f(x) = x^2$ is **increasing** on $[0, 4]$, the right endpoint is the *highest* point on each slice, so every rectangle overshoots the area beneath the curve. The estimate is **above** the true area. (The true value is $\int_0^4 x^2 dx = \frac{64}{3} \approx 21.33$, confirming 30 is an overestimate.)

9. Recognize each integral as a signed area.

(a) $\int_0^3 2 dx$ = area of a rectangle, width 3, height 2: $3 \times 2 = \boxed{6}$.

(b) $\int_0^2 x dx$ = area of a triangle under $y = x$, base 2, height 2: $\frac{1}{2} \cdot 2 \cdot 2 = \boxed{2}$.

(c) $\int_{-1}^1 x dx$: on $[-1, 0]$ the curve is below the axis (a triangle of signed area $-\frac{1}{2}$); on $[0, 1]$ it is above (a triangle of area $+\frac{1}{2}$). They cancel: $-\frac{1}{2} + \frac{1}{2} = \boxed{0}$. (The integrand is odd and the interval symmetric.)

Practice

10. Compute $\int_0^1 x^3 dx$ from the definition. Slice $[0, 1]$ into n pieces of width $\Delta x = \frac{1}{n}$; right endpoints $x_i = \frac{i}{n}$:

$$\sum_{i=1}^n \left(\frac{i}{n}\right)^3 \cdot \frac{1}{n} = \frac{1}{n^4} \sum_{i=1}^n i^3 = \frac{1}{n^4} \left[\frac{n(n+1)}{2} \right]^2 = \frac{1}{n^4} \cdot \frac{n^2(n+1)^2}{4} = \frac{(n+1)^2}{4n^2}.$$

As $n \rightarrow \infty$, $\frac{(n+1)^2}{4n^2} = \frac{1}{4} \left(1 + \frac{1}{n}\right)^2 \rightarrow \frac{1}{4} \cdot 1 = \frac{1}{4}$.

$$\boxed{\int_0^1 x^3 dx = \frac{1}{4}.}$$

11. Substitution.

(a) $u = x^2 + 1$, $du = 2x dx$. The numerator $2x dx$ is exactly du : $\int \frac{2x}{x^2 + 1} dx = \int \frac{1}{u} du = \ln|u| + C = \ln(x^2 + 1) + C$ (the argument is always positive, so the bars are optional). Check: $\frac{d}{dx} \ln(x^2 + 1) = \frac{2x}{x^2 + 1}$. $\boxed{\ln(x^2 + 1) + C}$

(b) $u = x^2 + 9$, $du = 2x dx$, so $x dx = \frac{1}{2} du$. $\int \frac{x}{\sqrt{x^2 + 9}} dx = \int \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du = \frac{1}{2} \int u^{-1/2} du = \frac{1}{2} \cdot 2u^{1/2} + C = \sqrt{x^2 + 9} + C$. Check: $\frac{d}{dx} \sqrt{x^2 + 9} = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}$. $\boxed{\sqrt{x^2 + 9} + C}$

(c) $u = \sin x$, $du = \cos x dx$. $\int \sin x \cos x dx = \int u du = \frac{u^2}{2} + C = \frac{\sin^2 x}{2} + C$.

Check: $\frac{d}{dx} \frac{\sin^2 x}{2} = \sin x \cos x$. $\boxed{\frac{\sin^2 x}{2} + C}$

(A valid alternative with $u = \cos x$ gives $-\frac{\cos^2 x}{2} + C$; the two answers differ by the constant $\frac{1}{2}$.)

12. Substitution with rearrangement.

(a) $u = \ln x$, $du = \frac{1}{x} dx$. $\int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$. Check: $\frac{d}{dx} \frac{(\ln x)^2}{2} = \ln x \cdot \frac{1}{x} = \frac{\ln x}{x}$. $\boxed{\frac{(\ln x)^2}{2} + C}$

(b) $u = x + 1$, so $x = u - 1$ and $du = dx$. Then $x\sqrt{x+1} = (u-1)\sqrt{u} = u^{3/2} - u^{1/2}$:

$$\int x\sqrt{x+1} dx = \int (u^{3/2} - u^{1/2}) du = \frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} + C = \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C.$$

Back-substitute: $\boxed{\frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + C}$. Check: $\frac{d}{dx} \left[\frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} \right] = (x+1)^{3/2} - (x+1)^{1/2} = (x+1)^{1/2} [(x+1) - 1] = x\sqrt{x+1}$. ✓

13. Initial value problem. Antidifferentiate $F'(x) = 3x^2 - 2$:

$$F(x) = \int (3x^2 - 2) dx = x^3 - 2x + C.$$

Apply $F(1) = 5$: $1 - 2 + C = 5 \Rightarrow -1 + C = 5 \Rightarrow C = 6$.

$$\boxed{F(x) = x^3 - 2x + 6.}$$

Check: $F'(x) = 3x^2 - 2$ ✓ and $F(1) = 1 - 2 + 6 = 5$ ✓.

14. Antidifferentiate the velocity $v(t) = -10t + 20$ to get position:

$$s(t) = \int (-10t + 20) dt = -10 \cdot \frac{t^2}{2} + 20t + C = -5t^2 + 20t + C.$$

Apply $s(0) = 0$: $-5(0) + 20(0) + C = 0 \Rightarrow C = 0$.

$$s(t) = -5t^2 + 20t \text{ (m).}$$

Check: $s'(t) = -10t + 20 = v(t)$ ✓ and $s(0) = 0$ ✓.

15. Set up the area under $f(x) = e^{-x^2}$ on $[0, 1]$ as a limit of Riemann sums. With $\Delta x = \frac{1}{n}$ and right endpoints $x_i = \frac{i}{n}$:

$$\int_0^1 e^{-x^2} dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n e^{-(i/n)^2} \cdot \frac{1}{n}.$$

Why you cannot finish with an antiderivative: the Fundamental-Theorem shortcut requires an antiderivative F with $F'(x) = e^{-x^2}$ that you can *write down*. But e^{-x^2} has **no elementary antiderivative** — no finite combination of powers, roots, exponentials, logarithms, and trig functions differentiates to it. The antiderivative exists as a genuine smooth function, but it is not expressible in elementary terms, so there is nothing to plug into and the only route to the number is the limit of sums above (computed numerically).

16. Sketch of $f(x) = x^2 - 1$ on $[0, 2]$: a parabola passing through $(0, -1)$, crossing the axis at $x = 1$, rising to $(2, 3)$.

(Picture: below the axis on $[0, 1]$ — a small region of area A_- ; above the axis on $(1, 2]$ — a larger region of area A_+ .)

The **signed** integral counts the part below the axis as **negative**:

$$\int_0^2 (x^2 - 1) dx = \underbrace{(+A_+)}_{\text{on } [1,2]} + \underbrace{(-A_-)}_{\text{on } [0,1]} = A_+ - A_-.$$

The **unsigned** area between curve and axis adds both as positive: $A_+ + A_-$. Since

$A_- > 0$, subtracting it gives less than adding it:

$$\int_0^2 (x^2 - 1) dx = A_+ - A_- < A_+ + A_- = (\text{unsigned area}).$$

The signed integral is smaller by $2A_-$ — the below-axis piece is subtracted, not added.

Concept & counterexample

17.

- (a) **True.** $\int_a^b f dx$ is the limit of Riemann sums — a single **number** (the signed area). $\int f dx$ has no limits; it denotes the family of antiderivatives $F(x) + C$ — a **function** (in fact a family of functions).
- (b) **False.** $F' = G'$ only forces F and G to differ by a *constant*: $F = G + C$. For example $F(x) = x^2 + 3$ and $G(x) = x^2$ have $F' = G' = 2x$ but $F \neq G$. (Counterexample.) It becomes true only if you additionally know F and G agree at one point.
- (c) **True.** “Real” (i.e. honest, ordinary) functions can dip below the axis, and the integral counts that area as negative. E.g. $\int_0^{2\pi} \sin x dx = 0$, or $\int_{-1}^0 x dx = -\frac{1}{2} < 0$. Signed area can be negative.
- (d) **False.** Every continuous function *has* an antiderivative, but it need not be **elementary**. e^{-x^2} is continuous everywhere yet has no elementary antiderivative (likewise $\sin(x^2)$, $\frac{\sin x}{x}$). The antiderivative exists; you just can’t always write it with elementary functions.

18. Why the $+C$. Suppose F and G are both antiderivatives of f on an interval, so $F' = f = G'$. Let $H = F - G$. Then $H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0$ for every x in the interval. By the **Mean Value Theorem**, a function with derivative 0 throughout an interval is constant: for any two points $a < b$ there is a c between them with $H(b) - H(a) = H'(c)(b - a) = 0 \cdot (b - a) = 0$, so H never changes value. Thus $F - G = H = C$, a constant, i.e. $F = G + C$. Because C can be *any* real number, the antiderivatives form a whole one-parameter **family** of vertically parallel curves — which is exactly why $\int f dx = F(x) + C$ carries the arbitrary constant.

19. Find the error. The student wrote $\int \sin x dx = \cos x + C$. **Differentiate the proposed answer:** $\frac{d}{dx}(\cos x + C) = -\sin x$, **not** $\sin x$. The sign is wrong: the

derivative of $\cos x$ carries a minus, so to land on $+\sin x$ the antiderivative must be $-\cos x$. Differentiating the answer immediately exposes the wrong sign. Correct result:

$$\boxed{\int \sin x \, dx = -\cos x + C} \quad \left(\text{since } \frac{d}{dx}(-\cos x) = \sin x\right).$$

20. Construct f on $[-1, 1]$ with $\int_{-1}^1 f \, dx = 0$ but f never zero. Use an **odd** function (symmetry $f(-x) = -f(x)$): the area on $[-1, 0]$ cancels the area on $[0, 1]$. Take

$$\boxed{f(x) = x^3} \quad (\text{or } f(x) = x, x^5, \dots).$$

It is never zero on $(-1, 1)$ except at the single point $x = 0$. To make it **never** zero at all, shift the idea to a sign-changing function with no root, e.g.

$$\boxed{f(x) = \begin{cases} -1, & x < 0 \\ 1, & x \geq 0 \end{cases}}$$

which is never 0 yet has $\int_{-1}^1 f \, dx = (-1)(1) + (1)(1) = 0$ — the negative block exactly cancels the positive block. Odd symmetry about the origin forces the cancellation.

21. When substitution helps. Substitution $u = g(x)$ works only when $g'(x)$ — the derivative of the inside — is **present in the integrand** (up to a constant factor), because the step $du = g'(x) \, dx$ must consume the rest of the integrand and leave a clean integral in u alone.

- $\int 2x(x^2 + 1)^5 \, dx$: inside $g = x^2 + 1$, $g' = 2x$ is right there. Set $u = x^2 + 1$, $du = 2x \, dx$, and the $2x \, dx$ is *exactly* du — the integral collapses to $\int u^5 \, du$. ✓
- $\int (x^2 + 1)^5 \, dx$ **as written**: inside is still $x^2 + 1$ with $g' = 2x$, but there is **no x factor** to absorb into du . Writing $du = 2x \, dx$ leaves a stray $\frac{1}{2x}$ that still contains x , so the integral does **not** reduce to a function of u only. The missing piece is the factor of x .

Substitution needs the inside's derivative (up to a constant) sitting in the integrand; $(x^2 + 1)^5$ lacks it.

(One would have to expand $(x^2 + 1)^5$ and integrate term by term instead.)

22. Is it possible $-\int_a^b f dx = 0$ with $a < b$ and $f > 0$ somewhere? **Yes.** The integral is *signed* area, so a positive bump can be cancelled by a negative dip. The function must be **negative elsewhere** by an amount whose area exactly offsets the positive part.

Yes — e.g. $f(x) = x$ on $[-1, 1]$: $f > 0$ on $(0, 1]$ yet $\int_{-1}^1 x dx = 0$.

What f must do: dip below the axis somewhere with total negative area equal to its total positive area. (If instead $f \geq 0$ *everywhere* and $f > 0$ somewhere — with f continuous — then $\int_a^b f dx > 0$; cancellation is impossible. The “somewhere positive” is only consistent with a zero integral when f also goes negative.)

◆ Diamond

23. ◆ Compute $\int_0^1 x^2 dx$ from the definition in full.

Set-up. Partition $[0, 1]$ into n subintervals of equal width $\Delta x = \frac{1-0}{n} = \frac{1}{n}$. The right endpoint of the i -th subinterval is $x_i = 0 + i \Delta x = \frac{i}{n}$, for $i = 1, \dots, n$.

The right Riemann sum.

$$S_n = \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} = \sum_{i=1}^n \frac{i^2}{n^3} = \frac{1}{n^3} \sum_{i=1}^n i^2,$$

where pulling out $\frac{1}{n^3}$ is legitimate because it is constant in the summation index i .

Apply the closed form $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ (Chapter 1):

$$S_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}.$$

Take the limit. Expand the numerator: $(n+1)(2n+1) = 2n^2 + 3n + 1$, so

$$S_n = \frac{2n^2 + 3n + 1}{6n^2} = \frac{2}{6} + \frac{3}{6n} + \frac{1}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

As $n \rightarrow \infty$ the last two terms vanish, leaving

$$\boxed{\int_0^1 x^2 dx = \lim_{n \rightarrow \infty} S_n = \frac{1}{3}.}$$

Every step is justified: the partition is uniform, the sum formula is exact for each finite n , and the limit of a sum of terms is the sum of the limits.

24. ♦ Prove linearity $\int_a^b (f + g) = \int_a^b f + \int_a^b g$ from the definition.

Fix a uniform partition of $[a, b]$ with $\Delta x = \frac{b-a}{n}$ and sample points x_i . The Riemann sum of $f + g$ splits because finite sums are additive:

$$\sum_{i=1}^n [f(x_i) + g(x_i)] \Delta x = \sum_{i=1}^n [f(x_i) \Delta x + g(x_i) \Delta x] = \sum_{i=1}^n f(x_i) \Delta x + \sum_{i=1}^n g(x_i) \Delta x.$$

Now take $n \rightarrow \infty$. Both sums on the right converge (to $\int_a^b f$ and $\int_a^b g$ respectively), and the limit of a sum of two convergent sequences is the sum of their limits:

$$\int_a^b (f+g) dx = \lim_{n \rightarrow \infty} \left[\sum f(x_i) \Delta x + \sum g(x_i) \Delta x \right] = \lim_{n \rightarrow \infty} \sum f(x_i) \Delta x + \lim_{n \rightarrow \infty} \sum g(x_i) \Delta x = \int_a^b f + \int_a^b g$$

$$\boxed{\int_a^b (f + g) dx = \int_a^b f dx + \int_a^b g dx.}$$

(The constant-multiple rule is identical: $\sum c f(x_i) \Delta x = c \sum f(x_i) \Delta x$, and constants pass through the limit, giving $\int_a^b c f = c \int_a^b f$.)

25. ♦ Derive substitution from the chain rule. Start from the chain rule for a composite $F(g(x))$:

$$\frac{d}{dx} F(g(x)) = F'(g(x)) g'(x).$$

This equation says the right-hand side is the derivative of $F(g(x))$ — i.e. $F(g(x))$ is an antiderivative of $F'(g(x)) g'(x)$. Antidifferentiate both sides (take $\int(\cdot) dx$): the left integrates to the function whose derivative it is, $F(g(x))$, up to a constant, so

$$\int F'(g(x)) g'(x) dx = F(g(x)) + C.$$

$$\boxed{\int F'(g(x)) g'(x) dx = F(g(x)) + C.}$$

Why it is the chain rule backward: the chain rule *manufactures* the factor $g'(x)$ when you differentiate $F(g(x))$ — the derivative of the inside is “left behind” by the differentiation. Substitution *recognizes* that leftover factor and undoes the operation, peeling the composition back to F evaluated at g . Writing $u = g(x)$, $du = g'(x) dx$ turns the left side into $\int F'(u) du = F(u) + C$ — the bookkeeping that makes “the chain rule read right-to-left” mechanical. Differentiating the answer with the chain rule returns the integrand exactly, confirming the two operations are inverse.

26. ♦ Prove: if $F'(x) = 0$ for all x on an interval I , then F is constant on I .

Take any two points $a < b$ in I . F is differentiable on I (its derivative exists and equals 0 there), hence continuous on $[a, b]$ and differentiable on (a, b) — the hypotheses of the **Mean Value Theorem**. The MVT gives a point $c \in (a, b)$ with

$$F(b) - F(a) = F'(c)(b - a).$$

But $F'(c) = 0$ by assumption, so $F(b) - F(a) = 0 \cdot (b - a) = 0$, i.e. $F(b) = F(a)$. Since a, b were arbitrary, F takes the same value at every pair of points: F is constant on I . ■

$F' \equiv 0$ on an interval $\implies F$ constant there — a one-line corollary of the MVT.

This is precisely the fact behind the $+C$: two antiderivatives of f have difference with zero derivative, hence differ by a constant.

27. ♦♦ The midpoint sum for $f(x) = x^2$ on $[0, 1]$ is sharp.

Set-up. With $\Delta x = \frac{1}{n}$, the i -th subinterval is $[\frac{i-1}{n}, \frac{i}{n}]$, whose **midpoint** is

$$m_i = \frac{1}{n} \left(i - \frac{1}{2} \right), \quad i = 1, \dots, n.$$

The midpoint sum.

$$M_n = \sum_{i=1}^n f(m_i) \Delta x = \sum_{i=1}^n \frac{1}{n^2} \left(i - \frac{1}{2} \right)^2 \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^n \left(i - \frac{1}{2} \right)^2.$$

Evaluate $\sum \left(i - \frac{1}{2}\right)^2$. Expand $\left(i - \frac{1}{2}\right)^2 = i^2 - i + \frac{1}{4}$:

$$\sum_{i=1}^n \left(i - \frac{1}{2}\right)^2 = \sum i^2 - \sum i + \sum \frac{1}{4} = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + \frac{n}{4}.$$

Put over a common denominator 12:

$$= \frac{2n(n+1)(2n+1) - 6n(n+1) + 3n}{12} = \frac{n[2(n+1)(2n+1) - 6(n+1) + 3]}{12}.$$

Inside: $2(n+1)(2n+1) = 4n^2 + 6n + 2$; subtract $6(n+1) = 6n + 6$ and add 3: $4n^2 + 6n + 2 - 6n - 6 + 3 = 4n^2 - 1$. Hence

$$\sum_{i=1}^n \left(i - \frac{1}{2}\right)^2 = \frac{n(4n^2 - 1)}{12} = \frac{4n^3 - n}{12}.$$

The sum and its limit.

$$M_n = \frac{1}{n^3} \cdot \frac{4n^3 - n}{12} = \frac{4n^3 - n}{12n^3} = \frac{1}{3} - \frac{1}{12n^2} \xrightarrow{n \rightarrow \infty} \frac{1}{3}.$$

So the midpoint sum also converges to $\frac{1}{3}$. ✓

$$\boxed{M_n = \frac{1}{3} - \frac{1}{12n^2}, \quad \lim_{n \rightarrow \infty} M_n = \frac{1}{3}.}$$

Why faster than the right sum. From #23 the right sum is $R_n = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}$, whose error is

$$R_n - \frac{1}{3} = \frac{1}{2n} + \frac{1}{6n^2} \sim \frac{1}{2n} \quad (\text{order } 1/n),$$

while the midpoint error is

$$M_n - \frac{1}{3} = -\frac{1}{12n^2} \quad (\text{order } 1/n^2).$$

The right-sum error shrinks like $1/n$; the midpoint-sum error shrinks like $1/n^2$ — far faster. Geometrically, the midpoint rectangle's overshoot on the rising left half of each slice nearly cancels its undershoot on the right half, so the leading $1/n$ error term is wiped out, leaving only the $1/n^2$ remainder. The midpoint rule approaches the true value of $\frac{1}{3}$ much more quickly.

28. ♦♦ Why a limit-definition can fail, and what rescues it.

Because $\int_a^b f \, dx$ is *defined* as a limit of Riemann sums, it exists only if that limit exists — and limits can fail. If f is badly behaved, different choices of sample points (or different partitions) can make the sums approach **different** values, or oscillate with no limit, so no single number deserves to be called the integral. The classic offender is **Dirichlet’s function**

$$f(x) = \begin{cases} 1, & x \text{ rational} \\ 0, & x \text{ irrational,} \end{cases}$$

which jumps infinitely often on every subinterval: sampling rationals makes every sum $= b - a$, sampling irrationals makes every sum $= 0$. The two limits disagree, so f is **not** Riemann integrable.

The property of f that guarantees the sums converge — and which *every continuous function on $[a, b]$ possesses* — is **(uniform) continuity** on the closed bounded interval. The relevant theorem:

Every function continuous on a closed bounded interval $[a, b]$ is Riemann integrable there.

The mechanism is uniform continuity: on a closed bounded interval continuity is automatically uniform, so as the mesh $\Delta x \rightarrow 0$ the oscillation of f within each subinterval is forced uniformly small, the gap between the upper and lower sums goes to 0, and all Riemann sums squeeze to a common limit. (More generally, boundedness together with at most a “measure-zero” set of discontinuities suffices; the full proof belongs to real analysis.)

Teach it

29. Model answer (rectangles picture). “Picture the region under the curve sliced into thin vertical strips. Each strip is almost a rectangle, so I estimate its area as (height of the curve) $\times$ (width of the strip), and I add up all the rectangles to estimate the whole region. The catch is that a rectangle’s flat top doesn’t follow the curve exactly, so there’s a little error on every strip. But as I use more and thinner rectangles, each flat top hugs the curve over a tinier span and the total error shrinks toward zero — so in the limit, as the strip width goes to 0, the staircase of rectangles melts onto the curve and the sum becomes the exact area. That limiting sum is the integral.”

30. Debug your classmate. They wrote $\int 2x(x^2+1)^5 \, dx = \frac{(x^2+1)^6}{6} \cdot 2x + C$ — keeping

the $2x$.

Where it goes wrong: in the substitution $u = x^2 + 1$, $du = 2x dx$, the factor $2x dx$ is *not* left hanging beside the answer — it **becomes** du and is **consumed** by the change of variable. The whole integral turns into $\int u^5 du$, a pure u -integral with no x left over. Their answer instead multiplies the u -result back by $2x$, double-counting it. (The tell: differentiate their expression and you get a product-rule mess, not $2x(x^2 + 1)^5$.)

Correct work: $u = x^2 + 1$, $du = 2x dx \Rightarrow \int 2x(x^2 + 1)^5 dx = \int u^5 du = \frac{u^6}{6} + C = \frac{(x^2 + 1)^6}{6} + C$.

$$\boxed{\int 2x(x^2 + 1)^5 dx = \frac{(x^2 + 1)^6}{6} + C.}$$

Check (their proofreader): $\frac{d}{dx} \frac{(x^2+1)^6}{6} = \frac{6(x^2+1)^5}{6} \cdot 2x = 2x(x^2 + 1)^5$ ✓ — the integrand exactly. Differentiating *their* answer would not have, which is how they'd have caught it.

31. Worked solution to $\int x e^{x^2} dx$ (narrated as a template).

Spot the inside. The exponent x^2 is a function tucked inside $e^{(\)}$. Its derivative is $2x$, and there is an x out front — so the inside's derivative is present *up to the constant factor 2*. That is the green light for substitution.

Name it. Let $u = x^2$. Then $du = 2x dx$.

Handle the stray constant. The integral has $x dx$, but $du = 2x dx$, i.e. $x dx = \frac{1}{2} du$. Solve for the piece you actually have and substitute it wholesale — don't leave a loose 2.

Rewrite entirely in u (no x may survive): $\int x e^{x^2} dx = \int e^u \cdot \frac{1}{2} du = \frac{1}{2} \int e^u du$.

Integrate from the table: $\frac{1}{2} \int e^u du = \frac{1}{2} e^u + C$.

Back-substitute $u = x^2$: $\frac{1}{2} e^{x^2} + C$.

Check by differentiating: $\frac{d}{dx} \frac{1}{2} e^{x^2} = \frac{1}{2} e^{x^2} \cdot 2x = x e^{x^2}$ ✓.

$$\int x e^{x^2} dx = \frac{1}{2} e^{x^2} + C.$$

Redo for $\int x \sin(x^2) dx$ (same template): $u = x^2$, $du = 2x dx$, $x dx = \frac{1}{2} du$;
 $\int x \sin(x^2) dx = \frac{1}{2} \int \sin u du = -\frac{1}{2} \cos u + C = -\frac{1}{2} \cos(x^2) + C$. Check: $\frac{d}{dx}(-\frac{1}{2} \cos x^2) = -\frac{1}{2}(-\sin x^2) \cdot 2x = x \sin(x^2)$ ✓.

32. Design a problem (model answer + full key).

Problem. Evaluate $\int 6x^2 \sqrt{x^3 + 1} dx$.

Why it requires substitution: the radicand $x^3 + 1$ is a composite inside the square root, and there is no elementary antiderivative by inspection; but its derivative $3x^2$ sits in the integrand (as $6x^2 = 2 \cdot 3x^2$), so substitution is the natural – and necessary – tool.

Solution key. 1. Let $u = x^3 + 1$. Then $du = 3x^2 dx$, so $6x^2 dx = 2 du$. 2. Rewrite: $\int 6x^2 \sqrt{x^3 + 1} dx = \int \sqrt{u} \cdot 2 du = 2 \int u^{1/2} du$. 3. Integrate: $2 \cdot \frac{u^{3/2}}{3/2} = 2 \cdot \frac{2}{3} u^{3/2} = \frac{4}{3} u^{3/2}$. 4. Back-substitute: $\frac{4}{3}(x^3 + 1)^{3/2} + C$.

$$\int 6x^2 \sqrt{x^3 + 1} dx = \frac{4}{3}(x^3 + 1)^{3/2} + C.$$

Differentiate to check: $\frac{d}{dx} \frac{4}{3}(x^3 + 1)^{3/2} = \frac{4}{3} \cdot \frac{3}{2}(x^3 + 1)^{1/2} \cdot 3x^2 = 2(x^3 + 1)^{1/2} \cdot 3x^2 = 6x^2 \sqrt{x^3 + 1}$ ✓ – the integrand exactly. (The clean coefficient $\frac{4}{3}$ confirms the constants were chosen to land tidily.)

33. Teach the difference (model answer). “The **definite** integral $\int_a^b f dx$ is a single **number**: the signed area trapped between the graph of f and the x -axis from $x = a$ to $x = b$, built as the limit of rectangle sums. The **indefinite** integral $\int f dx$ is not a number at all – it is a **family of functions**, all the antiderivatives $F(x) + C$ whose derivative is f , differing only by the constant C . One answers ‘how much area?’, the other answers ‘what function differentiates to this?’. They wear the **same elongated-S symbol** as a deliberate hint, because the Fundamental Theorem of Calculus (next chapter) reveals they are two sides of one coin: you compute the area-number by evaluating an antiderivative from the family at the two endpoints and subtracting.”

Looking ahead

34. Let $A(x) = \int_0^x t^2 dt$. The computation $\int_0^1 t^2 dt = \frac{1}{3}$ is the case $x = 1$. Scaling the same right-Riemann-sum argument from $[0, 1]$ to $[0, x]$ (slices of width $\frac{x}{n}$, right endpoints $\frac{ix}{n}$) gives $\sum (\frac{ix}{n})^2 \frac{x}{n} = \frac{x^3}{n^3} \sum i^2 \rightarrow \frac{x^3}{3}$, so

$$A(x) = \frac{x^3}{3}.$$

(Sanity check: $A(1) = \frac{1}{3}$ ✓.) Differentiate:

$$A'(x) = \frac{d}{dx} \frac{x^3}{3} = x^2.$$

What to notice: $A'(x) = x^2$ is exactly the **integrand** (with t renamed x). Differentiating the accumulated-area function gives back the function you were integrating. The area-accumulator and the derivative undo each other — this is the Fundamental Theorem of Calculus, discovered firsthand.

35. Predict $\int_a^b f dx$ in terms of an antiderivative F (with $F' = f$). If accumulating area is undone by differentiation (Problem 34), then the area function is an antiderivative, and the area over $[a, b]$ is the *net change* in that antiderivative from a to b :

$$\int_a^b f dx = F(b) - F(a).$$

Every area problem becomes a subtraction: find any antiderivative F , plug in the two endpoints, subtract. (Chapter 11 proves this exactly — the Fundamental Theorem. Quick test: $\int_0^1 x^2 dx$ with $F(x) = \frac{x^3}{3}$ gives $F(1) - F(0) = \frac{1}{3} - 0 = \frac{1}{3}$, matching the page of algebra in one line.)

36. Predict the role of $L(x) = \int_1^x \frac{1}{t} dt$.

- $L(1)$: the integral from 1 to 1 has no width, so $L(1) = 0$.
- **Increasing or decreasing for $x > 1$?** On $(1, x)$ the integrand $\frac{1}{t} > 0$, so accumulating it adds positive area: L is **increasing** for $x > 1$. (By Problem 34's pattern, $L'(x) = \frac{1}{x} > 0$ for $x > 0$, confirming L rises.)
- **What it is:** L vanishes at 1, is increasing, and has derivative $\frac{1}{x}$ — exactly

the defining properties of the **natural logarithm** $\ln x$. So $L(x) = \ln x$.

$$L(x) = \ln x : \quad L(1) = 0, \quad L \text{ increasing for } x > 1 \text{ (since } L'(x) = \frac{1}{x} > 0 \text{)}.$$

This is how Chapter 11 *defines* $\ln x := \int_1^x \frac{1}{t} dt$, finally paying back every logarithm and exponential borrowed since Chapter 5 — the $n = -1$ exception to the power rule was the logarithm in disguise all along.

Chapter 11

The Fundamental Theorem of Calculus

Fluency

1. $\int_0^2 x^3 dx$. Antiderivative $F(x) = \frac{x^4}{4}$ (check: $F' = x^3$). Then $\left[\frac{x^4}{4}\right]_0^2 = \frac{16}{4} - 0 = 4$.

4

2. $\int_1^4 \sqrt{x} dx = \int_1^4 x^{1/2} dx$. Antiderivative $\frac{x^{3/2}}{3/2} = \frac{2}{3}x^{3/2}$ (check: $\frac{d}{dx} \frac{2}{3}x^{3/2} = x^{1/2}$).

Then $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{2}{3}(4^{3/2} - 1) = \frac{2}{3}(8 - 1) = \frac{14}{3}$. $\frac{14}{3}$

3. $\int_0^\pi \sin x dx$. Antiderivative $-\cos x$. Then $[-\cos x]_0^\pi = -\cos \pi + \cos 0 = -(-1) + 1 = 2$. (Positive: $\sin$ is ≥ 0 on $[0, \pi]$.) 2

4. $\int_0^{\pi/2} \cos x dx = [\sin x]_0^{\pi/2} = \sin \frac{\pi}{2} - \sin 0 = 1 - 0 = 1$. 1

5. $\int_1^e \frac{1}{x} dx = [\ln x]_1^e = \ln e - \ln 1 = 1 - 0 = 1$. 1

6. $\int_0^1 e^x dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. $e - 1$

7. $\int_{-1}^1 (x^2 - 1) dx$. Antiderivative $\frac{x^3}{3} - x$. Then $\left[\frac{x^3}{3} - x\right]_{-1}^1 = \left(\frac{1}{3} - 1\right) - \left(-\frac{1}{3} + 1\right) = -\frac{2}{3} - \frac{2}{3} = -\frac{4}{3}$. **Sign:** the integral is *negative* because on $(-1, 1)$ the parabola $x^2 - 1$

lies *below* the x -axis; the signed area is negative. $-\frac{4}{3}$

8. $\int_1^2 (3x^2 - 2x) dx$. Antiderivative $x^3 - x^2$ (check: $\frac{d}{dx}(x^3 - x^2) = 3x^2 - 2x$). Then $[x^3 - x^2]_1^2 = (8 - 4) - (1 - 1) = 4 - 0 = 4$. 4

9. $\int_0^{\pi/4} \sec^2 x dx = [\tan x]_0^{\pi/4} = \tan \frac{\pi}{4} - \tan 0 = 1 - 0 = 1$. 1

10. $\int_1^3 \left(x + \frac{1}{x}\right) dx$. Antiderivative $\frac{x^2}{2} + \ln x$. Then $\left[\frac{x^2}{2} + \ln x\right]_1^3 = \left(\frac{9}{2} + \ln 3\right) - \left(\frac{1}{2} + 0\right) = 4 + \ln 3$. $\boxed{4 + \ln 3}$

11. FTC Part 1: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$ — just evaluate the integrand at the upper limit. (No antiderivative needed; none of these has an elementary one.)

(a) $\frac{d}{dx} \int_0^x \sin(t^2) dt = \sin(x^2)$. $\boxed{\sin(x^2)}$

(b) $\frac{d}{dx} \int_2^x e^{-t^2} dt = e^{-x^2}$. (The lower limit 2 is a constant, so it plays no role.)
 $\boxed{e^{-x^2}}$

(c) $\frac{d}{dx} \int_1^x \frac{1}{1+t^4} dt = \frac{1}{1+x^4}$. $\boxed{\frac{1}{1+x^4}}$

12. Integration by parts: $\int u dv = uv - \int v du$. Pick u = the factor whose derivative simplifies.

(a) $\int xe^x dx$: $u = x$, $dv = e^x dx \Rightarrow du = dx$, $v = e^x$. So $xe^x - \int e^x dx = xe^x - e^x + C$.
 Check: $\frac{d}{dx}(xe^x - e^x) = e^x + xe^x - e^x = xe^x$. $\boxed{(x-1)e^x + C}$

(b) $\int x \cos x dx$: $u = x$, $dv = \cos x dx \Rightarrow du = dx$, $v = \sin x$. So $x \sin x - \int \sin x dx = x \sin x + \cos x + C$. Check: $\frac{d}{dx}(x \sin x + \cos x) = \sin x + x \cos x - \sin x = x \cos x$. $\boxed{x \sin x + \cos x + C}$

(c) $\int x \sin x dx$: $u = x$, $dv = \sin x dx \Rightarrow du = dx$, $v = -\cos x$. So $-x \cos x - \int (-\cos x) dx = -x \cos x + \sin x + C$. Check: $\frac{d}{dx}(-x \cos x + \sin x) = -\cos x + x \sin x + \cos x = x \sin x$. $\boxed{-x \cos x + \sin x + C}$

(d) $\int xe^{-x} dx$: $u = x$, $dv = e^{-x} dx \Rightarrow du = dx$, $v = -e^{-x}$. So $-xe^{-x} - \int (-e^{-x}) dx = -xe^{-x} - e^{-x} + C$. Check: $\frac{d}{dx}(-xe^{-x} - e^{-x}) = -e^{-x} + xe^{-x} + e^{-x} = xe^{-x}$. $\boxed{-(x+1)e^{-x} + C}$

(e) $\int (x+1) \sin x dx$: $u = x+1$, $dv = \sin x dx \Rightarrow du = dx$, $v = -\cos x$. So $-(x+1) \cos x - \int (-\cos x) dx = -(x+1) \cos x + \sin x + C$. Check: $\frac{d}{dx}(-(x+1) \cos x + \sin x) = -\cos x + (x+1) \sin x + \cos x = (x+1) \sin x$.
 $\boxed{-(x+1) \cos x + \sin x + C}$

13. Integration by parts, harder.

(a) $\int \ln x dx$: the trick is $dv = dx$. Take $u = \ln x$, $dv = dx \Rightarrow du = \frac{1}{x} dx$, $v = x$. So $x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx = x \ln x - x + C$. Check: $\frac{d}{dx}(x \ln x - x) = \ln x + 1 - 1 = \ln x$. $\boxed{x \ln x - x + C}$

- (b) $\int x e^{2x} dx$: $u = x$, $dv = e^{2x} dx \Rightarrow du = dx$, $v = \frac{1}{2} e^{2x}$. So $\frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$. Check: $\frac{d}{dx} \left(\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} \right) = \frac{1}{2} e^{2x} + x e^{2x} - \frac{1}{2} e^{2x} = x e^{2x}$.

$$\boxed{\frac{1}{4}(2x - 1)e^{2x} + C}$$

- (c) $\int x^2 e^x dx$ (parts twice): First pass $u = x^2$, $dv = e^x dx \Rightarrow du = 2x dx$, $v = e^x$: $x^2 e^x - \int 2x e^x dx$. The leftover $\int 2x e^x dx = 2(xe^x - e^x)$ by part 12(a). So $x^2 e^x - 2xe^x + 2e^x + C$. Check: $\frac{d}{dx}((x^2 - 2x + 2)e^x) = (2x - 2)e^x + (x^2 - 2x + 2)e^x = x^2 e^x$. $\boxed{(x^2 - 2x + 2)e^x + C}$

- (d) $\int x^2 \cos x dx$ (parts twice): First pass $u = x^2$, $dv = \cos x dx \Rightarrow du = 2x dx$, $v = \sin x$: $x^2 \sin x - \int 2x \sin x dx$. The leftover $\int 2x \sin x dx = 2(-x \cos x + \sin x)$ by part 12(c). So $x^2 \sin x - 2(-x \cos x + \sin x) = x^2 \sin x + 2x \cos x - 2 \sin x + C$. Check: $\frac{d}{dx}(x^2 \sin x + 2x \cos x - 2 \sin x) = 2x \sin x + x^2 \cos x + 2 \cos x - 2x \sin x - 2 \cos x = x^2 \cos x$.

$$\boxed{x^2 \sin x + 2x \cos x - 2 \sin x + C}$$

- (e) $\int x \ln x dx$ (take $u = \ln x$): $u = \ln x$, $dv = x dx \Rightarrow du = \frac{1}{x} dx$, $v = \frac{x^2}{2}$. So $\frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2}{2} \ln x - \int \frac{x}{2} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$. Check: $\frac{d}{dx} \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) = x \ln x + \frac{x^2}{2} \cdot \frac{1}{x} - \frac{x}{2} = x \ln x$. $\boxed{\frac{x^2}{2} \ln x - \frac{x^2}{4} + C}$

Practice

14. $\int_0^1 x e^{x^2} dx$, two ways.

By substitution. Let $w = x^2$, $dw = 2x dx$, so $x dx = \frac{1}{2} dw$. Limits: $x : 0 \rightarrow 1$ becomes $w : 0 \rightarrow 1$. Then $\int_0^1 x e^{x^2} dx = \int_0^1 \frac{1}{2} e^w dw = \frac{1}{2} [e^w]_0^1 = \frac{1}{2}(e - 1)$.

By recognizing an antiderivative. Since $\frac{d}{dx} \left(\frac{1}{2} e^{x^2} \right) = \frac{1}{2} e^{x^2} \cdot 2x = x e^{x^2}$, an antiderivative is $\frac{1}{2} e^{x^2}$. Then $\left[\frac{1}{2} e^{x^2} \right]_0^1 = \frac{1}{2}(e^1 - e^0) = \frac{1}{2}(e - 1)$.

Both give the same value. $\boxed{\frac{e - 1}{2}}$

15. Area between $y = x$ (top) and $y = x^2$ (bottom) on $[0, 1]$. On $(0, 1)$, $x > x^2$, so x is the top curve. Slice into vertical strips of width dx and height (top - bottom) = $x - x^2$:

$$\text{Area} = \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

$$\boxed{\frac{1}{6}}$$

16. Region under $y = \frac{1}{x}$ on $[1, 4]$ spun about the x -axis. Slice into thin coins (disks) of radius $f(x) = \frac{1}{x}$ and thickness dx ; one disk has volume $\pi f(x)^2 dx = \pi \left(\frac{1}{x}\right)^2 dx = \frac{\pi}{x^2} dx$. Sum:

$$V = \int_1^4 \frac{\pi}{x^2} dx = \pi \left[-\frac{1}{x} \right]_1^4 = \pi \left(-\frac{1}{4} - (-1) \right) = \pi \cdot \frac{3}{4} = \frac{3\pi}{4}.$$

$$\boxed{\frac{3\pi}{4}}$$

17. Length of $y = x^2$ on $[0, 1]$ — set up only. Slice the curve into tiny nearly-straight pieces; one piece has length $\sqrt{dx^2 + dy^2} = \sqrt{1 + (y')^2} dx$. Here $y' = 2x$, so the integrand is $\sqrt{1 + (2x)^2} = \sqrt{1 + 4x^2}$:

$$L = \int_0^1 \sqrt{1 + 4x^2} dx.$$

Why no clean closed form: the integrand is the square root of a (quadratic) polynomial; the reverse power rule cannot touch a square root of a sum like this, and the antiderivative turns out to involve $\ln$ /inverse-hyperbolic terms ($\frac{x}{2}\sqrt{1 + 4x^2} + \frac{1}{4} \ln(2x + \sqrt{1 + 4x^2})$) rather than a tidy elementary value. Even for a humble parabola, arc length is naturally messy — exactly the chapter's point: set up the

slice, then often hand it to a Riemann sum.
$$L = \int_0^1 \sqrt{1 + 4x^2} dx$$

18. Find F with $F'(x) = \cos x$ and $F(0) = 3$. Antiderivatives of $\cos x$ are $\sin x + C$. Impose $F(0) = \sin 0 + C = C = 3$, so $F(x) = \sin x + 3$.
$$F(x) = \sin x + 3$$

Now $\int_0^\pi \cos x dx$ two ways. Using this F : $F(\pi) - F(0) = (\sin \pi + 3) - (\sin 0 + 3) = (0+3) - (0+3) = 0$. Directly (simplest antiderivative $\sin x$): $[\sin x]_0^\pi = \sin \pi - \sin 0 = 0 - 0 = 0$. They match — the constant 3 cancels in the subtraction, as FTC Part 2

guarantees for *any* antiderivative.
$$\int_0^\pi \cos x dx = 0$$

19. Show $\ln(2 \cdot 3) = \ln 2 + \ln 3$ from $\ln x = \int_1^x \frac{1}{t} dt$. By additivity of the integral

over adjacent intervals,

$$\ln 6 = \int_1^6 \frac{1}{t} dt = \int_1^2 \frac{1}{t} dt + \int_2^6 \frac{1}{t} dt = \ln 2 + \int_2^6 \frac{1}{t} dt.$$

In the second piece substitute $u = t/2$, so $t = 2u$, $dt = 2 du$, and $t : 2 \rightarrow 6$ becomes $u : 1 \rightarrow 3$:

$$\int_2^6 \frac{1}{t} dt = \int_1^3 \frac{1}{2u} (2 du) = \int_1^3 \frac{1}{u} du = \ln 3.$$

Therefore $\ln 6 = \ln 2 + \ln 3$. ■ (Numerically $\ln 6 \approx 1.7918 = \ln 2 + \ln 3 \approx 0.6931 + 1.0986$. ✓) $\ln 6 = \ln 2 + \ln 3$

Concept & counterexample

20. True/false with justification.

- (a) **True.** This is exactly FTC Part 1: if f is continuous, $\frac{d}{dx} \int_a^x f(t) dt = f(x)$. (Differentiating the area-so-far function returns the curve.) True
- (b) **True.** FTC Part 2 with F itself as the antiderivative of f' : $\int_a^b f'(x) dx = f(b) - f(a)$. (A net change equals the integral of the rate of change.) True
- (c) **True.** Both F and G are antiderivatives of f , so each computes the same definite integral by Part 2; equivalently F and G differ by a constant C , and $G(b) - G(a) = (F(b) + C) - (F(a) + C) = F(b) - F(a)$. The constant cancels. True
- (d) **True.** If f is continuous on an interval, the function $A(x) = \int_a^x f(t) dt$ is *defined* (the integral exists, Chapter 10) and, by FTC Part 1, satisfies $A' = f$. So A is an explicit antiderivative — even when no *elementary* formula exists. (Caveat: “continuous” is the operative hypothesis; a discontinuous f may fail to have one.) True

21. Why $\int_a^x f(t) dt$ is guaranteed to be an antiderivative even when no elementary one exists. The integral $A(x) = \int_a^x f(t) dt$ is a perfectly well-defined function: for each x it is a number, the signed area under f from a to x , and that area exists because f is continuous (Chapter 10 guarantees the Riemann limit exists). FTC Part 1 then says $A'(x) = f(x)$ directly from the moving-edge/strip argument — it never asks for a formula for A , only that f be continuous. So for $f(t) = e^{-t^2}$ the function erf-like $A(x) = \int_0^x e^{-t^2} dt$ is an antiderivative of e^{-t^2} , with $A'(x) = e^{-x^2}$,

even though no combination of polynomials, roots, exponentials, logs, and trig functions equals A . “Has an antiderivative” and “has an *elementary* antiderivative” are different claims; the integral supplies the former for free.

22. Find the error. The student wrote $\int_{-1}^1 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2$. **Error:** FTC Part 2 requires the integrand to be continuous on the *entire* closed interval $[-1, 1]$, but $\frac{1}{x^2}$ has an infinite discontinuity at $x = 0$, which lies inside $[-1, 1]$. The antiderivative $-\frac{1}{x}$ is likewise broken at 0, so plugging in the two endpoints across the gap is meaningless — that is exactly how a positive function produces a “negative area.” **Fix:** the integral is *improper* and must be split at 0 and taken as a limit:

$$\int_0^1 \frac{1}{x^2} dx = \lim_{c \rightarrow 0^+} \left[-\frac{1}{x}\right]_c^1 = \lim_{c \rightarrow 0^+} \left(-1 + \frac{1}{c}\right) = +\infty.$$

So $\int_{-1}^1 \frac{1}{x^2} dx$ **diverges** (is $+\infty$), consistent with the positive integrand. The bogus answer -2 came from applying FTC where its continuity hypothesis fails.

Discontinuity at $x = 0$; the integral diverges to $+\infty$.

23. What “undo” means in each part. - **Part 1** does **integration first, then differentiation**. Start with a continuous f , build $A(x) = \int_a^x f(t) dt$ (integrate), then differentiate: $A'(x) = f(x)$. You get back the *original integrand* f . In symbols, $\frac{d}{dx} \int_a^x f = f$. - **Part 2** does **differentiation first, then integration**. Start with a function F , differentiate it to get $f = F'$, then integrate that derivative over $[a, b]$: $\int_a^b F'(x) dx = F(b) - F(a)$. You get back the *net change of the original function*. In symbols, $\int_a^b F' = F(b) - F(a)$.

So differentiation and integration are inverse operations, with the two parts saying so in the two possible orders: integrate-then-differentiate returns the function; differentiate-then-integrate returns the (net change of the) function.

24. Is it possible? No. Suppose $\int_a^x f(t) dt = \int_a^x g(t) dt$ for all x . Differentiate both sides with respect to x ; by FTC Part 1 the left gives $f(x)$ and the right gives $g(x)$, so $f(x) = g(x)$ for every x . Hence $f = g$, contradicting $f \neq g$. *Two continuous functions with the same area-so-far function for all x must be identical.*

Not possible (for continuous f, g).

◆ Diamond

25. ◆ FTC Part 1, rigorously (strip argument). Let f be continuous and $A(x) = \int_a^x f(t) dt$. Fix x and take $h > 0$ (the case $h < 0$ is symmetric). By additivity, $A(x+h) - A(x) = \int_x^{x+h} f(t) dt$. On the closed interval $[x, x+h]$ the continuous function f attains a minimum m_h and a maximum M_h (Extreme Value Theorem, Chapter 2). Since $m_h \leq f(t) \leq M_h$ on that interval, the integral is squeezed between the areas of the short and tall rectangles of width h :

$$m_h \cdot h \leq \int_x^{x+h} f(t) dt = A(x+h) - A(x) \leq M_h \cdot h.$$

Divide by $h > 0$:

$$m_h \leq \frac{A(x+h) - A(x)}{h} \leq M_h.$$

As $h \rightarrow 0$, the interval $[x, x+h]$ shrinks to the single point x ; because f is continuous at x , both $m_h \rightarrow f(x)$ and $M_h \rightarrow f(x)$. The middle quantity, trapped between two things converging to $f(x)$, must converge to $f(x)$ as well (Squeeze Theorem, Chapter 2). But that middle quantity is precisely the difference quotient of A , so its limit is $A'(x)$. Therefore $A'(x) = f(x)$, and A is differentiable. ■

26. ◆ FTC Part 2 from Part 1. Let f be continuous and let F be *any* antiderivative, $F' = f$. By Part 1, $A(x) = \int_a^x f(t) dt$ is *also* an antiderivative ($A' = f$). Two antiderivatives of the same function on an interval differ by a constant (Chapter 6, a consequence of the Mean Value Theorem: $A' - F' = 0$ forces $A - F$ constant). So $F(x) = A(x) + C$ for some constant C . Evaluate the net change:

$$F(b) - F(a) = (A(b) + C) - (A(a) + C) = A(b) - A(a).$$

Now use $A(a) = \int_a^a f(t) dt = 0$ (an integral over zero width): $A(b) - A(a) = A(b) - 0 = A(b) = \int_a^b f(t) dt$. Hence

$$\int_a^b f(x) dx = F(b) - F(a). \quad \blacksquare$$

Where $A(a) = 0$ is used: at the very last step, to drop the $A(a)$ term so that $F(b) - F(a)$ equals the *full* integral $\int_a^b f$ rather than $\int_a^b f - \int_a^a f$. (It is also what makes the cancellation land on the integral and not on some unknown baseline.)

27. ♦ Integration by parts from the product rule; evaluate $\int \ln x \, dx$. Start from the product rule for two differentiable functions $u(x), v(x)$:

$$\frac{d}{dx}(uv) = u'v + uv'.$$

Integrate both sides over x . The left side integrates to uv (it is an exact derivative). Rearranging,

$$uv = \int u'v \, dx + \int uv' \, dx \implies \int uv' \, dx = uv - \int u'v \, dx,$$

i.e. $\int u \, dv = uv - \int v \, du$. This is integration by parts — the product rule read right to left.

Apply to $\int \ln x \, dx$. There is no obvious product, so we *invent* one by choosing $dv = dx$. Take $u = \ln x$ (whose derivative $\frac{1}{x}$ is simple) and $dv = dx$. Then $du = \frac{1}{x} dx$ and $v = x$. **Why $dv = dx$:** $\ln x$ has no elementary antiderivative we can write down by inspection, but its *derivative* $\frac{1}{x}$ is elementary, so we want $\ln x$ to be the differentiated factor u ; that forces the remaining “ dx ” to be dv . Then

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Check: $\frac{d}{dx}(x \ln x - x) = \ln x + 1 - 1 = \ln x$. ✓ $\int \ln x \, dx = x \ln x - x + C$

28. ♦♦ Power law $\ln(x^r) = r \ln x$ for r a positive integer, from $\ln x = \int_1^x \frac{1}{t} dt$. We use the product law $\ln(ab) = \ln a + \ln b$ (proved in the chapter from the definition), and induct on r .

Base case $r = 1$: $\ln(x^1) = \ln x = 1 \cdot \ln x$. ✓

Inductive step: assume $\ln(x^k) = k \ln x$ for some positive integer k . Then, writing $x^{k+1} = x^k \cdot x$ and applying the product law,

$$\ln(x^{k+1}) = \ln(x^k \cdot x) = \ln(x^k) + \ln x = k \ln x + \ln x = (k + 1) \ln x.$$

By induction, $\ln(x^r) = r \ln x$ holds for every positive integer r . ■

(*Direct substitution alternative for completeness:* in $\ln(x^r) = \int_1^{x^r} \frac{1}{t} dt$ substitute $t = s^r$, $dt = rs^{r-1} ds$, with $t : 1 \rightarrow x^r$ giving $s : 1 \rightarrow x$; then $\int_1^{x^r} \frac{1}{t} dt = \int_1^x \frac{1}{s^r} rs^{r-1} ds = r \int_1^x \frac{1}{s} ds = r \ln x$.) $\ln(x^r) = r \ln x$

29. ♦♦ e (the number with $\ln e = 1$) exists and is unique. Work from $\ln x = \int_1^x \frac{1}{t} dt$ on $(0, \infty)$.

Existence (Intermediate Value Theorem). $\ln$ is continuous (it is differentiable with $\ln' x = \frac{1}{x}$ by FTC Part 1, and differentiable $\Rightarrow$ continuous). We have $\ln 1 = 0 < 1$. And $\ln$ grows without bound: e.g. comparing areas, $\ln 2^n = n \ln 2$ (power law, #28) with $\ln 2 = \int_1^2 \frac{1}{t} dt > 0$, so $\ln 2^n \rightarrow \infty$; thus $\ln$ exceeds 1 at some large input, say $\ln N > 1$. Since $\ln$ is continuous on $[1, N]$ with $\ln 1 = 0 < 1 < \ln N$, the IVT (Chapter 2) guarantees a point $e \in (1, N)$ with $\ln e = 1$. So e **exists**.

Uniqueness (strict monotonicity). $\ln' x = \frac{1}{x} > 0$ for $x > 0$, so $\ln$ is *strictly increasing*; a strictly increasing function takes each value at most once. Hence there is exactly one x with $\ln x = 1$. So e is **unique**.

Discharging the Chapter 1 loan. In Chapter 1 we merely *asserted* that the limit defining e converges — that a number e exists at all. Here that assertion becomes a theorem: e is pinned down as the unique right-edge where the area under $\frac{1}{t}$ first equals 1, its existence delivered by the IVT and its uniqueness by monotonicity, with nothing assumed beyond the integral definition. The promissory note is paid.

■

30. ♦♦ The inverse of $\ln$ is its own derivative. Define $E(x)$ as the inverse function of $\ln$, so $\ln(E(x)) = x$ for all x (and $E(\ln y) = y$ for $y > 0$). Since $\ln$ is differentiable with $\ln' > 0$ everywhere, the inverse-function rule (Chapter 4) applies:

$$E'(x) = \frac{1}{\ln'(E(x))}.$$

But $\ln'(t) = \frac{1}{t}$, so $\ln'(E(x)) = \frac{1}{E(x)}$. Substituting,

$$E'(x) = \frac{1}{1/E(x)} = E(x).$$

Thus $E'(x) = E(x)$: the inverse of the log-integral is unchanged by differentiation. A function equal to its own derivative, with $E(0) = 1$ (since $\ln 1 = 0$), is exactly the exponential — so $E(x)$ deserves the name e^x . ■

Teach it

31. (Model answer — moving-edge picture, FTC Part 1.) Picture the area under the curve f as a shaded region whose right edge sits at x ; call that running area $A(x)$. Slide the right edge a hair to the right, by h : the shaded region gains a thin vertical sliver of width h and height essentially $f(x)$, so the extra area is about $f(x) \cdot h$. The rate at which area grows is therefore $\frac{A(x+h)-A(x)}{h} \approx f(x)$, and as the sliver thins to nothing ($h \rightarrow 0$) this becomes exact: $A'(x) = f(x)$. In one line — *how fast the area grows is just how tall the curve is right there* — which is why differentiating an area gives back the curve.

32. Debug your classmate. They wrote $\int_0^\pi \sin x \, dx = [\cos x]_0^\pi = \cos \pi - \cos 0 = -2$. **Two errors:** 1. *Wrong antiderivative.* The antiderivative of $\sin x$ is $-\cos x$, **not** $\cos x$ (since $\frac{d}{dx}(-\cos x) = \sin x$, whereas $\frac{d}{dx} \cos x = -\sin x$). They dropped the minus sign. 2. *Sign of the result.* Because of that, their numerical value comes out -2 , but $\sin x \geq 0$ on $[0, \pi]$, so the area must be **positive** — a negative answer is an immediate red flag.

Correct computation: $\int_0^\pi \sin x \, dx = [-\cos x]_0^\pi = -\cos \pi - (-\cos 0) = -(-1) + 1 = 2$. 2

33. Worked solution to $\int x^2 e^x \, dx$ (parts twice), narrated.

We integrate by parts, $\int u \, dv = uv - \int v \, du$. The integrand is a *polynomial* times e^x . The key insight: choose u = the polynomial, because differentiating it (the du step) *lowers its degree*, while e^x integrates back to e^x and never gets worse. Each pass therefore trades the current integral for one with the power reduced by one — so after enough passes the polynomial disappears.

Pass 1. Let $u = x^2$, $dv = e^x \, dx$. Then $du = 2x \, dx$, $v = e^x$:

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx.$$

The power dropped from 2 to 1.

Pass 2 (on the leftover $\int 2x e^x \, dx$). Let $u = 2x$, $dv = e^x \, dx$. Then $du = 2 \, dx$, $v = e^x$:

$$\int 2x e^x \, dx = 2x e^x - \int 2 e^x \, dx = 2x e^x - 2e^x.$$

The power dropped from 1 to 0 — now it's a bare $\int e^x$, which we just integrate.

Assemble:

$$\int x^2 e^x dx = x^2 e^x - (2x e^x - 2e^x) + C = (x^2 - 2x + 2)e^x + C.$$

Check: $\frac{d}{dx}((x^2 - 2x + 2)e^x) = (2x - 2)e^x + (x^2 - 2x + 2)e^x = x^2 e^x$. ✓

$$\boxed{(x^2 - 2x + 2)e^x + C}$$

For a peer doing $\int x^3 e^x dx$: same move, **three** passes — the power falls $3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ — giving $\int x^3 e^x dx = (x^3 - 3x^2 + 6x - 6)e^x + C$. (Check by differentiating, or note each pass peels off one factor exactly as above.)

34. Teach the slicing principle. Every “total” you want to measure — an area trapped between two curves, the volume of a spun solid, the length of a curve — is computed by the same two-step move: ask *what one infinitesimally thin piece contributes*, then *add up all the pieces with an integral*. For area between curves, one vertical strip of width dx contributes (top – bottom) dx , so the total is $\int_a^b (\text{top} - \text{bottom}) dx$. For a solid of revolution, one coin-shaped disk of radius $f(x)$ and thickness dx contributes $\pi f(x)^2 dx$, so the volume is $\int_a^b \pi f(x)^2 dx$. For arc length, one tiny near-straight segment contributes $\sqrt{1 + (f'(x))^2} dx$ by the Pythagorean theorem, so the length is $\int_a^b \sqrt{1 + (f'(x))^2} dx$. The three “formulas” are not three things to memorize — they are one idea (*slice into measurable pieces, sum them*) applied to three shapes, and once you can write down what one slice contributes you can rederive any of them on the spot.

35. Teach the capstone. (*Model answer.*) In Chapter 5 we simply *announced* that $\ln' x = \frac{1}{x}$ and that logs turn products into sums — true statements, but unearned: they presuppose that a function called $\ln$ already exists with all the right properties, which is precisely what was never established. *Defining* $\ln x = \int_1^x \frac{1}{t} dt$ flips this around. The right-hand side is a genuine object built only from the integral, which we constructed honestly in Chapter 10 — so $\ln$ now *exists* by construction, no faith required. And every property we previously assumed becomes a *consequence*: $\ln' x = \frac{1}{x}$ falls out in one line from FTC Part 1; $\ln 1 = 0$ is an integral over zero width; the product law comes from splitting the integral and substituting; and even the number e (the input where the area first hits 1) is pinned down by the Intermediate Value Theorem. The assertion gave us rules to use; the definition gives us the *object those rules describe* and *proves the rules from scratch* — that is what makes it honest, and it is why the book can say at the summit that every loan is repaid.

Looking ahead

36. $\int_1^\infty \frac{1}{x} dx$ — finite or infinite? Compute the finite version first and take the limit:

$$\int_1^b \frac{1}{x} dx = [\ln x]_1^b = \ln b - \ln 1 = \ln b.$$

As $b \rightarrow \infty$, $\ln b \rightarrow \infty$ (the log grows without bound, #29). So

$$\int_1^\infty \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln b = +\infty \quad (\mathbf{infinite}).$$

The area under $\frac{1}{x}$ never stops accumulating. By the picture Chapter 12 will make exact, this matches the **divergence** of the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \cdots = +\infty$.

Infinite (diverges); harmonic series diverges too.

37. $\int_1^\infty \frac{1}{x^2} dx$, compared with #36. Compute:

$$\int_1^b \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_1^b = -\frac{1}{b} - (-1) = 1 - \frac{1}{b}.$$

As $b \rightarrow \infty$, $\frac{1}{b} \rightarrow 0$, so

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b}\right) = 1 \quad (\mathbf{finite}).$$

Contrast: $\int_1^\infty \frac{1}{x^2} = 1$ is finite, while $\int_1^\infty \frac{1}{x} = \infty$ is infinite — the extra power of x in the denominator makes the tail shrink fast enough to enclose a finite area.

Prediction: $\sum \frac{1}{n^2}$ **converges** (its matching area is finite) while $\sum \frac{1}{n}$ **diverges** (its matching area is infinite).

The rule being sensed: a sum $\sum f(n)$ and the area $\int_1^\infty f(x) dx$ of the same positive, decreasing f live or die together — both converge or both diverge. That is exactly Chapter 12's **integral test**, built from this chapter's area pictures (a sum of rectangles bracketed above and below by the curve's area).

$\sum 1/n^2$ converges, $\sum 1/n$ diverges.

Chapter 12

Adding Up Forever

Fluency

1. Geometric series $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ when $|r| < 1$; diverges when $|r| \geq 1$.

(a) $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$: $a = 1$, $r = \frac{1}{3}$, $|r| < 1$. Sum = $\frac{1}{1 - \frac{1}{3}} = \frac{1}{2/3} = \boxed{\frac{3}{2}}$.

(b) $\sum_{n=0}^{\infty} \left(\frac{3}{2}\right)^n$: $r = \frac{3}{2}$, $|r| = \frac{3}{2} \geq 1$. Terms grow, partial sums run away. **Diverges**.

(c) $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^n$: starts at $n = 1$, so first term is $a = \frac{1}{5}$, $r = \frac{1}{5}$. Sum = $\frac{1/5}{1 - 1/5} = \frac{1/5}{4/5} = \boxed{\frac{1}{4}}$.

(d) $\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n$: $a = 1$, $r = -\frac{1}{2}$, $|r| = \frac{1}{2} < 1$. Sum = $\frac{1}{1 - (-\frac{1}{2})} = \frac{1}{3/2} = \boxed{\frac{2}{3}}$.

2. Term test: if $a_n \not\rightarrow 0$ the series **diverges**; if $a_n \rightarrow 0$ the test is **inconclusive**.

(a) $a_n = \frac{n}{2n+1} \rightarrow \frac{1}{2} \neq 0$. **Diverges** by the term test.

(b) $a_n = \frac{1}{n} \rightarrow 0$. **Inconclusive** — term test can't rule it out. (In fact it diverges, harmonic series, but not by *this* test.)

(c) $a_n = \frac{n^2}{n^2+1} \rightarrow 1 \neq 0$. **Diverges** by the term test.

(d) $a_n = \frac{1}{\sqrt{n}} \rightarrow 0$. **Inconclusive**. (In fact it diverges — a p -series with $p = \frac{1}{2} \leq 1$ — but not by the term test.)

3. p -series $\sum \frac{1}{n^p}$ converges $\iff p > 1$.

(a) $\sum \frac{1}{n^3}$: $p = 3 > 1$. **Converges**.

(b) $\sum \frac{1}{n^{1/2}}$: $p = \frac{1}{2} \leq 1$. **Diverges**.

(c) $\sum \frac{1}{n}$: $p = 1$. **Diverges** (the harmonic series).

(d) $\sum \frac{1}{n^{1.01}}$: $p = 1.01 > 1$. **Converges** (barely, but $p > 1$ is all that matters).

4. Comparison. All terms positive.

(a) $\frac{1}{2^n + n} \leq \frac{1}{2^n}$ for all $n \geq 1$, and $\sum \frac{1}{2^n}$ is a convergent geometric series ($r = \frac{1}{2}$). By comparison, **converges**.

(b) $\frac{1}{n^2 + 1} \leq \frac{1}{n^2}$, and $\sum \frac{1}{n^2}$ is a convergent p -series ($p = 2$). By comparison, **converges**.

(c) For $n \geq 2$, $n^2 - n = n(n - 1) \geq \frac{n^2}{2}$ once $n \geq 2$ (since $n - 1 \geq \frac{n}{2}$ there), so $\frac{1}{n^2 - n} \leq \frac{2}{n^2}$, and $\sum \frac{2}{n^2}$ converges ($p = 2$). By comparison, **converges**.
(Indeed it telescopes: $\frac{1}{n^2 - n} = \frac{1}{n-1} - \frac{1}{n}$, so $\sum_{n=2}^{\infty} = 1$.)

5. Radius via ratio of coefficients: if $\left| \frac{c_{n+1}}{c_n} \right| \rightarrow L$ then $R = \frac{1}{L}$ ($R = \infty$ if $L = 0$, $R = 0$ if $L = \infty$).

(a) $\sum \frac{x^n}{n!}$: $\frac{c_{n+1}}{c_n} = \frac{1/(n+1)!}{1/n!} = \frac{1}{n+1} \rightarrow 0 = L$. $R = \frac{1}{0} = \boxed{\infty}$.

(b) $\sum x^n$: $c_n = 1$, ratio $= 1 = L$. $R = \boxed{1}$.

(c) $\sum \frac{x^n}{2^n}$: $\frac{c_{n+1}}{c_n} = \frac{1/2^{n+1}}{1/2^n} = \frac{1}{2} = L$. $R = \frac{1}{1/2} = \boxed{2}$.

(d) $\sum n! x^n$: $\frac{c_{n+1}}{c_n} = \frac{(n+1)!}{n!} = n+1 \rightarrow \infty = L$. $R = \boxed{0}$ — converges only at $x = 0$.

6. $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so $S_N = 1 - \frac{1}{N+1}$ (everything cancels).

$$S_1 = 1 - \frac{1}{2} = \frac{1}{2}, \quad S_2 = 1 - \frac{1}{3} = \frac{2}{3}, \quad S_3 = 1 - \frac{1}{4} = \frac{3}{4}, \quad S_4 = 1 - \frac{1}{5} = \frac{4}{5}.$$

The pattern $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots = 1 - \frac{1}{N+1}$ climbs toward 1. **Limit** = 1.

Practice

7. $f(x) = \frac{1}{x \ln x}$ is positive, continuous, and decreasing for $x \geq 2$ (both x and $\ln x$ increase), so the **integral test** applies. Substitute $u = \ln x$, $du = \frac{dx}{x}$:

$$\int_2^\infty \frac{dx}{x \ln x} = \int_{\ln 2}^\infty \frac{du}{u} = \left[\ln u \right]_{\ln 2}^\infty = \lim_{b \rightarrow \infty} (\ln(\ln b) - \ln(\ln 2)) = \infty.$$

The integral diverges, so the series **diverges**. (Even though $\frac{1}{n \ln n} \rightarrow 0$ — another “terms vanish, sum infinite” case, just slower than harmonic.)

8. $f(x) = \frac{1}{x^2}$ is positive, continuous, decreasing on $[1, \infty)$.

$$\int_1^\infty \frac{dx}{x^2} = \left[-\frac{1}{x} \right]_1^\infty = \lim_{b \rightarrow \infty} \left(-\frac{1}{b} + 1 \right) = 1 < \infty.$$

Finite integral $\Rightarrow \sum \frac{1}{n^2}$ **converges**. For $\sum \frac{1}{n}$, take $f(x) = \frac{1}{x}$:

$$\int_1^\infty \frac{dx}{x} = \left[\ln x \right]_1^\infty = \lim_{b \rightarrow \infty} \ln b = \infty.$$

Infinite integral $\Rightarrow \sum \frac{1}{n}$ **diverges**. (The two integrals — finite vs. infinite — decide the two sums, exactly as Chapter 11 foretold.)

9. Use $R = 1/L$, $L = \lim |c_{n+1}/c_n|$.

(a) $\sum \frac{x^n}{n}$: $\frac{c_{n+1}}{c_n} = \frac{n}{n+1} \rightarrow 1 = L$, so $R = 1$. Interval $\boxed{(-1, 1)}$.

(b) $\sum \frac{x^n}{n^2}$: $\frac{c_{n+1}}{c_n} = \frac{n^2}{(n+1)^2} \rightarrow 1 = L$, so $R = 1$. Interval $\boxed{(-1, 1)}$.

(c) $\sum \frac{(x-2)^n}{3^n}$: this is geometric in $(x-2)$ with ratio $\frac{x-2}{3}$; it converges iff $\left| \frac{x-2}{3} \right| < 1$, i.e. $|x-2| < 3$, so $R = 3$. Centered at 2, the interval is $\boxed{(-1, 5)}$.

10. In $\frac{1}{1-u} = \sum_{n=0}^\infty u^n$ (valid for $|u| < 1$), substitute $u = -x^2$:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^\infty (-x^2)^n = \sum_{n=0}^\infty (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \dots,$$

valid where $| -x^2 | < 1$, i.e. $|x| < 1$. **Connection:** since $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$, integrating this series term by term gives

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots,$$

Chapter 7's arctangent series — and it inherits the same radius $R = 1$.

11. Drop height $H = 2$ m; each bounce rises to $r = \frac{3}{4}$ of the previous height. - Initial **fall:** 2 m. - After bounce k the ball rises $2 \cdot (\frac{3}{4})^k$ and then falls the same distance — each rise height is traveled **twice** (up then down).

$$\text{Total} = 2 + 2 \sum_{k=1}^{\infty} 2\left(\frac{3}{4}\right)^k = 2 + 4 \cdot \frac{3/4}{1-3/4} = 2 + 4 \cdot \frac{3/4}{1/4} = 2 + 4 \cdot 3 = \boxed{14 \text{ m}}.$$

(Geometric series with $r = \frac{3}{4} < 1$, so the total path is finite even though the ball bounces forever.)

12. Decide and name the principle.

- (a) $\sum \frac{n}{n^3+1}$: for $n \geq 1$, $\frac{n}{n^3+1} \leq \frac{n}{n^3} = \frac{1}{n^2}$; $\sum \frac{1}{n^2}$ converges ($p = 2$). **Converges** — comparison to a p -series.
- (b) $\sum \frac{1}{2^n}$: geometric with $r = \frac{1}{2} < 1$. **Converges** — geometric series.
- (c) $\sum \frac{1}{\sqrt{n}}$: p -series with $p = \frac{1}{2} \leq 1$. **Diverges** — p -series.
- (d) $\sum \frac{n+1}{n}$: $a_n = 1 + \frac{1}{n} \rightarrow 1 \neq 0$. **Diverges** — term test.

Concept & counterexample

13. True/false.

- (a) **False.** $a_n \rightarrow 0$ does not imply convergence — the harmonic series $\sum \frac{1}{n}$ has $a_n \rightarrow 0$ yet diverges. (The converse of the term test fails.)
- (b) **True.** This is the term test: convergence forces $a_n = S_n - S_{n-1} \rightarrow S - S = 0$.
- (c) **True.** Its terms $\frac{1}{n} \rightarrow 0$, yet by Oresme's blocking the partial sums grow without bound. This is the headline counterexample to (a).
- (d) **True.** $\sum r^n = \frac{1}{1-r}$ exactly when $|r| < 1$; that is the convergence condition

for a geometric series.

14. “Terms $\rightarrow 0$ ” is **necessary**: the term test says any convergent series must have $a_n \rightarrow 0$, so a series whose terms *don’t* die cannot converge. But it is **not sufficient**: the harmonic series $\sum \frac{1}{n}$ has terms $\frac{1}{n} \rightarrow 0$ and still diverges. The reason is that *how fast* the terms shrink decides the outcome — group $\frac{1}{n}$ into blocks $(\frac{1}{3} + \frac{1}{4}), (\frac{1}{5} + \dots + \frac{1}{8}), \dots$, each exceeding $\frac{1}{2}$; infinitely many half-units accumulate to ∞ . Vanishing terms are a prerequisite, never a guarantee.

15. Take $\sum \frac{1}{\sqrt{n}}$ (a p -series with $p = \frac{1}{2} \leq 1$). Its terms $\frac{1}{\sqrt{n}} \rightarrow 0$, yet it **diverges** by the p -series rule. (Check via the integral test: $\int_1^\infty x^{-1/2} dx = [2\sqrt{x}]_1^\infty = \infty$.) So it is a divergent series with vanishing terms, distinct from the harmonic series.

16. The error: the student used the term test *backwards*. The term test only says convergence $\Rightarrow a_n \rightarrow 0$; it does **not** say $a_n \rightarrow 0 \Rightarrow$ convergence. Showing $\frac{1}{n} \rightarrow 0$ merely fails to rule out convergence — it is **inconclusive**, not a proof. **Correct conclusion:** $\sum \frac{1}{n}$ **diverges** (harmonic series; proved by Oresme’s blocking or the integral test $\int_1^\infty \frac{dx}{x} = \infty$).

17. The picture works only if each term $f(n)$ is the height of a single rectangle that the curve cleanly under- or over-shoots. - **Positive:** if f dipped negative, “area under the curve” could cancel against positive parts, and a rectangle of negative height no longer bounds the corresponding term — the squeeze $\sum f(n) \leq \int f \leq \sum f(n)$ collapses. - **Decreasing:** the inequalities $f(n+1) \leq \int_n^{n+1} f \leq f(n)$ rely on f falling monotonically across each unit interval. If f **wandered up and down**, on some intervals the rectangle of height $f(n)$ would sit *below* the curve and on others *above*, so neither the under-estimate nor the over-estimate would hold uniformly, and the integral would no longer trap the sum. Monotonicity is what guarantees a consistent comparison on every interval.

18. Yes, it’s possible. For $R = 0$ the coefficients must grow so fast that for any fixed $x \neq 0$ the terms $c_n x^n$ eventually blow up. Concretely, $\left| \frac{c_{n+1}}{c_n} \right| \rightarrow \infty$ forces $L = \infty$ and $R = \frac{1}{L} = 0$. **Example:** $\sum n! x^n$ has $\frac{c_{n+1}}{c_n} = n+1 \rightarrow \infty$, so $R = 0$: at any $x \neq 0$ the terms $n! x^n \rightarrow \infty$ (factorial growth beats every geometric decay), failing the term test, while at $x = 0$ every term past the first is 0 and the series trivially converges to c_0 . So it converges **only at its center**.

◆ Diamond

19. Let $S_N = \sum_{n=0}^N r^n = 1 + r + \dots + r^N$. Multiply by r : $rS_N = r + r^2 + \dots + r^{N+1}$. Subtract:

$$S_N - rS_N = 1 - r^{N+1} \implies (1-r)S_N = 1 - r^{N+1} \implies S_N = \frac{1 - r^{N+1}}{1-r} \quad (r \neq 1).$$

Now let $N \rightarrow \infty$. **This is exactly where $|r| < 1$ is used:** when $|r| < 1$, $r^{N+1} \rightarrow 0$, so

$$\sum_{n=0}^{\infty} r^n = \lim_{N \rightarrow \infty} \frac{1 - r^{N+1}}{1-r} = \frac{1-0}{1-r} = \frac{1}{1-r}. \quad \blacksquare$$

If $|r| \geq 1$ then r^{N+1} does not tend to 0 (it stays ≥ 1 in size or oscillates), the limit fails, and the series diverges.

20. Suppose $\sum a_n$ converges, so the partial sums $S_n \rightarrow S$ for some finite S . Then also $S_{n-1} \rightarrow S$. Since

$$a_n = S_n - S_{n-1},$$

taking limits gives $\lim_{n \rightarrow \infty} a_n = \lim S_n - \lim S_{n-1} = S - S = 0$. Hence $a_n \rightarrow 0$. $\blacksquare$

Why the converse fails (one sentence): the harmonic series $\sum \frac{1}{n}$ has $a_n \rightarrow 0$ but diverges, so $a_n \rightarrow 0$ never implies convergence.

21. Group the harmonic series into blocks whose lengths double:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\text{block 1}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{\text{block 2}} + \underbrace{\left(\frac{1}{9} + \dots + \frac{1}{16}\right)}_{\text{block 3}} + \dots$$

The k -th block runs from $\frac{1}{2^{k+1}}$ to $\frac{1}{2^{k+1}}$ and has exactly 2^k terms, each $\geq \frac{1}{2^{k+1}}$ (the smallest, last one). Explicitly: - Block 1: $\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = 2 \cdot \frac{1}{4} = \frac{1}{2}$. - Block 2: $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > 4 \cdot \frac{1}{8} = \frac{1}{2}$. - Block 3: $\frac{1}{9} + \dots + \frac{1}{16} > 8 \cdot \frac{1}{16} = \frac{1}{2}$.

In general block k contributes $> 2^k \cdot \frac{1}{2^{k+1}} = \frac{1}{2}$. The partial sums therefore exceed $1 + \frac{1}{2} \cdot (\text{number of completed blocks})$, which grows without bound. So $\sum \frac{1}{n} = \infty$. $\blacksquare$

22. Let f be positive, decreasing, continuous on $[1, \infty)$. On each interval $[n, n+1]$, monotonicity gives, for all x in $[n, n+1]$, $f(n+1) \leq f(x) \leq f(n)$. Integrate over

the unit-width interval:

$$f(n+1) \leq \int_n^{n+1} f(x) dx \leq f(n).$$

Over-estimate (left rectangles, height $f(n)$): sum the right inequality from $n = 1$ to ∞ :

$$\int_1^{\infty} f(x) dx = \sum_{n=1}^{\infty} \int_n^{n+1} f \leq \sum_{n=1}^{\infty} f(n).$$

Under-estimate (right rectangles, height $f(n+1)$): sum the left inequality:

$$\sum_{n=1}^{\infty} f(n+1) \leq \int_1^{\infty} f(x) dx, \quad \text{i.e.} \quad \sum_{n=2}^{\infty} f(n) \leq \int_1^{\infty} f(x) dx.$$

Combining, $\sum_{n=2}^{\infty} f(n) \leq \int_1^{\infty} f \leq \sum_{n=1}^{\infty} f(n)$. The outer two differ by exactly the finite term $f(1)$, so the sum and the integral are **finite together or infinite together**:
 - if $\int_1^{\infty} f < \infty$, the right inequality bounds $\sum_{n=2}^{\infty} f(n)$ (hence the whole sum) above — the series converges;
 - if $\int_1^{\infty} f = \infty$, the left inequality forces $\sum_{n=1}^{\infty} f(n) = \infty$ — the series diverges.

Thus $\sum f(n)$ and $\int_1^{\infty} f$ converge together. ■

23. Take $f(x) = x^{-p}$, positive, continuous, and decreasing on $[1, \infty)$ for $p > 0$, so the integral test applies. For $p \neq 1$ (Chapter 11):

$$\int_1^{\infty} x^{-p} dx = \left[\frac{x^{1-p}}{1-p} \right]_1^{\infty} = \frac{1}{1-p} \left(\lim_{b \rightarrow \infty} b^{1-p} - 1 \right).$$

- If $p > 1$: $1-p < 0$, so $b^{1-p} \rightarrow 0$ and the integral $= \frac{1}{p-1} < \infty$. **Converges.**
 - If $0 < p < 1$: $1-p > 0$, so $b^{1-p} \rightarrow \infty$ and the integral diverges. **Diverges.**
 - $p = 1$ **separately:** $\int_1^{\infty} x^{-1} dx = [\ln x]_1^{\infty} = \infty$, diverges (the harmonic series).
 - (If $p \leq 0$ the terms don't even tend to 0, so it diverges by the term test.)

By the integral test the series shares each verdict, so $\sum \frac{1}{n^p}$ **converges iff $p > 1$** .
 ■

24. Fix any real x . Apply the ratio test to the terms $a_n = \frac{x^n}{n!}$ (for $x \neq 0$; at $x = 0$

the sum is trivially 1):

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \frac{|x|}{n+1} \xrightarrow{n \rightarrow \infty} 0 < 1.$$

Since the limiting ratio is $0 < 1$ for **every** x , the series converges (absolutely) for all x : the radius is $R = \infty$. ■ **Reconciliation with Chapter 7.** There we bounded the Taylor remainder $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} x^{n+1}$; since $|f^{(n+1)}| = e^\xi \leq e^{|x|}$ is bounded on the interval, $|R_n(x)| \leq e^{|x|} \frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$, so the *partial sums of the Taylor series reach* e^x . The ratio test instead shows directly that *the series converges to some number*. Both rest on the same engine — $\frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$, factorial growth crushing any power — viewed once as “the leftover vanishes” and once as “the next term shrinks geometrically.” Same fact, two doors.

Teach it

25. To add up infinitely many numbers, you never actually add forever — you add the first N of them, getting a *partial sum* S_N , and watch where those partial sums go as N grows. The series **converges** to S exactly when that sequence of partial sums closes in on S , nothing more mysterious than the limits from Chapter 1. For $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ the partial sums are $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \dots$, each halfway closer to 1, so the running total settles onto 1 — that’s all “the sum is 1” means.

26. The one fact: **the harmonic series** $1 + \frac{1}{2} + \frac{1}{3} + \dots$ **diverges to infinity even though its terms shrink to zero** — so “terms heading to zero” never proves convergence; it’s necessary, not sufficient. (Group the terms into doubling blocks each exceeding $\frac{1}{2}$, and infinitely many half-units add to ∞ .)

27. Worked solution — does $\sum \frac{1}{n^2 + 1}$ **converge**? The terms are positive, so we use the **comparison test**. For every $n \geq 1$,

$$n^2 + 1 > n^2 \implies \frac{1}{n^2 + 1} < \frac{1}{n^2}.$$

Our comparison series $\sum \frac{1}{n^2}$ is a **p -series with $p = 2 > 1$, hence convergent**.

Since $0 < \frac{1}{n^2 + 1} < \frac{1}{n^2}$ term by term and the larger series converges, by compari-

son the smaller one converges too.

$$\sum \frac{1}{n^2 + 1} \text{ converges.}$$

(Template for a peer doing $\sum \frac{1}{n^3+5}$: note $\frac{1}{n^3+5} < \frac{1}{n^3}$, a convergent p -series with $p = 3$, so it converges by comparison.)

28. Picture the terms $f(1), f(2), f(3), \dots$ as the heights of unit-width rectangles, and lay them against the graph of a positive, decreasing curve $y = f(x)$. Drawn one way (left edges), each rectangle pokes *above* the curve, so the rectangles' total area — the sum — is at least the area under the curve; drawn the other way (right edges), each tucks *below*, so the sum is at most that same area (plus one stray term). The integral is thus squeezed between the sum and the sum-minus-a-term, differing by only the finite $f(1)$. Because that gap is finite, the sum and the integral **must be finite together or infinite together** — so you can settle an infinite sum just by computing an area you already know how to integrate.

29. Designed problem: Decide whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. **Solution key.**

The terms are positive, and $f(x) = \frac{1}{x(\ln x)^2}$ is continuous and decreasing for $x \geq 2$ (both x and $(\ln x)^2$ increase), so the **integral test** applies. Substitute $u = \ln x$, $du = \frac{dx}{x}$:

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^{\infty} \frac{du}{u^2} = \left[-\frac{1}{u} \right]_{\ln 2}^{\infty} = 0 - \left(-\frac{1}{\ln 2} \right) = \frac{1}{\ln 2} < \infty.$$

The integral is finite, so by the integral test the series **converges**. (Teaching point: contrast with $\sum \frac{1}{n \ln n}$, which has the same flavor but gives $\int \frac{du}{u} = \infty$ and diverges — the extra power of $\ln n$ is exactly what tips it into convergence.)

Looking ahead

30. Prediction: waves are better building blocks for a *repeating* signal because $\sin x$ and $\cos x$ are themselves **periodic** — they return to the same value every 2π and stay bounded forever — whereas x^n does the opposite: a polynomial term grows without bound and never repeats, so reproducing a signal that cycles (a sound, a heartbeat) with powers fights the function's nature, while a sum of $\sin$

and $\cos$ already *has* the repetition built in. A periodic signal of period T is naturally matched by sines and cosines whose periods divide T ($\sin x, \sin 2x, \dots$), so few terms capture it; a Taylor (power) series, accurate only near one point and blowing up far away, would need to fight to stay bounded over every cycle.

31. Speculation: the constant π slips into $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ because, although the *left* side is built from pure number-fractions, the *machinery* that evaluates it is circular. Book 3 will expand a function in **waves** $\sin nx, \cos nx$ — and waves are circular motion in disguise (a point going around a circle traces a sine), so π , the circle's own number, is baked into every frequency. Matching such a wave expansion to a simple function and reading off coefficients converts an integral of $\sin/\cos$ (which carries π) into the sum $\sum \frac{1}{n^2}$ — so the π on the right is the fingerprint of the circles hiding inside the sines used to compute it. Pure fractions know about circles because the *only known way* to add them up runs through the geometry of waves. The surprise is real, and it is exactly the bridge into Book 3.